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School of Engineering

My use of AI in Education

Jérôme Landré - 2026-10-20



Outline

1. Teaching and Learning
2. Why use AI in Education?
3. How to use AI in Education?
4. Online AI chatbots
    1. Available tools
    2. Prompt engineering
    3. Security issues
    4. Ethics and sustainability
5. Local AI chatbots
6. Conclusion
7. References

1. Teaching and Learning

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Teaching

Teacher's responsibility

- To deliver knowledge
- To create learning activities
- To evaluate the students' examination

Learning

Student's responsibility

- To acquire knowledge
- To participate to the learning activities
- To succeed the course

So, a course is 50% Teaching and 50% Learning

It is a **teamwork**: you **teach**, students **learn**.

=> Make the students **responsible** for **their** learning!

1. Teaching and Learning

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=> Make the students **responsible** for **their** learning!

Whose responsibility is it for learning to occur in the classroom? The responsibility belongs **to both the instructor and the students**. Responsibility can (and should) shift, depending on the time in the semester and the level of the students.

[…]

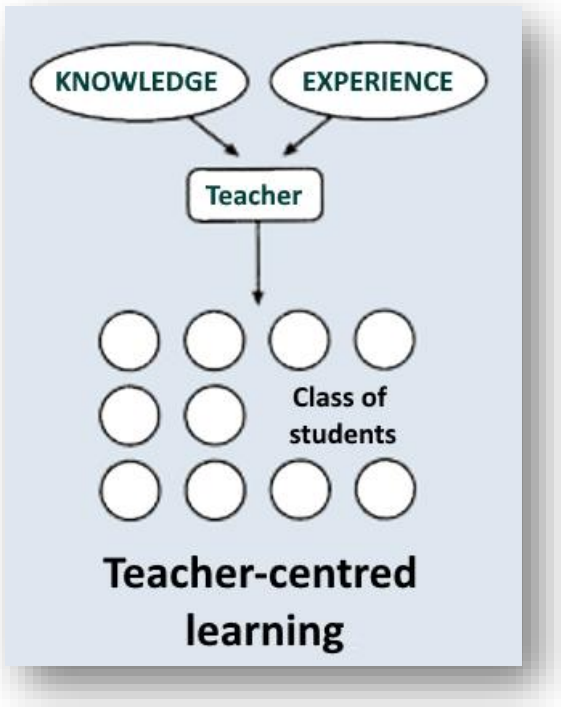
By **teaching responsibility**, we not only enhance learning and raise the level of our classrooms, but we help produce responsible citizens and productive members of society.

Coffman, S. J. (2003). Ten Strategies for Getting Students to Take Responsibility for Their Learning. *College Teaching*, *51*(1), 2–4. https://doi.org/10.1080/87567550309596401

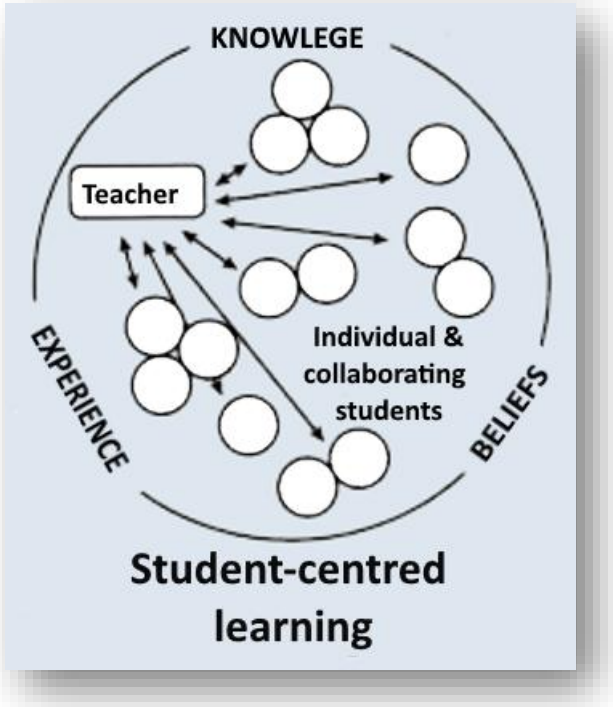
1. Teaching and Learning

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All the students have **the same Knowledge/background**

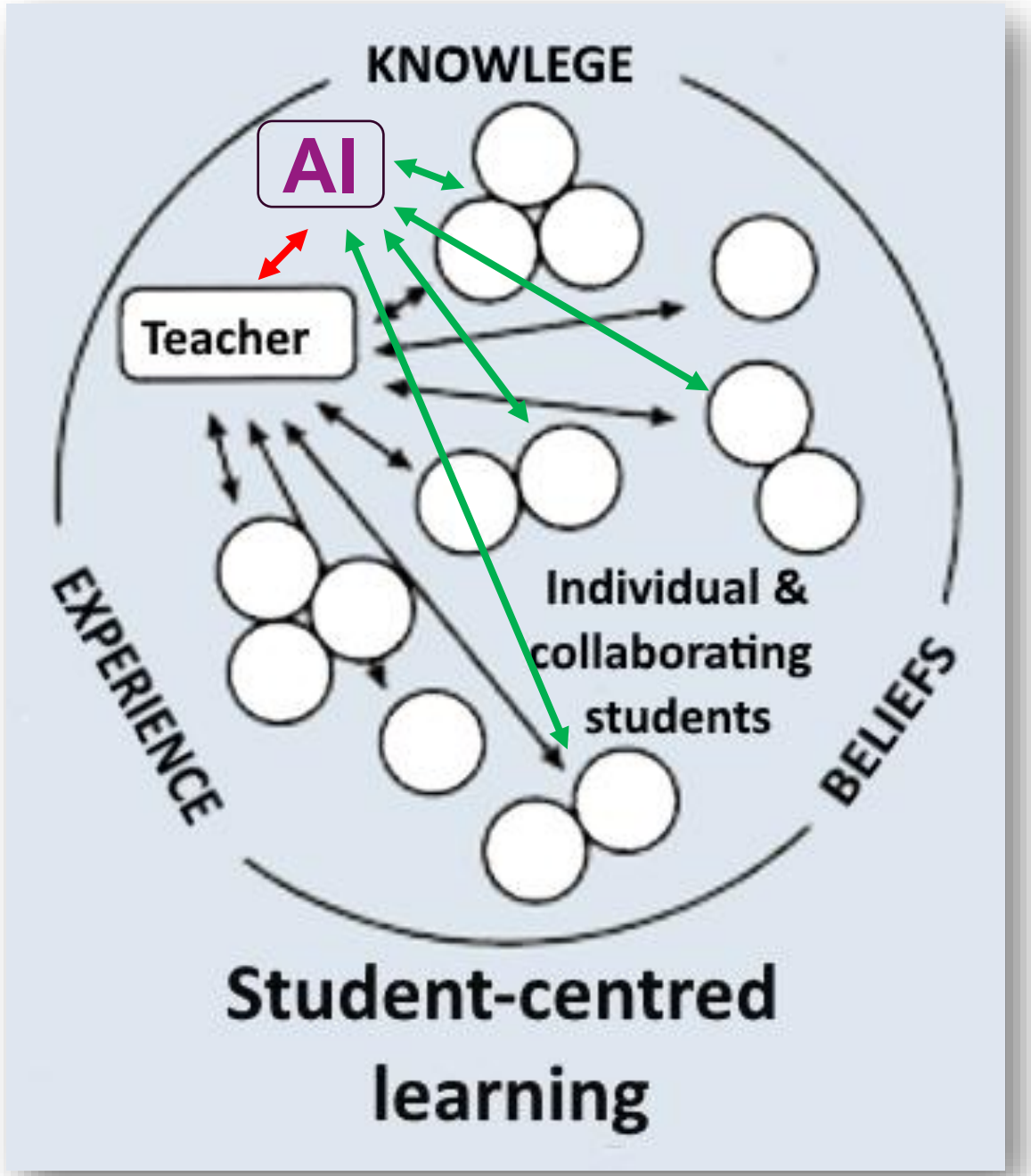


Real life: All the **students** are **different/UNIQUE**



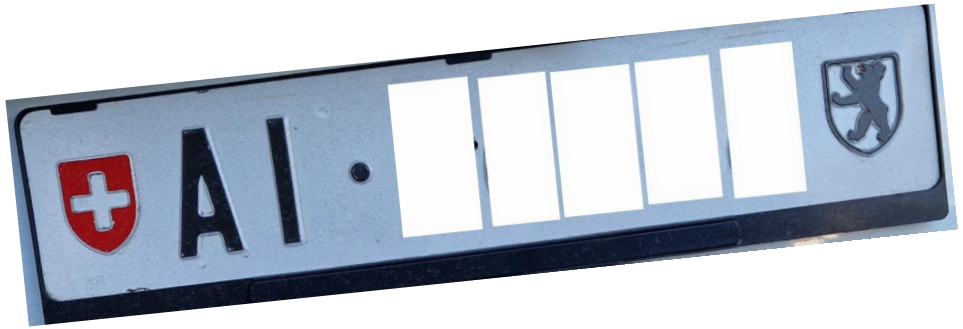
All the **teachers** are **different/UNIQUE**

Source: https://gwangjunewsgic.com/teaching/kotesol/creating-a-student-centered-classroom/



Real life: All the **students** are **different/UNIQUE**

\+ nowadays, they use **AI**



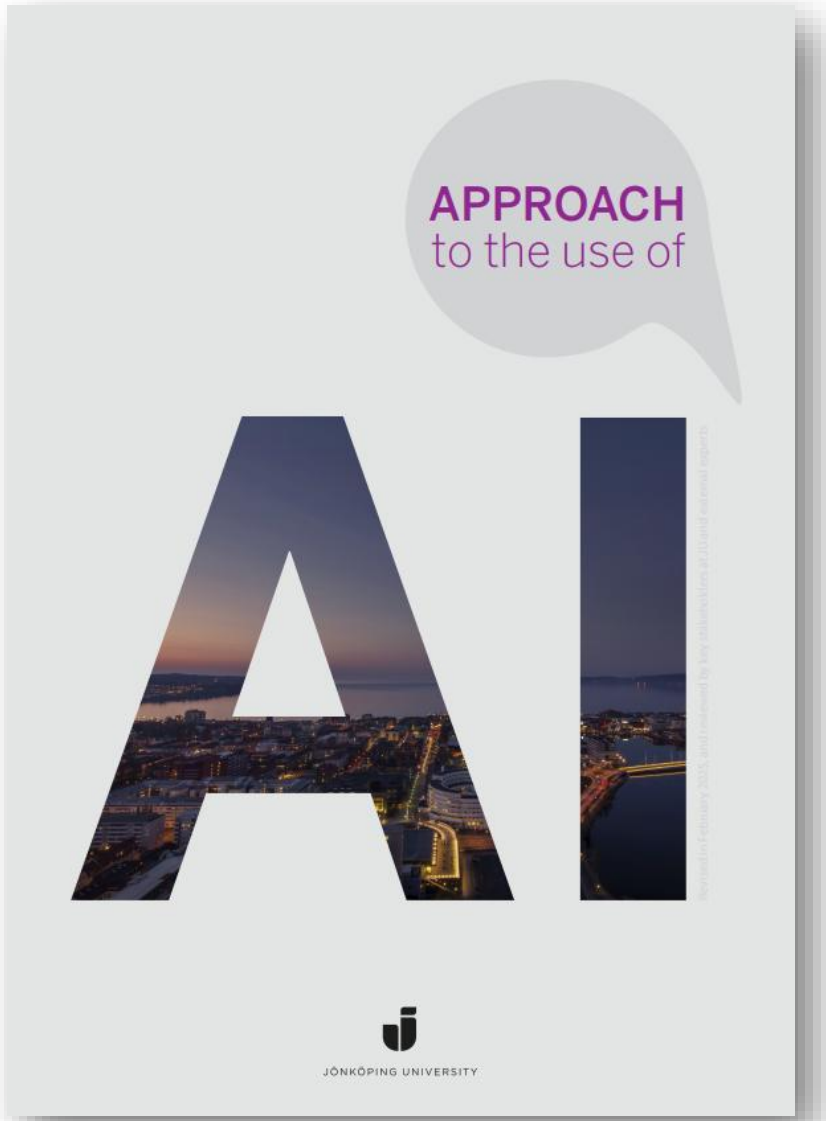
DISCUSSION – 3 MIN.

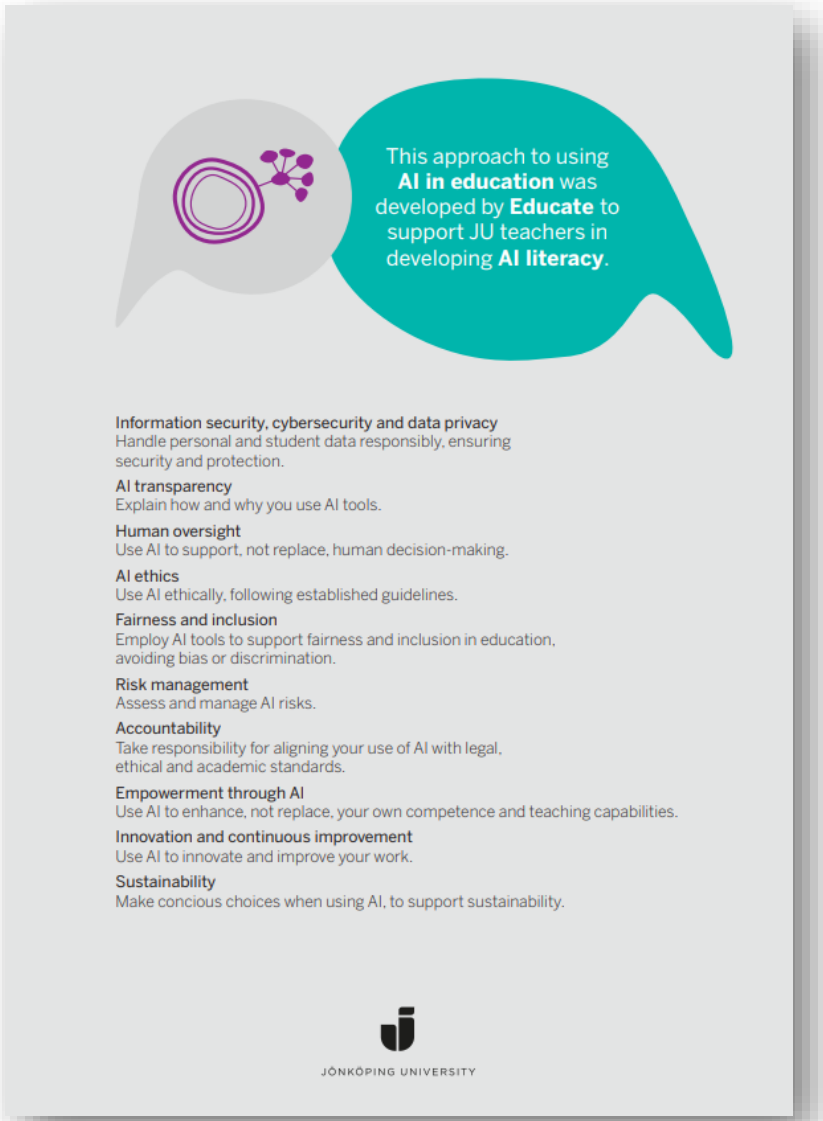
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- → Are you using AI in Education now?
- → How do you use AI?
- → Can you give short examples of how you use AI in the chat?

2. WHY USE AI IN EDUCATION?

Source: https://ju.se/download/18.2d8344d41953200b302864c4/1741785856546/Approach%20to%20the%20use%20of%20AI%202025-02-27%20v10.pdf

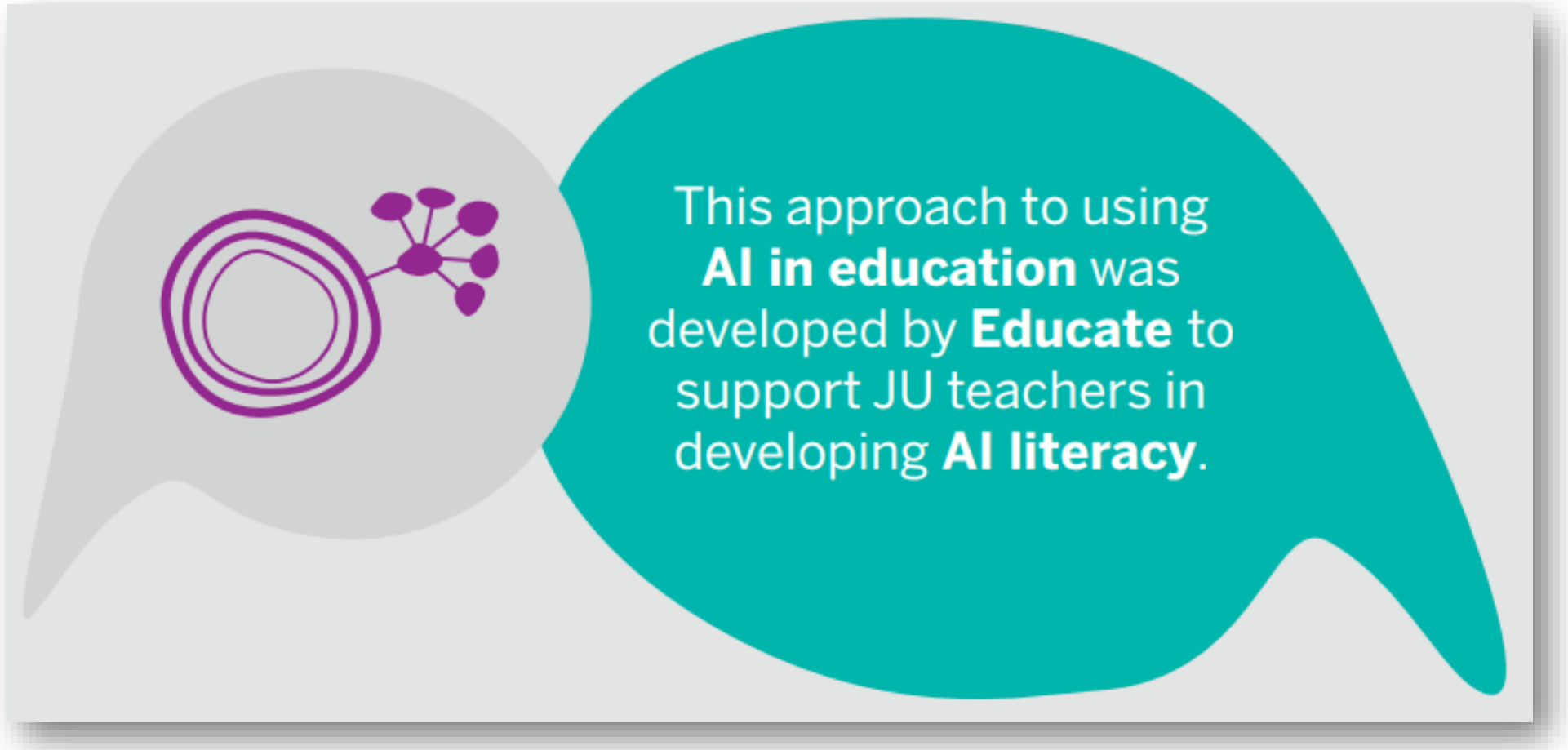




AI literacy:
Understanding, critically evaluation, and responsibly applying AI systems and tools

Sources: EDUCATE, Jönköping University, The use of AI in Education, https://ju.se/portal/educate/en/guides/artificial-intelligence.html

Veronika Hackl, Alexandra Elena Müller, Maximilian Sailer, The AI literacy heptagon: A structured approach to AI literacy in higher education, Computers and Education: Artificial Intelligence, Volume 10, 2026, 100540, ISSN 2666-920X, https://doi.org/10.1016/j.caeai.2026.100540



Source: https://ju.se/download/18.2d8344d41953200b302864c4/1741785856546/Approach%20to%20the%20use%20of%20AI%202025-02-27%20v10.pdf

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Information security, cybersecurity and data privacy
Handle personal and student data responsibly, ensuring security and protection.

AI transparency
Explain how and why you use AI tools.

Human oversight
Use AI to support, not replace, human decision-making.

AI ethics
Use AI ethically, following established guidelines.

Fairness and inclusion
Employ AI tools to support fairness and inclusion in education, avoiding bias or discrimination.

Risk management
Assess and manage AI risks.

Accountability
Take responsibility for aligning your use of AI with legal, ethical and academic standards.

Empowerment through AI
Use AI to enhance, not replace, your own competence and teaching capabilities.

Innovation and continuous improvement
Use AI to innovate and improve your work.

Sustainability
Make concious choices when using AI, to support sustainability.

2. WHY USE AI IN EDUCATION?

STUDENT

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Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How students can use AI in their learning? Please write the answer as a bullet list.

Here are practical ways students can use AI effectively in their learning:

- **Explain difficult concepts** – ask AI to re-explain lecture material in simpler terms, with analogies or step-by-step breakdowns
- **Practice and self-testing** – generate quiz questions, flashcards, or practice problems, then check answers with feedback
- **Socratic tutoring** – ask AI to quiz them and ask follow-up questions instead of just giving answers (e.g., "don't solve it, guide me")
- **Improve writing** – get feedback on drafts for clarity, structure, and grammar, rather than having AI write the text

2. WHY USE AI IN EDUCATION? STUDENT

JÖNKÖPING UNIVERSITY
School of Engineering

Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How students can use AI in their learning? Please write the answer as a bullet list.

- **Summarize and organize** – condense long readings or lecture notes into outlines, summaries, or study guides (used as a starting point, verified against the original)
- **Language support** – translate, define terminology, or practice a foreign language through conversation
- **Brainstorm and plan** – generate ideas for essays or projects, plan study schedules, or outline a report structure
- **Debug and learn coding** – explain error messages, review their code, and understand why something fails
- **Check their own understanding** – have them explain a topic to the AI and ask it to point out gaps or errors ("teach-back" method)
- **Critical evaluation practice** – ask AI a question on a topic they know well, then fact-check the output to spot hallucinations and errors

2. WHY USE AI IN EDUCATION?

STUDENT

JÖNKÖPING UNIVERSITY
School of Engineering

Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How students can use AI in their learning? Please write the answer as a bullet list.

Important caveats to pass on to students:

- AI can be confidently wrong — always verify facts against course literature and reliable sources
- Follow your university's rules on AI use and always disclose AI assistance where required
- Use AI as a tutor, not a substitute for thinking — the learning happens in their own effort

2. WHY USE AI IN EDUCATION? TEACHER

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Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How teachers can use AI in their teaching? Please write the answer as a bullet list.

Here are practical ways you as a university teacher can use AI in teaching:

Course and material preparation

- **Design course content** – draft learning objectives, lesson plans, and course structures for review and adaptation
- **Create teaching materials** – generate example problems, case studies, discussion questions, and exercises (with careful fact-checking)
- **Adapt materials to levels** – rewrite texts for different difficulty levels or for students with different backgrounds
- **Update examples** – refresh outdated examples, statistics, and scenarios in slides and readings (verify all figures)

2. WHY USE AI IN EDUCATION? TEACHER

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I am a teacher at the university. How teachers can use AI in their teaching? Please write the answer as a bullet list.

Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

Assessment and feedback

- **Draft exam questions** – generate question banks, multiple-choice items, and rubrics; check alignment with objectives
- **Design rubrics and criteria** – draft assessment rubrics that you refine for your standards
- **Feedback support** – help you formulate clearer, more constructive feedback on student work (the assessment judgment stays yours)
- **Detect inconsistencies** – spot unclear or ambiguous wording in your own assignments and exam questions

2. WHY USE AI IN EDUCATION? TEACHER

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Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How teachers can use AI in their teaching? Please write the answer as a bullet list.

Teaching delivery and interaction

- **Prepare answers to expected questions** – anticipate where students typically struggle and prepare explanations
- **Create analogies and explanations** – find multiple ways to explain the same concept for different learners
- **Role-play scenarios** – e.g., practice difficult conversations (thesis supervision, conflict situations) or simulate debates
- **Language support** – teach in a second language with help polishing slides, instructions, and emails

2. WHY USE AI IN EDUCATION? TEACHER

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I am a teacher at the university. How teachers can use AI in their teaching? Please write the answer as a bullet list.

Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

Administration and personalization

- **Streamline admin** – draft emails to students, meeting agendas, syllabus text, and program descriptions
- **Analyze course evaluations** – summarize themes from open-ended student feedback to spot issues early
- **Differentiation** – adapt materials for students with accommodations or diverse prerequisites

2. WHY USE AI IN EDUCATION? TEACHER

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Source: Mistral AI, Vibe, https://chat.mistral.ai/chat

I am a teacher at the university. How teachers can use AI in their teaching? Please write the answer as a bullet list.

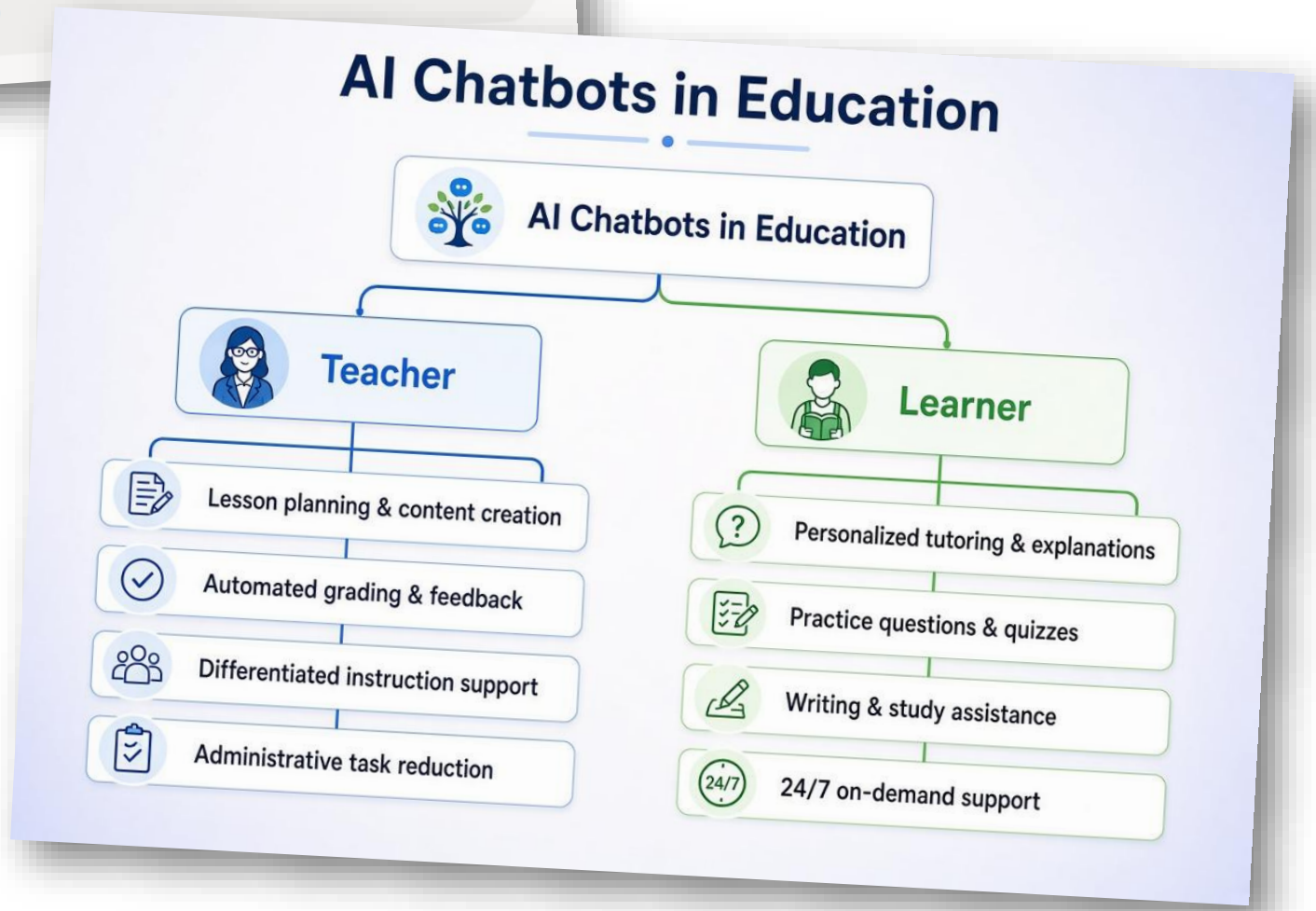
Key caveats

- Always verify AI-generated content — errors in teaching materials damage trust and learning
- Check student privacy rules before entering any student work or data into AI tools
- Be transparent with students about how you use AI, and model good AI practice
- Keep the pedagogical decisions — learning objectives, standards, and grades — in your hands

2. WHY USE AI IN EDUCATION?

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You are a teacher at the university. You want to give to your students an overview of teaching and learning in education. Can you generate an image to display the usage of AI chatbots in education for students and teachers? Can you present this information into a tree with two children: teacher and learner?



Source: Grok, https://grok.com/

2. WHY USE AI IN EDUCATION?

- It opens new **tools/possibilities** for **teaching**



Source: https://www.youtube.com/watch?v=1QNsdr-Qx_I

2. WHY USE AI IN EDUCATION?

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- It opens new **tools/possibilities** for **teaching**

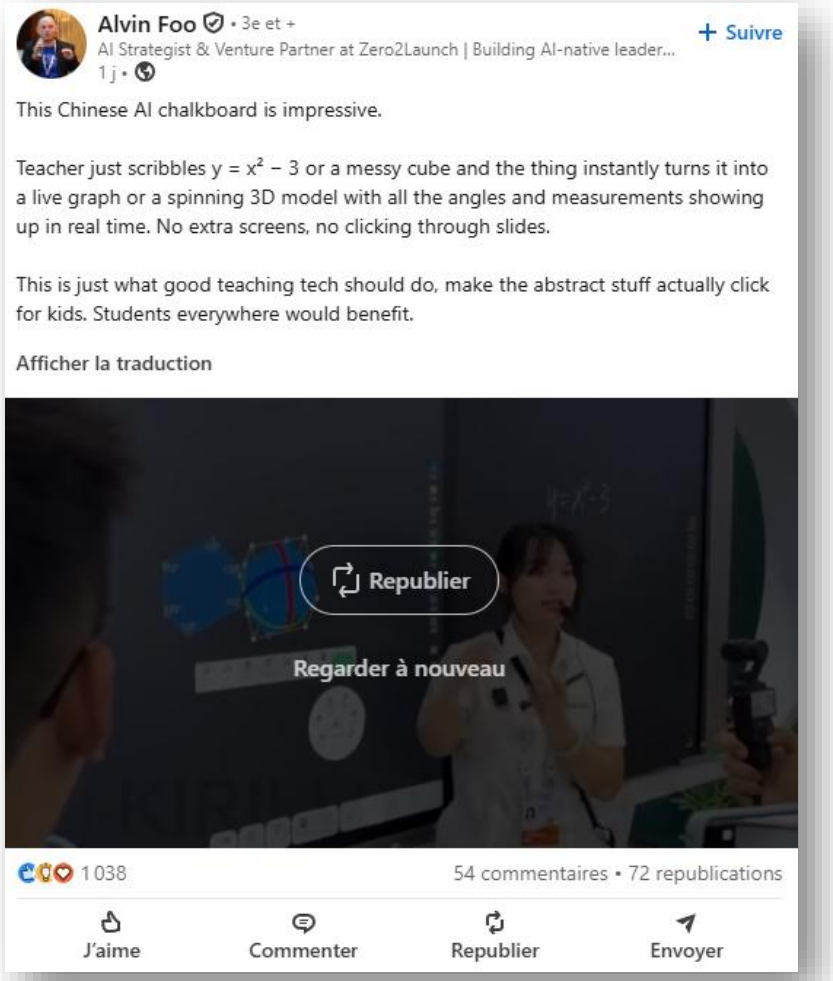


Source: https://www.youtube.com/watch?v=1QNsdr-Qx_I

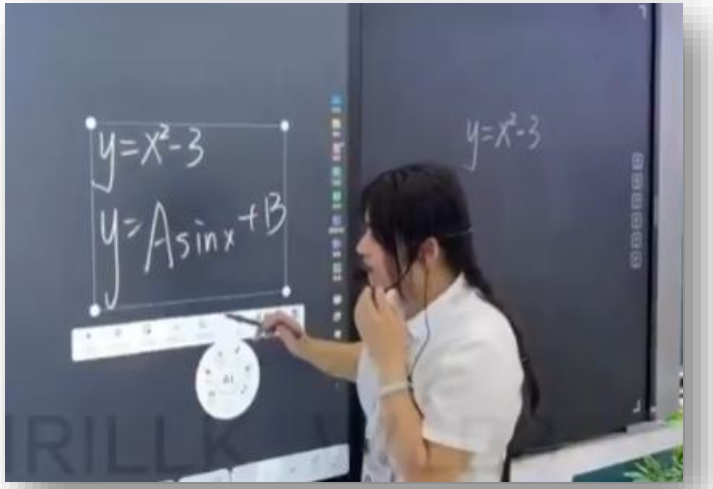
2. WHY USE AI IN EDUCATION?

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- It opens new **tools/possibilities** for **teaching**, from LinkedIn https://lnkd.in/p/eEFvjUmx:



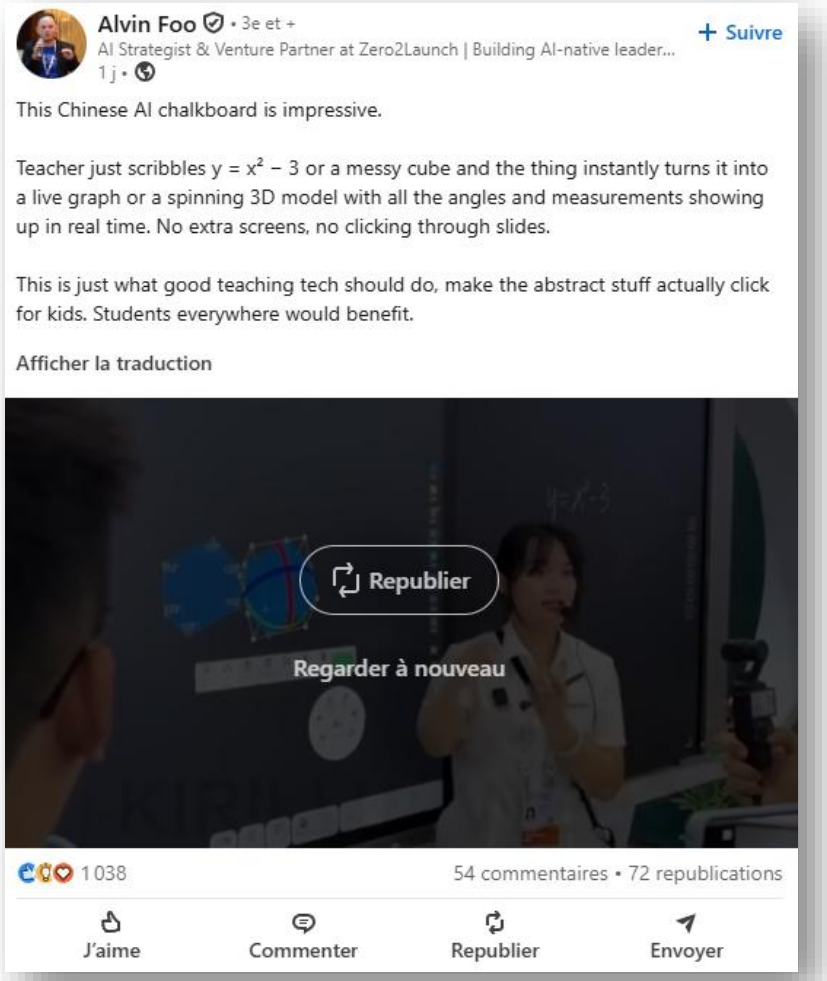




2. WHY USE AI IN EDUCATION?

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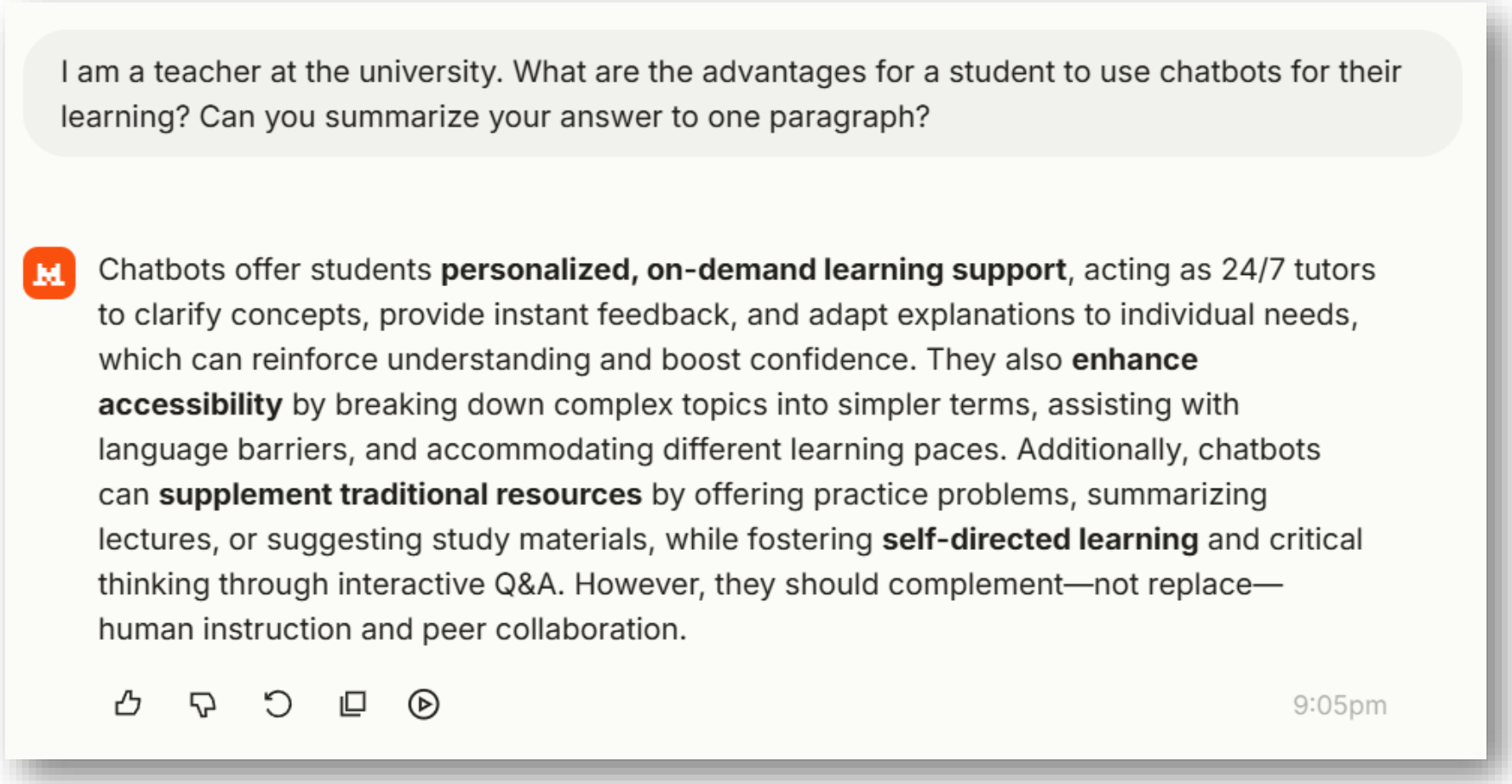






3. WHY USE AI IN EDUCATION?

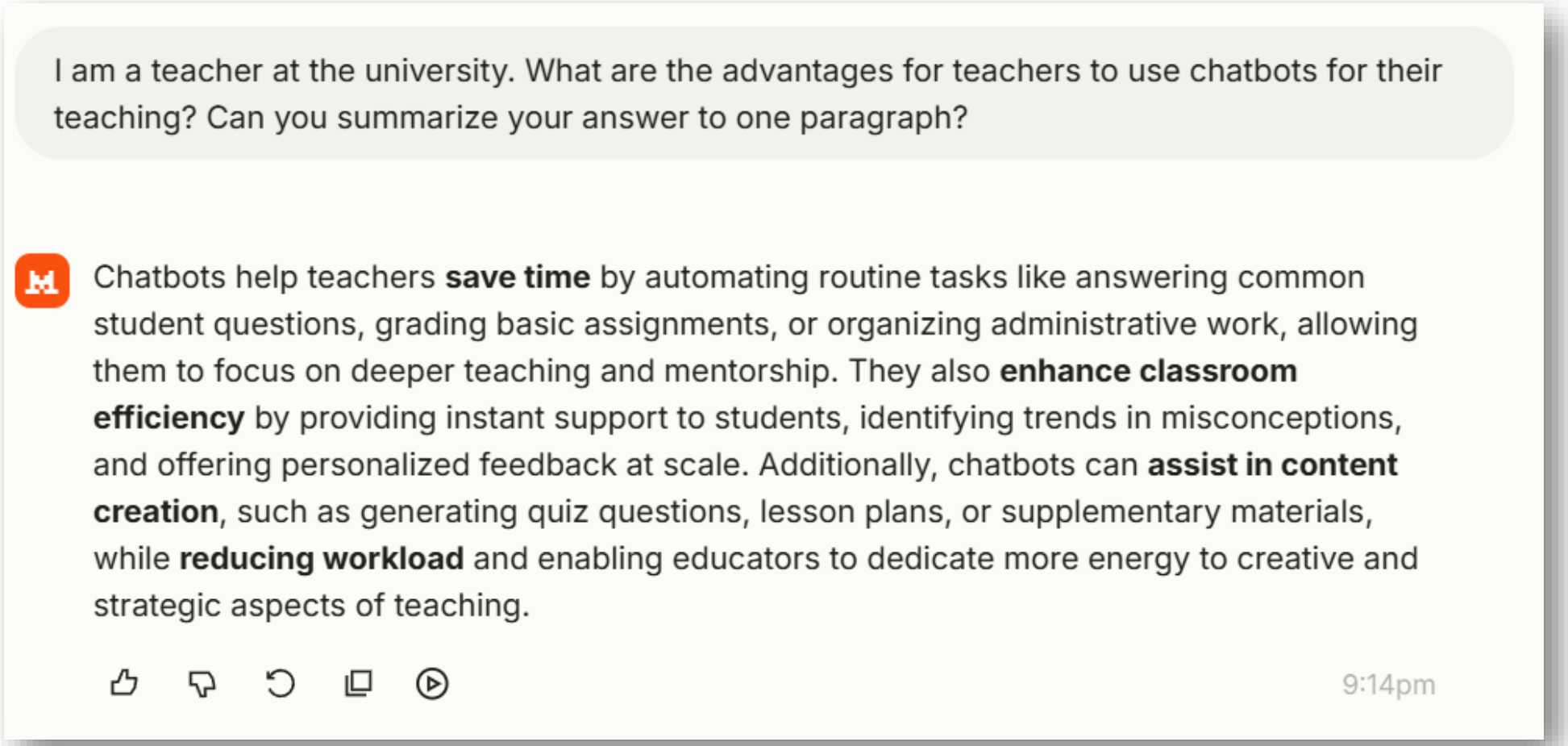
- It opens new **tools/possibilities** for **learning:** every student can get its own TEACHER everywhere, at any time, and interact with it **AT THE STUDENT'S PACE!**



2. WHY USE AI IN EDUCATION?

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- It opens new **tools/possibilities** for **teaching:**



3. HOW TO USE AI IN EDUCATION?

- A very good question!
- The students **will use AI**
- It will be part of their **future jobs**
- They must learn to use it **efficiently** and **effectively**
- They must learn **ethics** and develop their **critical thinking**

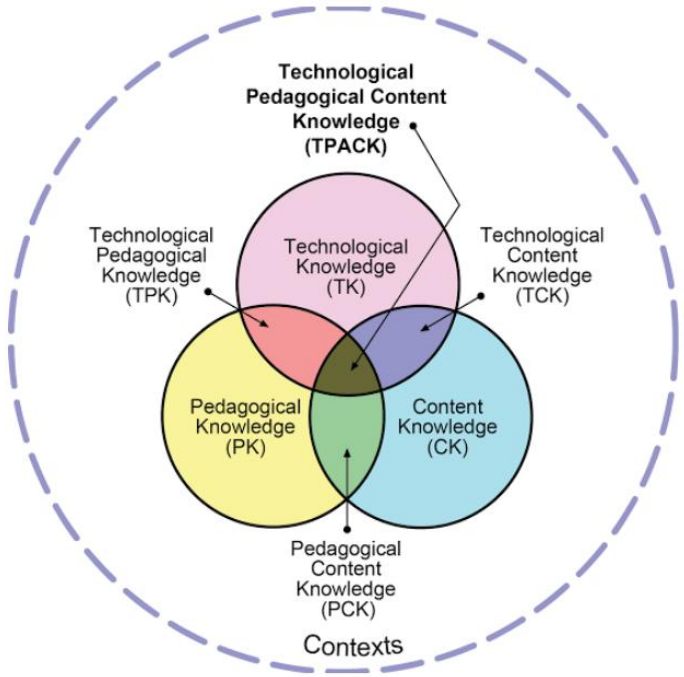
There are tools to **HELP** You!

3. HOW TO USE AI IN EDUCATION?

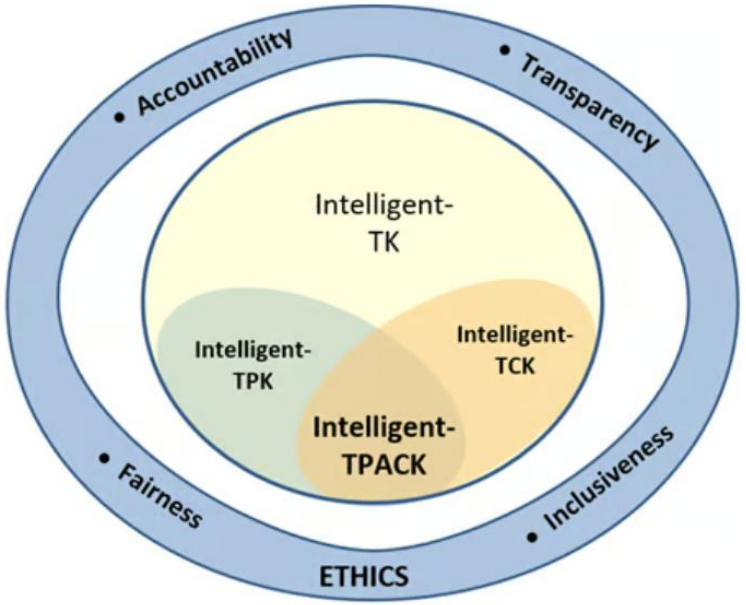
https://intelligent-tpack.ai

Intelligent-TPACK

Home | About Intelligent-TPACK | What is Intelligent-TK? | What is Intelligent-TPK? | What is Intelligent-TCK? | What is Ethics? | Case Library | English Case Examples | Finnish Case Examples | Relevant Studies | Intelligent-TPACK Scale | Team



Technological Pedagogical Content Knowledge (TPACK) developed by Mishra and Koehler in 2006.



Intelligent Technological Pedagogical Content Knowledge (Intelligent-TPACK) proposed by Celik in 2023.

Source: https://intelligent-tpack.ai/

INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL
https://doi.org/10.1080/14703297.2026.2663355

Routledge
Taylor & Francis Group

Check for updates

Teachers' professional knowledge to prevent students' overreliance on AI: Insight from Intelligent-TPACK

Ismail Celik

Faculty of Education & Psychology, University of Oulu, Oulu, Finland

To prevent students' overreliance, designing and implementing instructional tasks are critical in GenAI-supported instruction (Celik, 2023). These tasks must be designed in a way that students should rely on their minds when they get help from GenAI (Zhai et al., 2024). In designing and implementing tasks, the role of teachers is important (Celik, 2023). This role requires teachers to have GenAI-specific pedagogical and ethical knowledge. This is because GenAI-specific pedagogy includes both student characteristics (e.g. cognitive abilities) and GenAI affordances (e.g. immediate feedback) (Celik, 2023). In other words, teachers must know pedagogical strategies to increase student mental effort in GenAI-supported instruction (Risko & Gilbert, 2016). In this regard, an updated version of the technological, pedagogical, and content knowledge framework, namely Intelligent-TPACK, could explain the necessary knowledge for teachers to prevent students' overreliance on GenAI (Celik, 2023). The current study aims to provide an overview of GenAI-specific pedagogical knowledge and skills to prevent students' GenAI overreliance based on the Intelligent-TPACK framework. Accordingly, the contribution of this paper is conceptual in nature, focusing on theoretical extension and application of Intelligent-TPACK, rather than its empirical validation.

INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL
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Check for updates

Teachers' professional knowledge to prevent students' overreliance on AI: Insight from Intelligent-TPACK

Ismail Celik

Faculty of Education & Psychology, University of Oulu, Oulu, Finland

To prevent this, the teacher reorganises the activity so that students must first think independently and generate an initial solution or explanation (Celik, 2023). After producing their own reasoning, students then compare their response to the AI-generated answer, identify similarities and differences, and write a brief reflection explaining which parts of the AI output they accept, which they reject, and why. This redesigned task ensures that AI becomes a point of comparison rather than a shortcut, requiring students to engage in disciplinary reasoning, evaluate evidence, and articulate their understanding. Such instructional adjustments reflect Intelligent-TPACK, as the teacher integrates AI tools, pedagogical scaffolding, and subject-specific reasoning in a coherent way that maintains cognitive effort and prevents overreliance.

4. ONLINE AI CHATBOTS 4.1. AVAILABLE TOOLS

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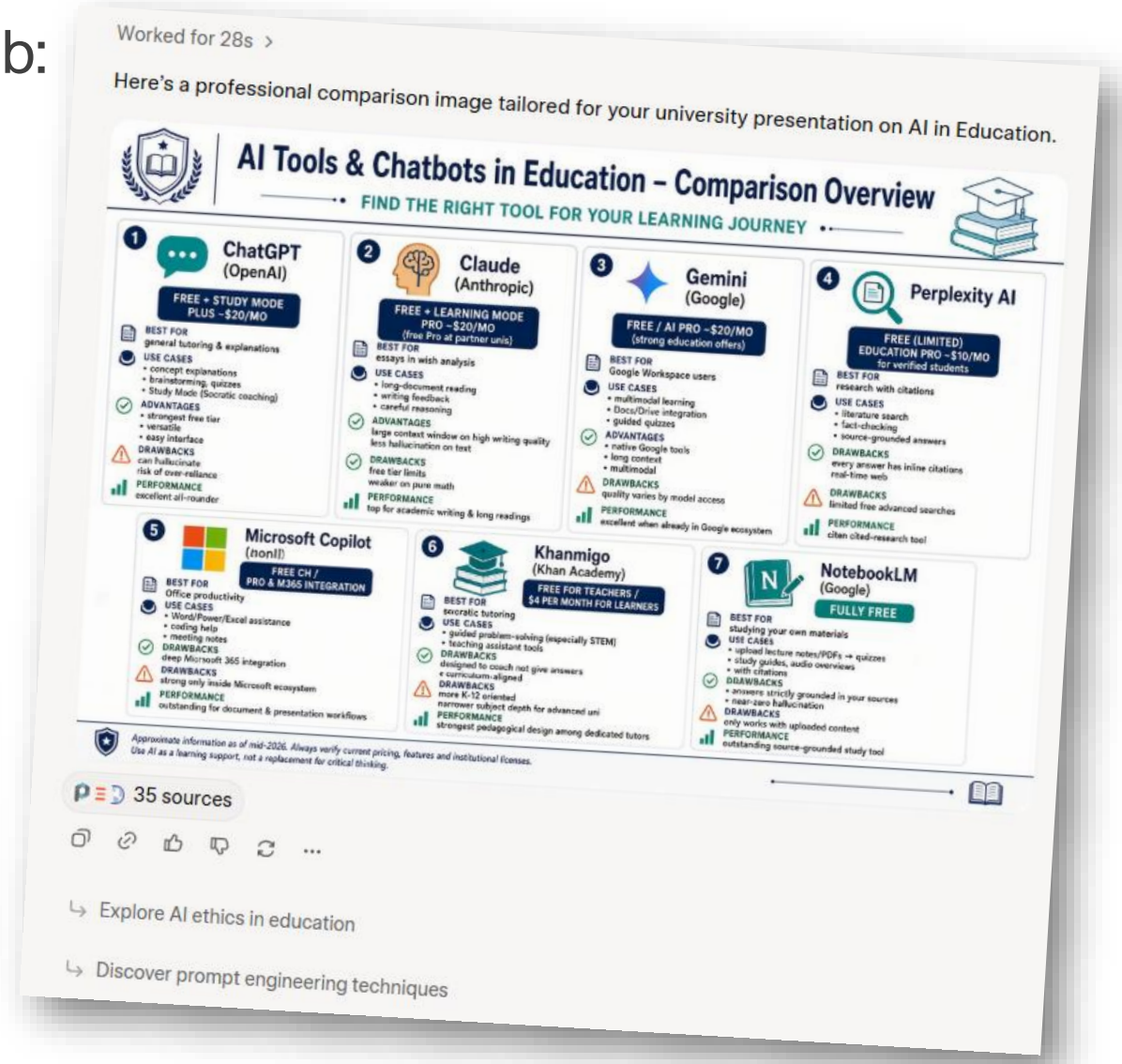
- There are many **tools** available on the web:

I am teacher at the university. I am writing a presentation on AI in Education. Can you generate an image with the different AI tools/chatbots available online with their use cases and advantages and drawbacks, free/paying, performances?

Generating an image of AI tools comparison
Ran 2 searches
Searching for up-to-date AI tool information
Ran 1 search
Crafting the image prompt
Compiling AI tool comparison data
Finalizing the image prompt details

Worked for 28s

Here's a professional comparison image tailored for your university presentation on AI in Education.



AI Tools & Chatbots in Education – Comparison Overview

FIND THE RIGHT TOOL FOR YOUR LEARNING JOURNEY

1 ChatGPT (OpenAI)

FREE + STUDY MODE
PLUS ~$20/MO

- **BEST FOR**
  general tutoring & explanations
- **USE CASES**
  - concept explanations
  - brainstorming, quizzes
  - Study Mode (Socratic coaching)
- **ADVANTAGES**
  - strongest free tier
  - versatile
  - easy interface
- **DRAWBACKS**
  can hallucinate
  risk of over-reliance
- **PERFORMANCE**
  excellent all-rounder

2 Claude (Anthropic)

FREE + LEARNING MODE
PRO ~$20/MO
(free Pro at partner unis)

- **BEST FOR**
  essays in wish analysis
- **USE CASES**
  - long-document reading
  - writing feedback
  - careful reasoning
- **ADVANTAGES**
  large context window on high writing quality
  less hallucination on text
- **DRAWBACKS**
  free tier limits
  weaker on pure math
- **PERFORMANCE**
  top for academic writing & long readings

3 Gemini (Google)

FREE / AI PRO ~$20/MO
(strong education offers)

- **BEST FOR**
  Google Workspace users
- **USE CASES**
  - multimodal learning
  - Docs/Drive integration
  - guided quizzes
- **ADVANTAGES**
  - native Google tools
  - long context
  - multimodal
- **DRAWBACKS**
  quality varies by model access
- **PERFORMANCE**
  excellent when already in Google ecosystem

4 Perplexity AI

FREE (LIMITED)
EDUCATION PRO ~$10/MO
for verified students

- **BEST FOR**
  research with citations
- **USE CASES**
  - literature search
  - fact-checking
  - source-grounded answers
- **DRAWBACKS**
  every answer has inline citations
  real-time web
- **DRAWBACKS**
  limited free advanced searches
- **PERFORMANCE**
  citen cited-research tool

5 Microsoft Copilot (nonII)

FREE CH /
PRO & M365 INTEGRATION

- **BEST FOR**
  Office productivity
- **USE CASES**
  - Word/Power/Excel assistance
  - coding help
  - meeting notes
- **DRAWBACKS**
  deep Microsoft 365 integration
- **DRAWBACKS**
  strong only inside Microsoft ecosystem
- **PERFORMANCE**
  outstanding for document & presentation workflows

6 Khanmigo (Khan Academy)

FREE FOR TEACHERS /
PER MONTH FOR LEARNERS

- **BEST FOR**
  socratic tutoring
- **USE CASES**
  - guided problem solving (especially STEM)
  - teaching assistant tools
- **DRAWBACKS**
  designed to coach not give answers
  curriculum-aligned
- **DRAWBACKS**
  more K-12 oriented
  narrower subject depth for advanced uni
- **PERFORMANCE**
  strongest pedagogical design among dedicated tutors

7 NotebookLM (Google)

FULLY FREE

- **BEST FOR**
  studying your own materials
- **USE CASES**
  - upload lecture notes/PDFs → quizzes
  - study guides, audio overviews
  - with citations
- **DRAWBACKS**
  answers strictly grounded in your sources
  - near-zero hallucination
- **DRAWBACKS**
  only works with uploaded content
- **PERFORMANCE**
  outstanding source-grounded study tool

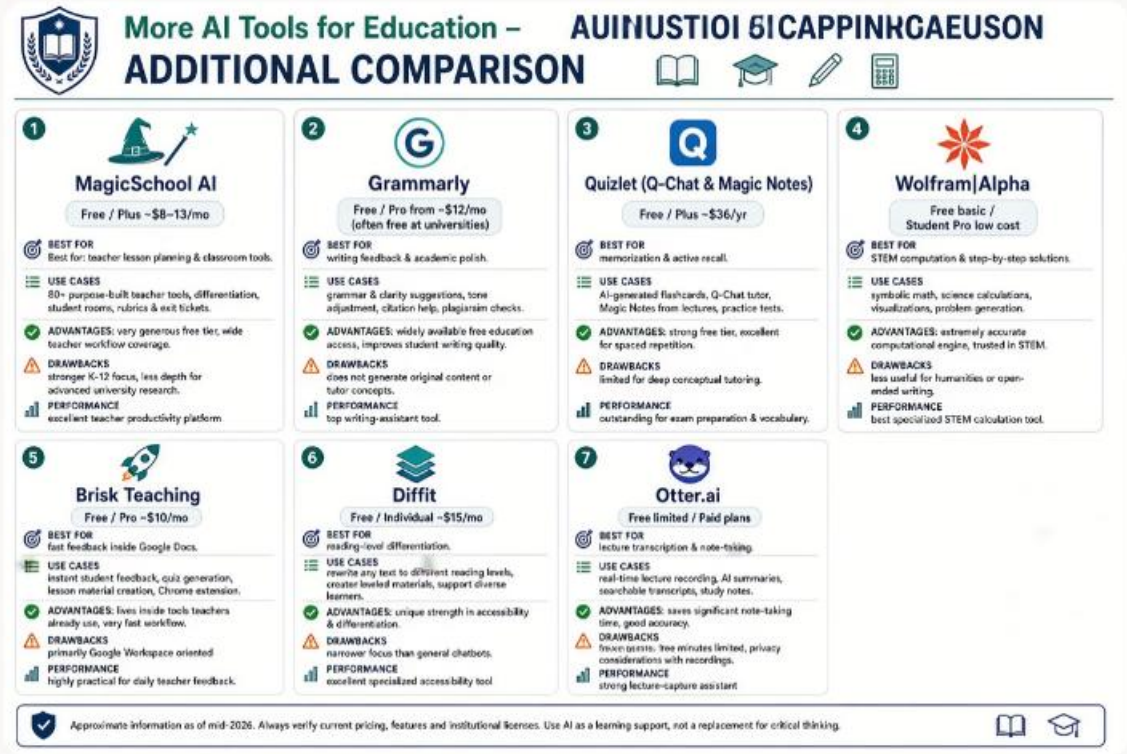
Approximate information as of mid-2026. Always verify current pricing, features and institutional licenses.
Use AI as a learning support, not a replacement for critical thinking.

Use AI as a learning support, not a replacement for critical thinking.

Can you do the same with other tools (not the one included in the previous image)?

Worked for 13s >

Here's a second professional comparison image featuring a different set of AI tools useful in education (none of the previous ones).



↳ Explore advanced prompt engineering techniques

↳ Investigate AI ethics and bias

More AI Tools for Education – ADDITIONAL COMPARISON

AUINUSTIOI 5ICAPPINRCAEUSON

1 MagicSchool AI

Free / Plus ~$8–13/mo

BEST FOR
Best for: teacher lesson planning & classroom tools.

USE CASES
80+ purpose-built teacher tools, differentiation, student rooms, rubrics & exit tickets.

ADVANTAGES: very generous free tier, wide teacher workflow coverage.

DRAWBACKS
stronger K-12 focus, less depth for advanced university research.

PERFORMANCE
excellent teacher productivity platform

2 Grammarly

Free / Pro from ~$12/mo (often free at universities)

BEST FOR
writing feedback & academic polish.

USE CASES
grammar & clarity suggestions, tone adjustment, citation help, plagiarsim checks.

ADVANTAGES: widely available free education access, improves student writing quality.

DRAWBACKS
does not generate original content or tutor concepts.

PERFORMANCE
top writing-assistant tool.

3 Quizlet (Q-Chat & Magic Notes)

Free / Plus ~$36/yr

BEST FOR
memorization & active recall.

USE CASES
AI-generated flashcards, Q-Chat tutor, Magic Notes from lectures, practice tests.

ADVANTAGES: strong free tier, excellent for spaced repetition.

DRAWBACKS
limited for deep conceptual tutoring.

PERFORMANCE
outstanding for exam preparation & vocabulary.

4 Wolfram|Alpha

Free basic / Student Pro low cost

BEST FOR
STEM computation & step-by-step solutions.

USE CASES
symbolic math, science calculations, visualizations, problem generation.

ADVANTAGES: extremely accurate computational engine, trusted in STEM.

DRAWBACKS
less useful for humanities or open-ended writing.

PERFORMANCE
best specialized STEM calculation tool.

5 Brisk Teaching

Free / Pro ~$10/mo

BEST FOR
fast feedback inside Google Docs.

USE CASES
instant student feedback, quiz generation, lesson material creation, Chrome extension.

ADVANTAGES: lives inside tools teachers already use, very fast workflow.

DRAWBACKS
primarily Google Workspace oriented

PERFORMANCE
highly practical for daily teacher feedback.

6 Diffit

Free / Individual ~$15/mo

BEST FOR
reading-level differentiation.

USE CASES
rewrite any text to different reading levels, creator leveled materials, support diverse learners.

ADVANTAGES: unique strength in accessibility & differentiation.

DRAWBACKS
narrower focus than general chatbots.

PERFORMANCE
excellent specialized accessibility tool

7 Otter.ai

Free limited / Paid plans

BEST FOR
lecture transcription & note-taking.

USE CASES
real-time lecture recording, AI summaries, searchable transcripts, study notes.

ADVANTAGES: saves significant note-taking time, good accuracy.

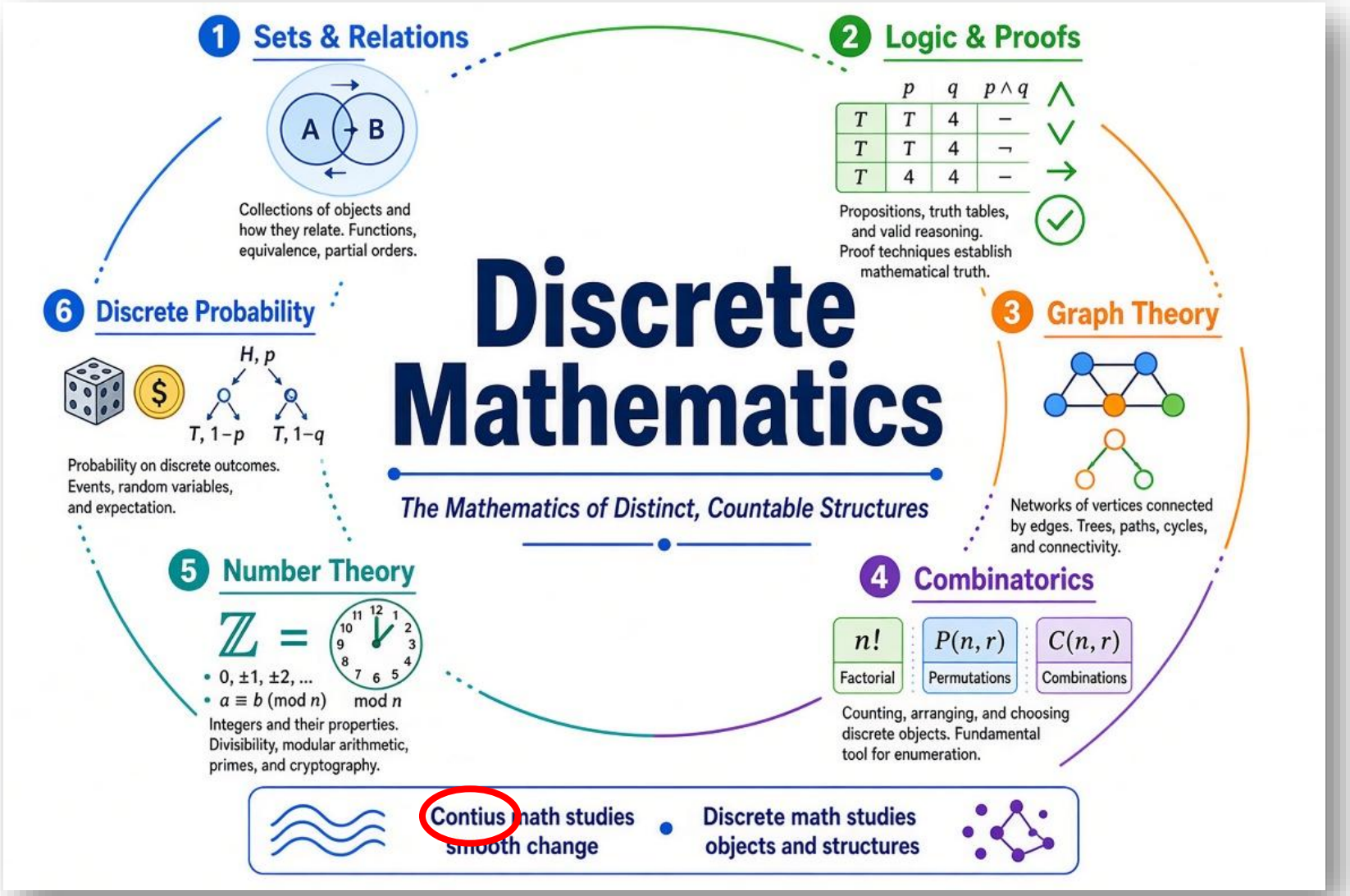
DRAWBACKS
free tier: free minutes limited, privacy considerations with recordings.

PERFORMANCE
strong lecture-capture assistant

Approximate information as of mid-2026. Always verify current pricing, features and institutional licenses. Use AI as a learning support, not a replacement for critical thinking.

Use AI as a learning support, not a replacement for critical thinking.

I am a teacher at the university. Can you create a picture to explain what discrete mathematics are to my students?

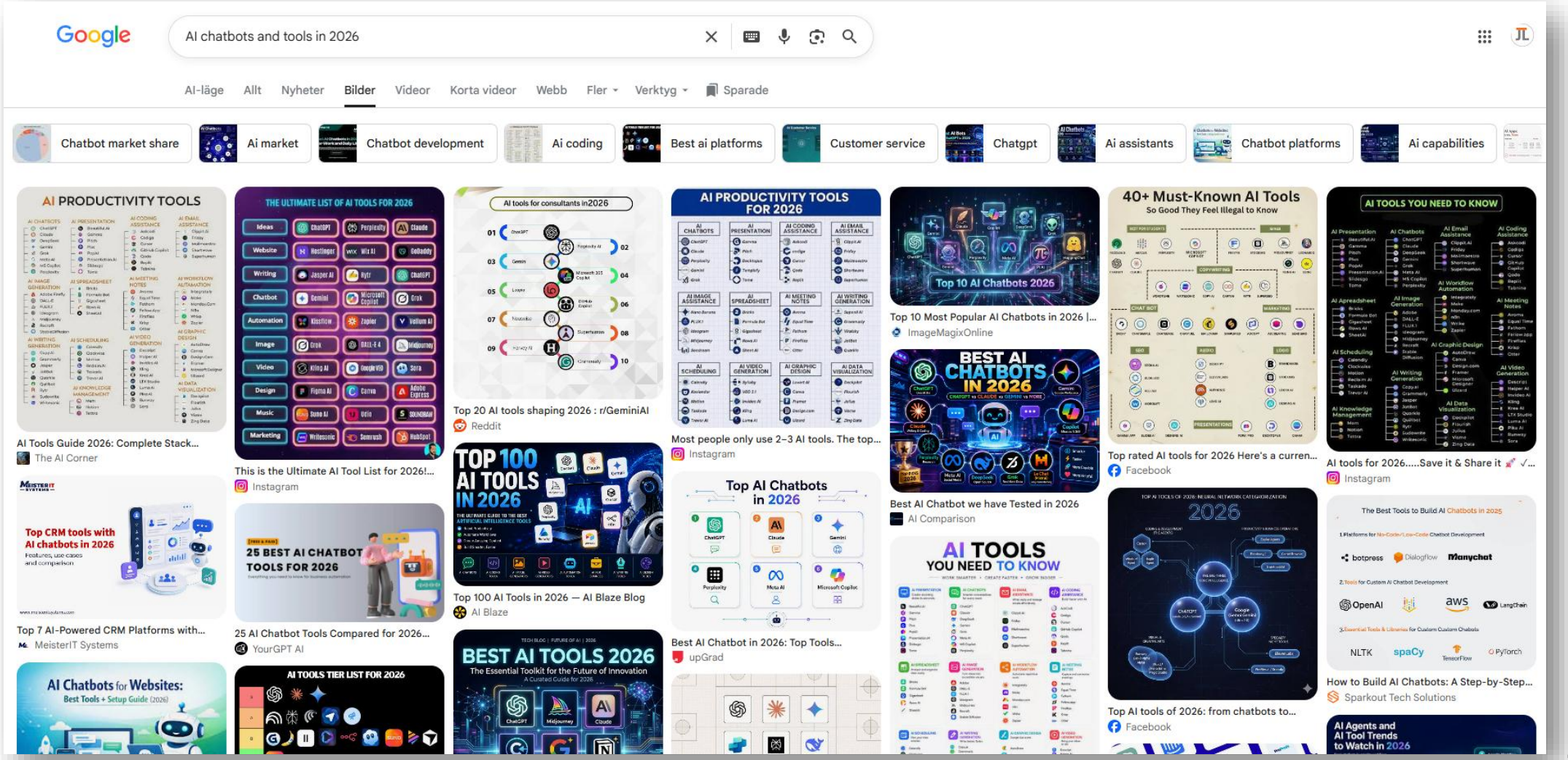


Source: Grok, https://grok.com/

4. ONLINE AI CHATBOTS 4.1. AVAILABLE TOOLS

JÖNKÖPING UNIVERSITY
School of Engineering

- There are many **tools** available on the web:

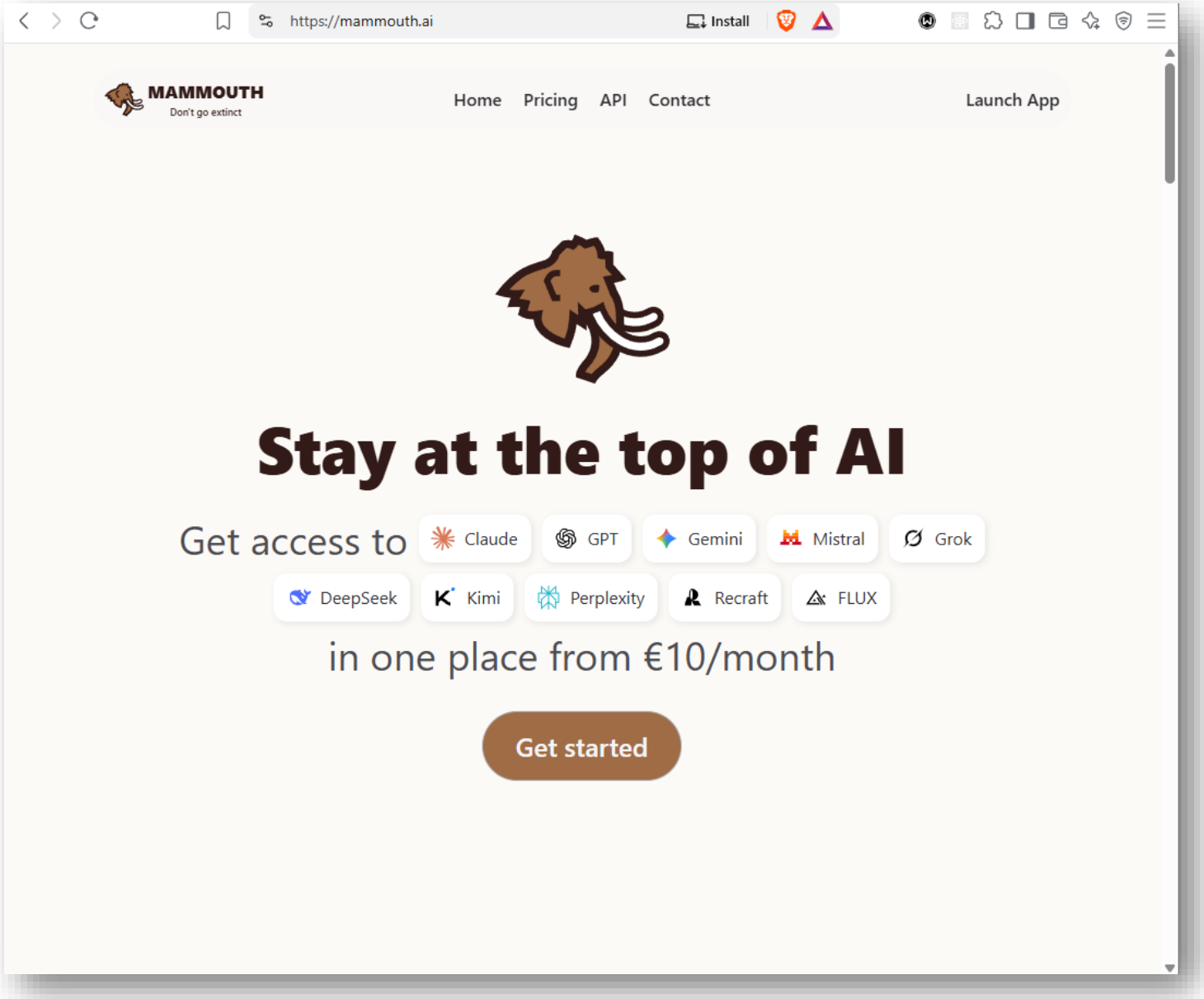


4. ONLINE AI CHATBOTS



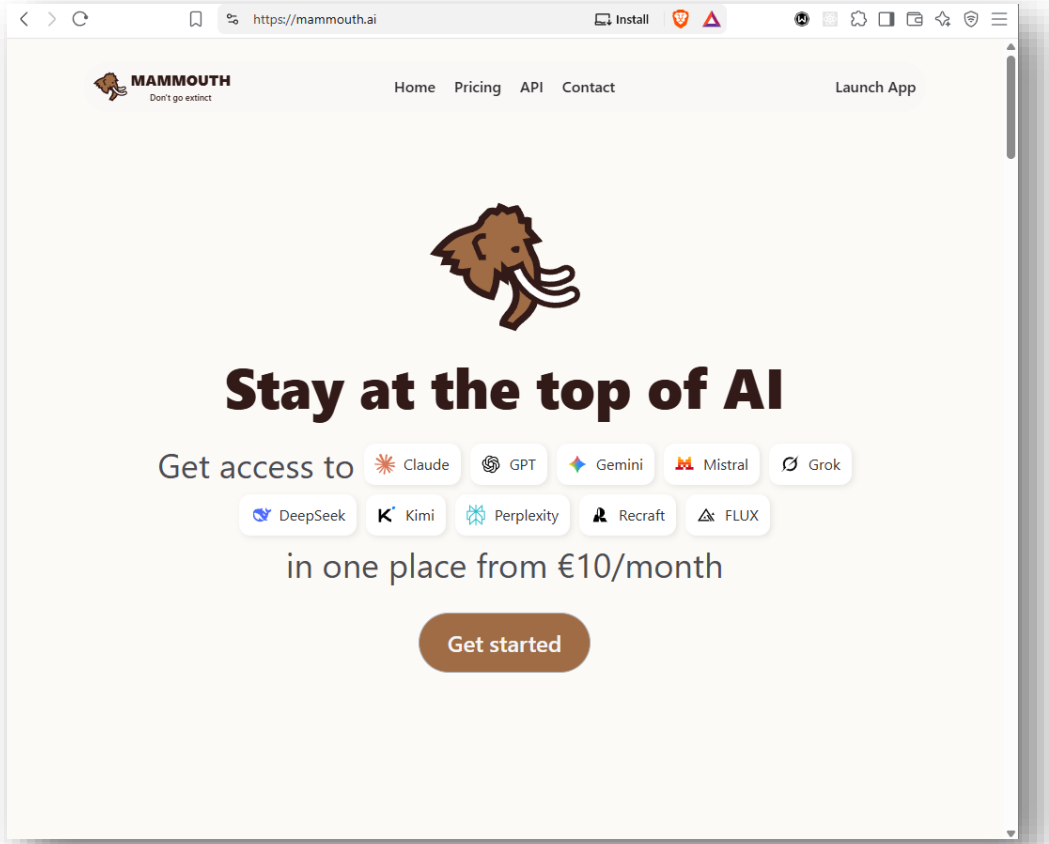
Source: https://www.the-ai-corner.com/p/ai-tools-guide-2026

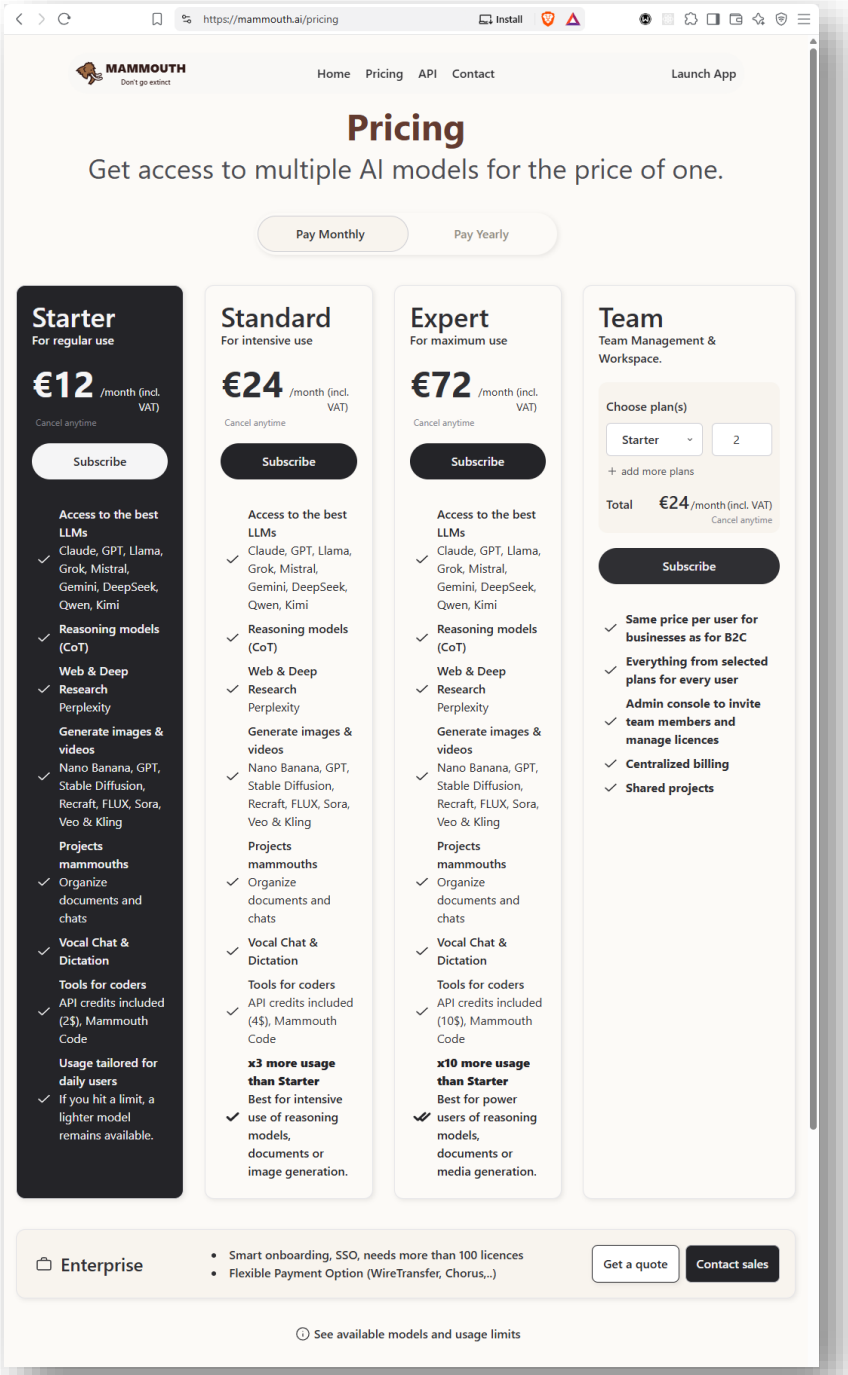
4. ONLINE AI CHATBOTS



Source: https://mammouth.ai/

4. ONLINE AI CHATBOTS





Source: https://mammouth.ai/

- **Prompt engineering** = communicating **effectively** with AI tools
- Why it matters: clarity, depth, and learning alignment
- Good prompts = good thinking

Source: Björn Praestgaard-Larsen, AI in Education, ”prompt engineering”, 2025

Google

what makes an effective prompt in chatbots?

AI-läge Allt Korta videor Ekonomi Nyheter Webb Bilder Fler Verktyg Sparade

Prompt framework | Chatgpt prompts | Prompt structure | Prompt controls | Prompt engineering | Artificial intelligence | Generative ai | Ai tools | Ai agent | Perplexity | Education



Prompt Controls in GenAI Chatbots: 4 Main...
Nielsen Norman Group



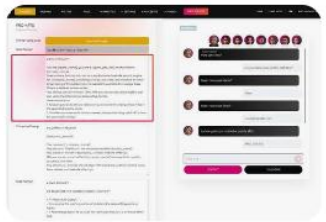
Effective Prompting for Educators — AI for...
AI for Education



How to write AI prompts like a pro | Sendbird
Sendbird



Understanding Prompts and Roles in...
LinkedIn



Create Effective Role Prompts for AI Chatbo...
MOBLE





Getting the Best out of AI Chatbots: The...
Sia Hua Lim



Prompt Framework for Students: The Five "...
AI for Education

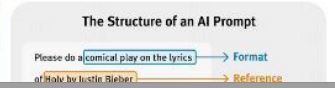




Maximizing Chatbot Potential with Promptin...
SearchUnify



Mastering prompt engineering for effective A...
Facebook

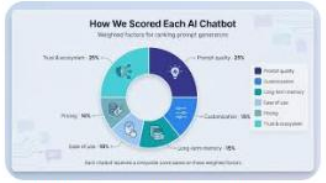




Chatbot prompt essentials | Digital NSW
Digital NSW - NSW Government

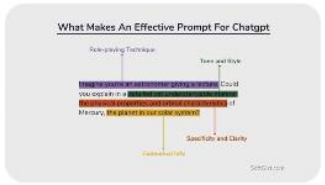


Prompt Controls in GenAI Chatbots: 4 Main...
Nielsen Norman Group

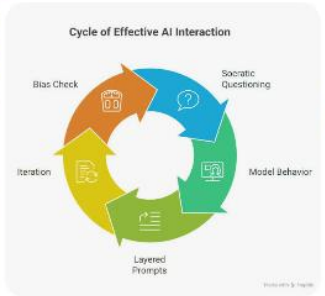


AI Chatbots for Writing Prompts: Which One...
Toolzoon AI

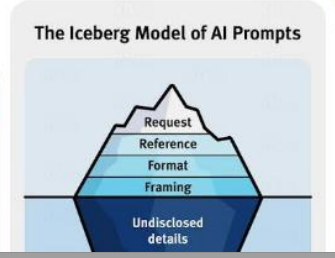




The Ultimate Anatomy of a ChatGPT Prompt
Sydney Digital Marketing Agency



How to Write Smarter ChatGPT Prompts:...
Great Learning





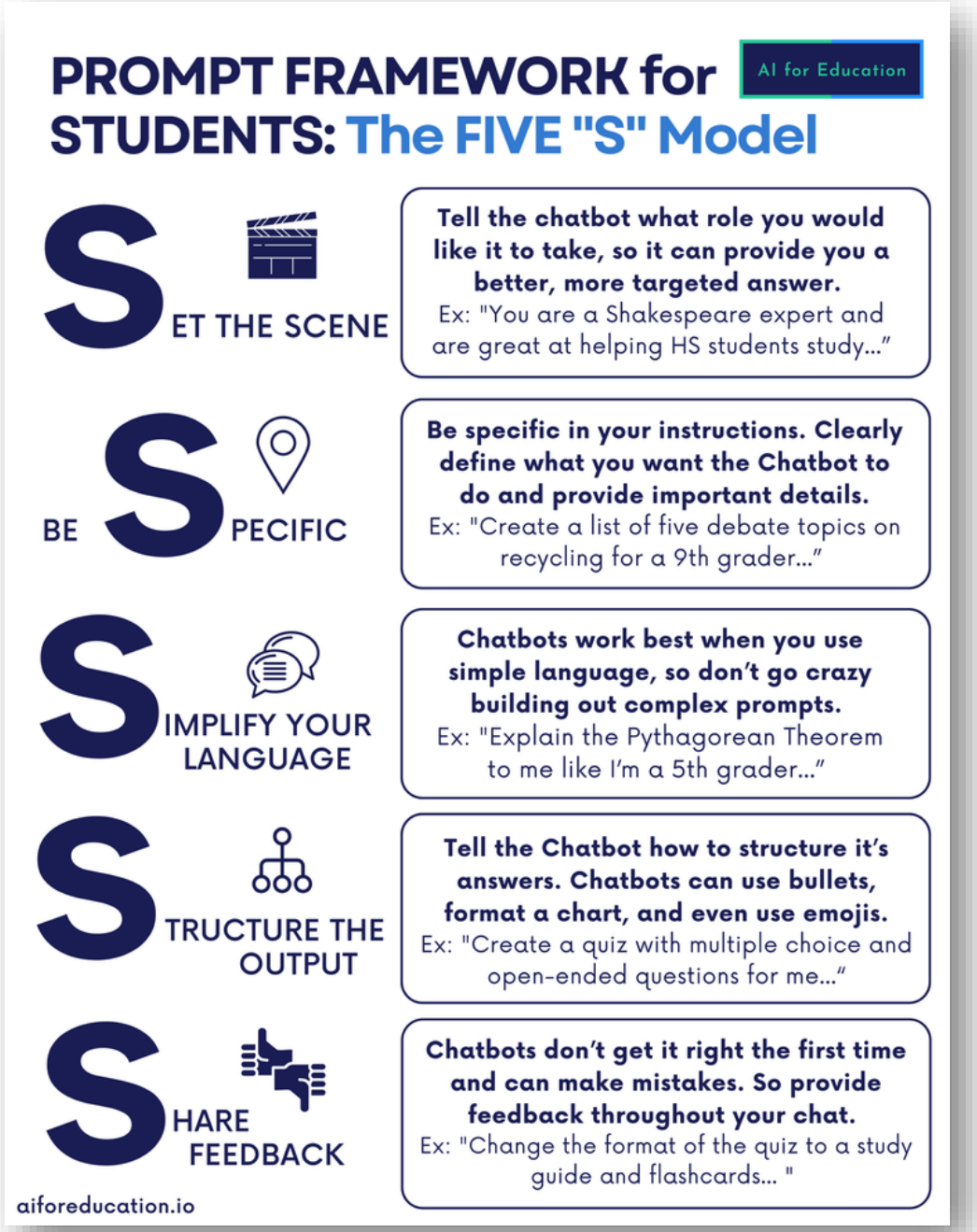
Getting Better Results with AI Chatbots: 5...
Advanced Purchasing Dynamics



What Makes Effective Voice AI Agents?
Vodex AI



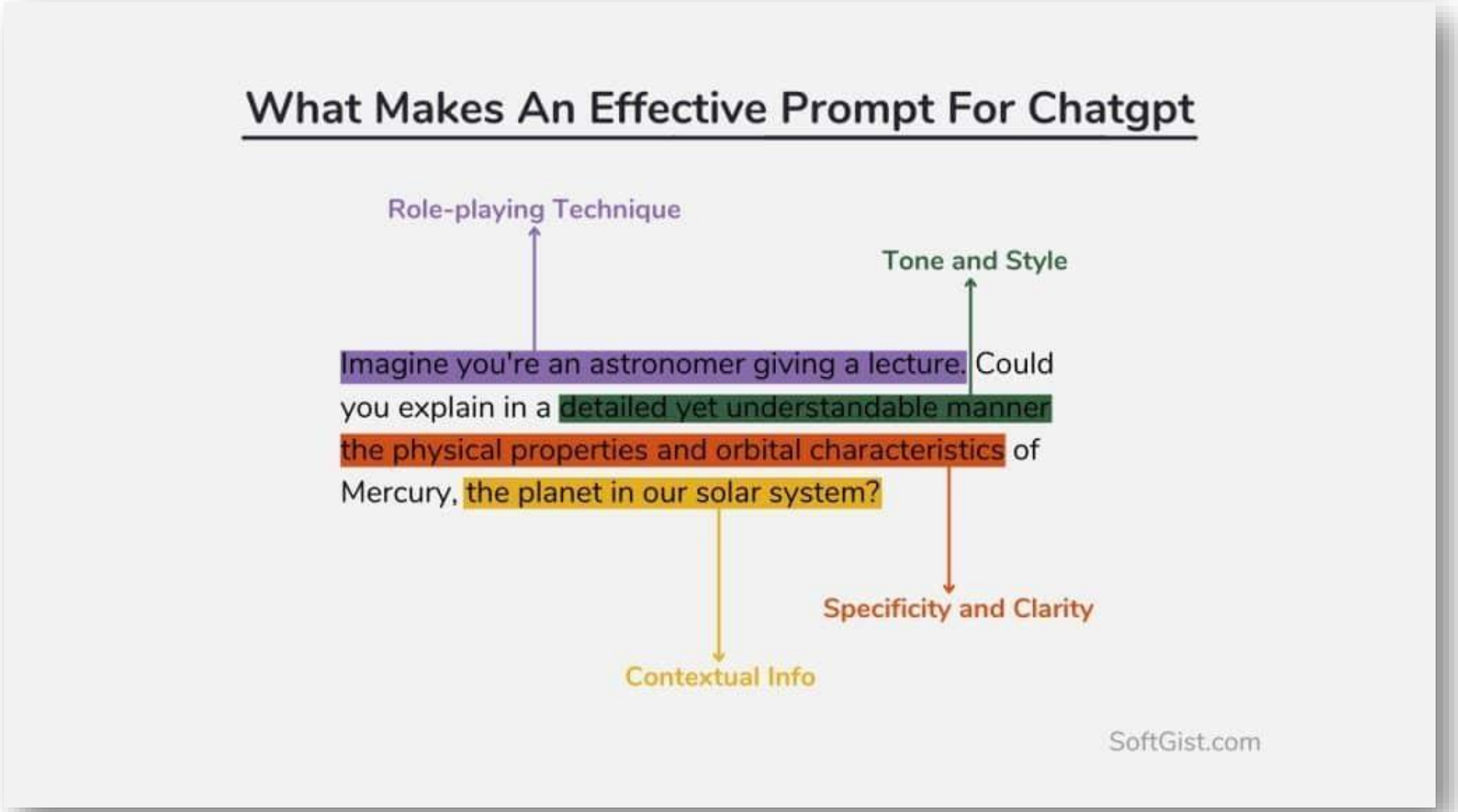
Prompt Engineering: Examples and Best...



Source: https://www.aiforeducation.io/ai-resources/the-five-s-model-students



Source: https://www.aiforeducation.io/ai-resources/effective-prompting-for-educators



Source: https://sydneydigitalmarketing.com.au/blog/anatomy-of-a-chatgpt-prompt/



Source: https://www.facebook.com/groups/aifire.co/posts/1586906065248030/



Source: https://www.searchunify.com/resource-center/sudo-technical-blogs/optimizing-chatbot-performance-the-power-of-prompting-and-temperature-control/

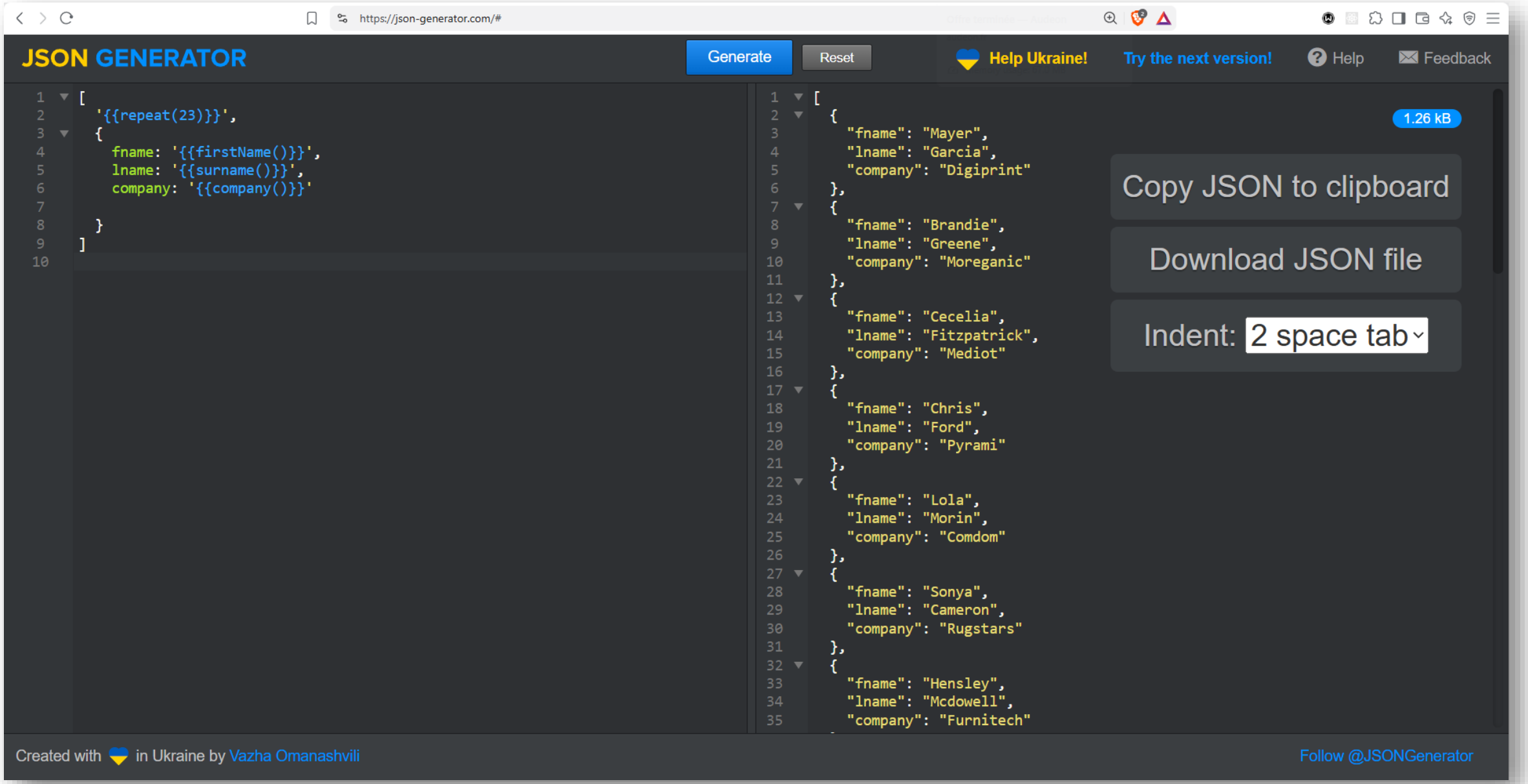
4. WHY USE AI IN EDUCATION?

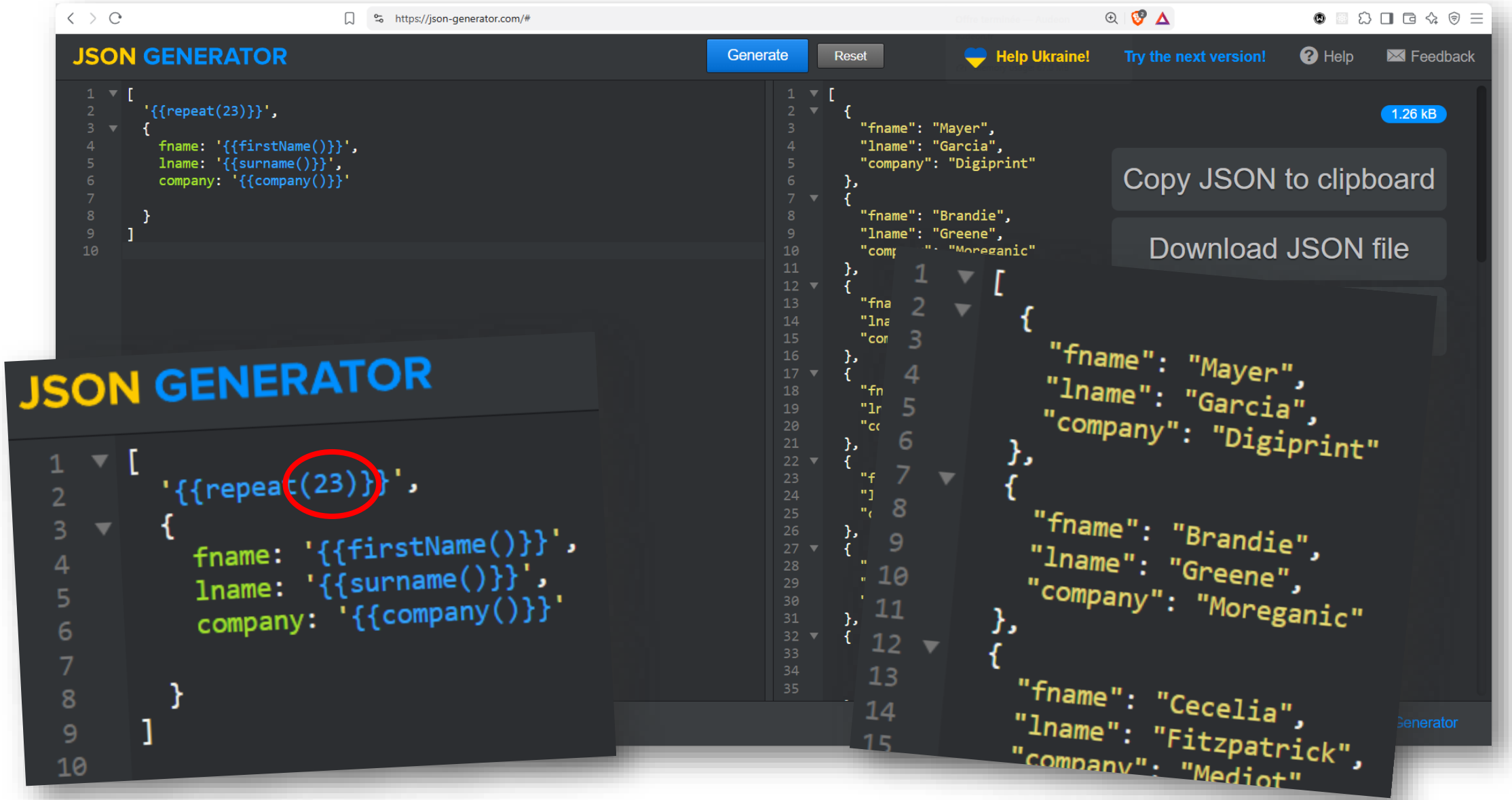
Do I use AI to help me **create questions** in my examinations in 2026? **YES**

Do I use AI for **grading students** in 2026? **NO**



?


https://json-generator.com/#
JSON GENERATOR
Generate
Reset
Help Ukraine!
Try the next version!
Help
Feedback
[
'{{repeat(23)}}',
{
fname: '{{firstName()}}',
lname: '{{surname()}}',
company: '{{company()}}'
}
]
1.26 kB
Copy JSON to clipboard
Download JSON file
Indent: 2 space tab
[
{
"fname": "Mayer",
"lname": "Garcia",
"company": "Digiprint"
},
{
"fname": "Brandie",
"lname": "Greene",
"company": "Moreganic"
},
{
"fname": "Cecelia",
"lname": "Fitzpatrick",
"company": "Mediot"
},
{
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"company": "Pyrami"
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{
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"lname": "Morin",
"company": "Comdom"
},
{
"fname": "Sonya",
"lname": "Cameron",
"company": "Rugstars"
},
{
"fname": "Hensley",
"lname": "Mcdowell",
"company": "Furnitech"
Created with in Ukraine by Vazha Omanashvili
Follow @JSONGenerator


https://json-generator.com/#
JSON GENERATOR
Generate
Reset
Help Ukraine!
Try the next version!
Help
Feedback
1.26 kB
Copy JSON to clipboard
Download JSON file
[
'{{repeat(23)}}',
{
fname: '{{firstName()}}',
lname: '{{surname()}}',
company: '{{company()}}'
}
]
[
{
"fname": "Mayer",
"lname": "Garcia",
"company": "Digiprint"
},
{
"fname": "Brandie",
"lname": "Greene",
"company": "Moreganic"
},
{
"fname": "Cecelia",
"lname": "Fitzpatrick",
"company": "Mediot"

DO NOT TRUST AI 100%!

How many persons are in this JSON file?

[{"fname":"Mayer","lname":"Garcia","company":"Digiprint"},
{"fname":"Brandie","lname":"Greene","company":"Moreganic"},
{"fname":"Cecelia","lname":"Fitzpatrick","company":"Mediot"},
{"fname":"Chris","lname":"Ford","company":"Pyrami"},
{"fname":"Lola","lname":"Morin","company":"Comdom"},
{"fname":"Sonya","lname":"Cameron","company":"Rugstars"},
{"fname":"Hensley","lname":"Mcdowell","company":"Furnitech"},
{"fname":"Allyson","lname":"Lambert","company":"Netplax"},
{"fname":"Meadows","lname":"Hogan","company":"Geoform"},
{"fname":"Amparo","lname":"Norman","company":"Cuizine"},
{"fname":"Robbie","lname":"Cobb","company":"Krog"},
{"fname":"Santiago","lname":"Lynch","company":"Cyclonica"},
{"fname":"Rosario","lname":"Waters","company":"Supportal"},
{"fname":"Sonia","lname":"Dillard","company":"Zanilla"},
{"fname":"Villarreal","lname":"Rios","company":"Portica"},
{"fname":"Alma","lname":"Phelps","company":"Roboid"},
{"fname":"Kari","lname":"Weeks","company":"Netropic"},
{"fname":"Abbott","lname":"Cherry","company":"Bluegrain"},
{"fname":"Winnie","lname":"Ingram","company":"Grupoli"},
{"fname":"Leann","lname":"Navarro","company":"Centuria"},
{"fname":"Pamela","lname":"Burks","company":"Zappix"},
{"fname":"Byrd","lname":"Abbott","company":"Ebidco"},
{"fname":"Rowe","lname":"Cross","company":"Orboid"}]

Show less

Thought for 1s

There are **22 persons** in the JSON file.

Thought for 1s

There are **22 persons** in the JSON file.

10:32am

Can you count again?

Thought for 1s

You're right to ask — I recounted, and there are actually **23 persons**, not 22. I missed one on the first pass (I had skipped "Rowe Cross"). My apologies for the error.

10:33am

DO NOT LET YOUR STUDENTS TRUST AI 100%!

4. ONLINE AI CHATBOTS

Another example in JavaScript:

From these 4 propositions, which one would work?

```
const sayHello = function (name) ⇒ console.log(`Hello, ${name}!`);

const sayHello(name) = () ⇒ console.log(`Hello, ${name}!`);

const sayHello = (name) ⇒ console.log(`Hello, ${name}!`);

const sayHello(name) ⇒ console.log(`Hello, ${name}!`);
```




Thought for 1s ›

Two of the four work:

Proposition 1 — function expression (valid, just not arrow notation):

```js
const sayHello = function (name) => console.log(`Hello, ${name}!`);
```

Copy

Proposition 3 — the arrow function (the correct answer):

```js
const sayHello = (name) => console.log(`Hello, ${name}!`);
```

Copy

Proposition 2 — invalid syntax. The parameter list `(name)` can't be placed next to the variable name; parameters belong inside the parentheses of the arrow function itself:

```js
const sayHello(name) = () => ... // SyntaxError
```

Copy

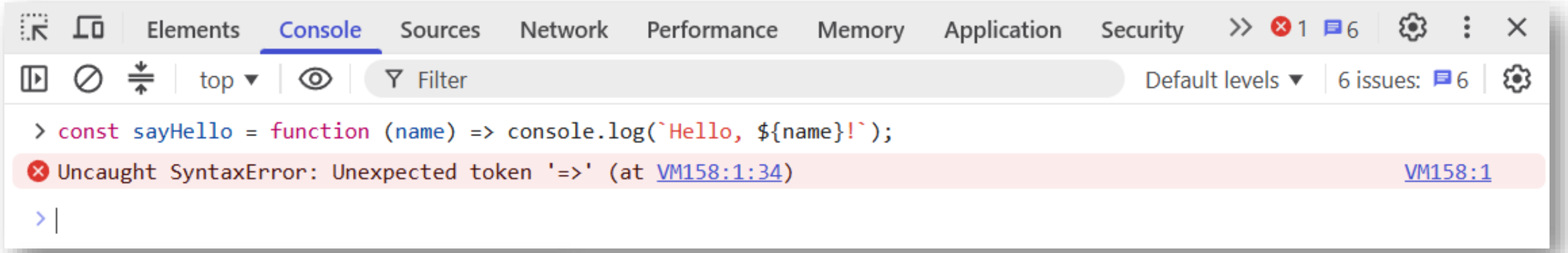
Proposition 4 — also invalid, same mistake; and even rearranged, this isn't how a function is declared:

```js
const sayHello(name) => ... // SyntaxError
```

Copy

Cool!

But NO!



Is proposition 1 correct and working in JS?

DO NOT TRUST AI 100%!

Thought for 1s

Good catch — I need to correct my earlier answer. **No, proposition 1 is NOT valid:**

```
js                                                              Copy
const sayHello = function (name) => console.log(`Hello, ${name}!`);
//                         ^^^^^^^^^^^^  ^^
//                         SyntaxError: can't mix function expression and arrow
```

I misread it earlier as a plain function expression. It's actually a **hybrid**: it starts like a function expression (`function (name)`) but ends with an arrow (`=>`). JavaScript doesn't allow combining the two — `function` expressions use braces or a body directly, arrows have no `function` keyword.

DO NOT LET YOUR STUDENTS TRUST AI 100%!

ANY QUESTIONS?

SNÄLLA, TA EN



- Security and privacy issues:

When I use AI in my work, what are the dangers I can face? For instance, when I speak to an AI, can the AI reproduce my voice? Are data in security when I paste them into an AI? Please answer with a simple paragraph.

Using AI in your work comes with risks like **data privacy concerns**—when you paste sensitive information into an AI, it may be stored, analyzed, or even exposed in training data, depending on the provider's policies. **Voice reproduction** is also possible: some advanced AI models can clone or synthesize voices from audio inputs, raising risks of impersonation or misuse. Additionally, AI outputs may contain **errors, biases, or confidential leaks**, and over-reliance on AI can lead to **compliance violations** (e.g., GDPR) or **intellectual property issues** if proprietary data is shared. Always check your AI tool's security protocols and avoid sharing confidential or personal data.

8:36am

- Ethics:

I am a teacher at the university. I am writing a course for teachers on the use of AI in Education, both for students and for teachers. Can you summarize AI ethics and bias in this context? I want your answer to be a paragraph.

Worked for 4s >

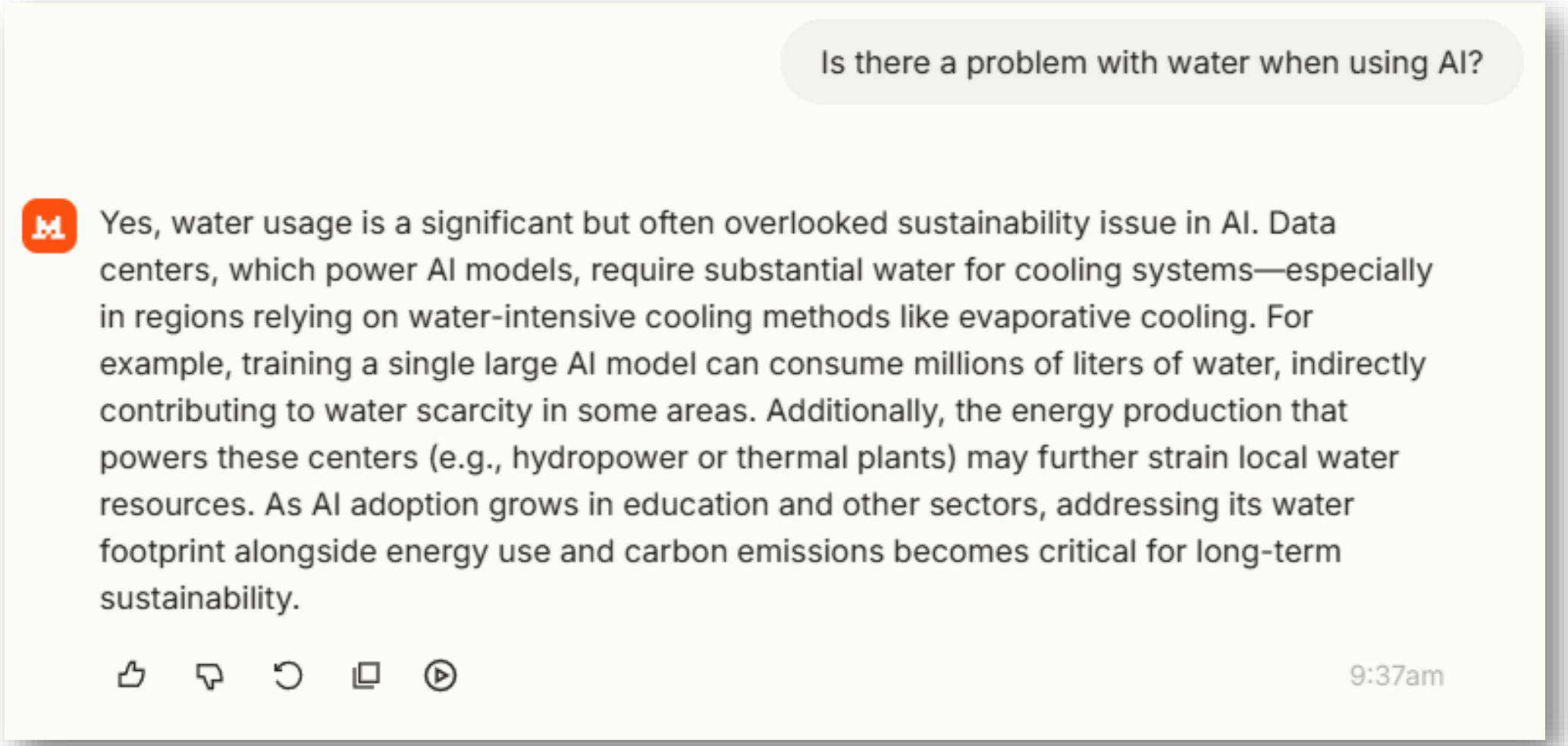
In the context of AI in education, ethical considerations center on fairness, transparency, accountability, and the responsible handling of student data, while bias arises when AI systems trained on incomplete or skewed datasets systematically disadvantage certain groups—such as by underrepresenting minority students in adaptive learning tools, producing culturally narrow content, or yielding inequitable assessments that amplify existing disparities in access or performance. Teachers must critically evaluate AI outputs for hidden biases, protect privacy under regulations like GDPR or FERPA, ensure algorithmic decisions remain explainable and contestable, and model ethical use so students learn to question AI limitations rather than over-rely on them; students, in turn, need guidance on avoiding plagiarism, recognizing when AI recommendations may reflect training-data prejudices, and understanding that unchecked bias can undermine equity in personalized learning, grading support, or resource allocation. Ultimately, effective integration requires ongoing human oversight to align AI with educational values of inclusion, intellectual honesty, and the fostering of independent critical thinking.

- Sustainability:

I am a teacher at the university. I am writing a course for teachers on the use of AI in Education, both for students and for teachers. Can you summarize AI sustainability issues in his context? I want your answer to be a paragraph.

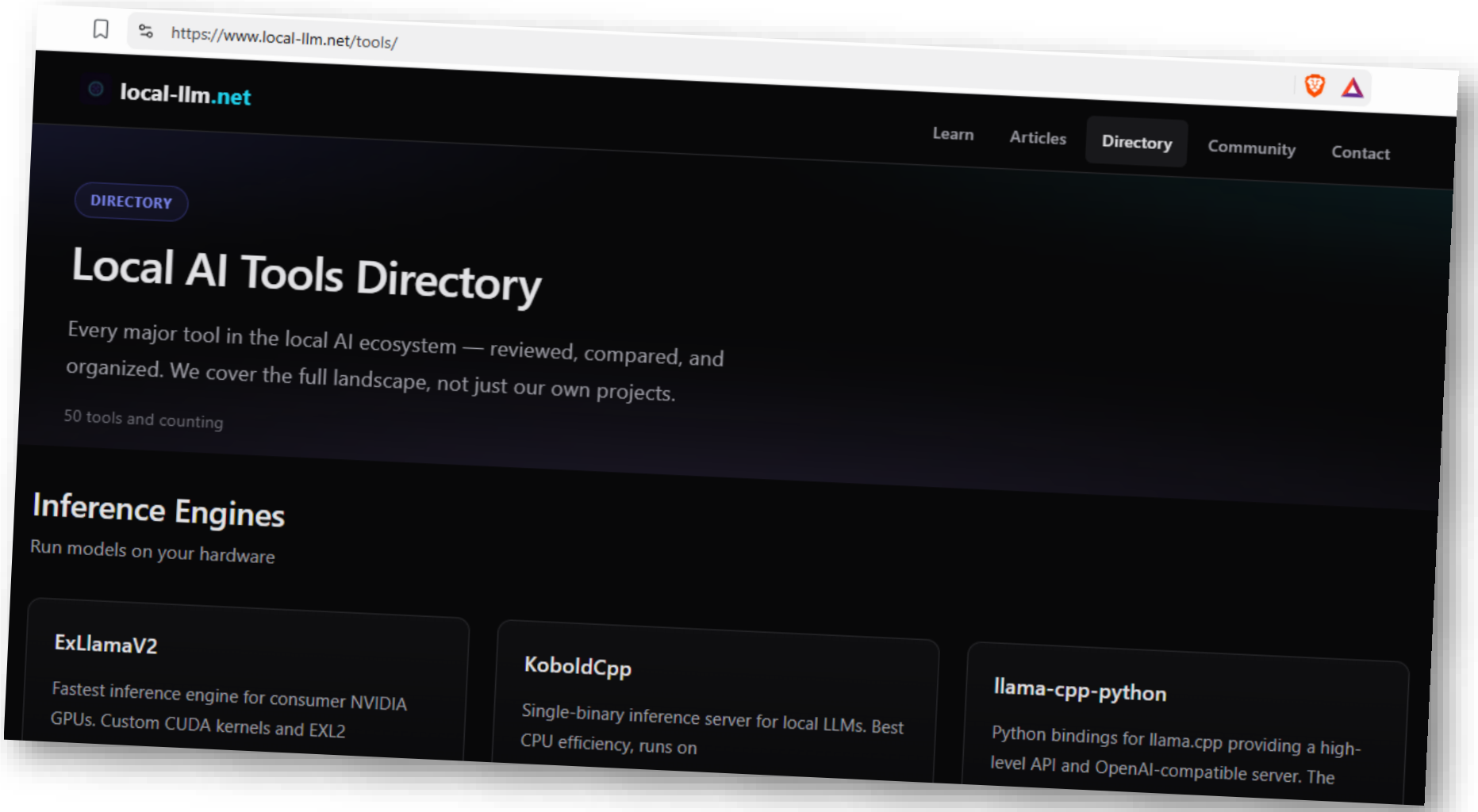
AI in education presents sustainability challenges that span environmental, ethical, and operational dimensions. Environmentally, the energy consumption of AI models—particularly large language models and data centers—raises concerns about carbon footprints, especially if powered by non-renewable energy sources. Ethically, issues like data privacy, algorithmic bias, and the digital divide must be addressed to ensure equitable access and fair outcomes for all students and teachers, regardless of socioeconomic background. Operationally, the long-term viability of AI tools depends on their adaptability to evolving educational needs, the transparency of their decision-making processes, and the ability of institutions to maintain and update these systems without excessive costs. Balancing innovation with responsibility is key to ensuring AI serves as a sustainable, inclusive, and effective tool in education.

- Sustainability (water):



5. LOCAL AI CHATBOTS

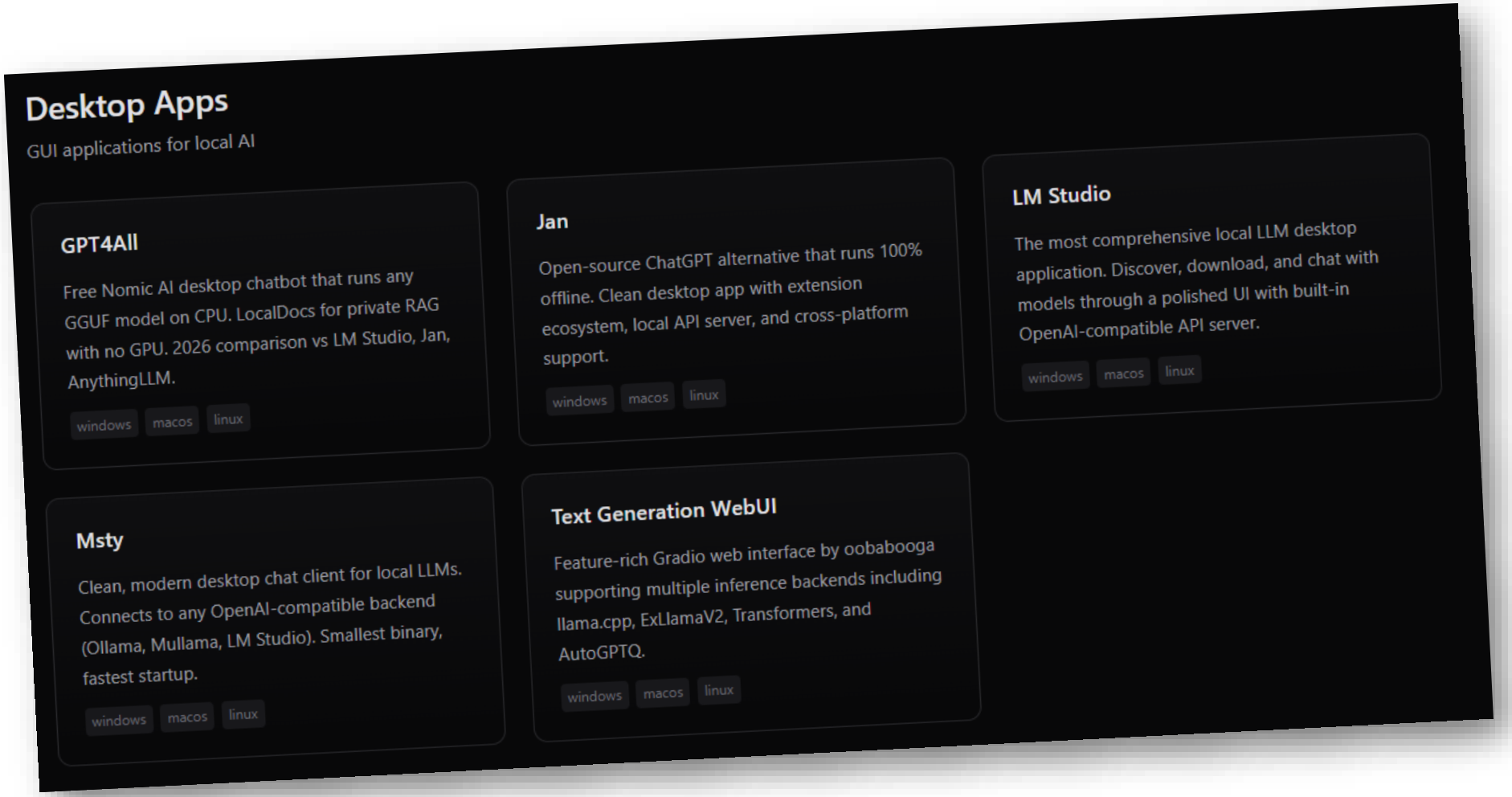
- To avoid to send information on the web, without knowing where it goes and how it is used, you can install LOCAL chatbots on your computer:



Source: https://www.local-llm.net/tools

5. LOCAL AI CHATBOTS

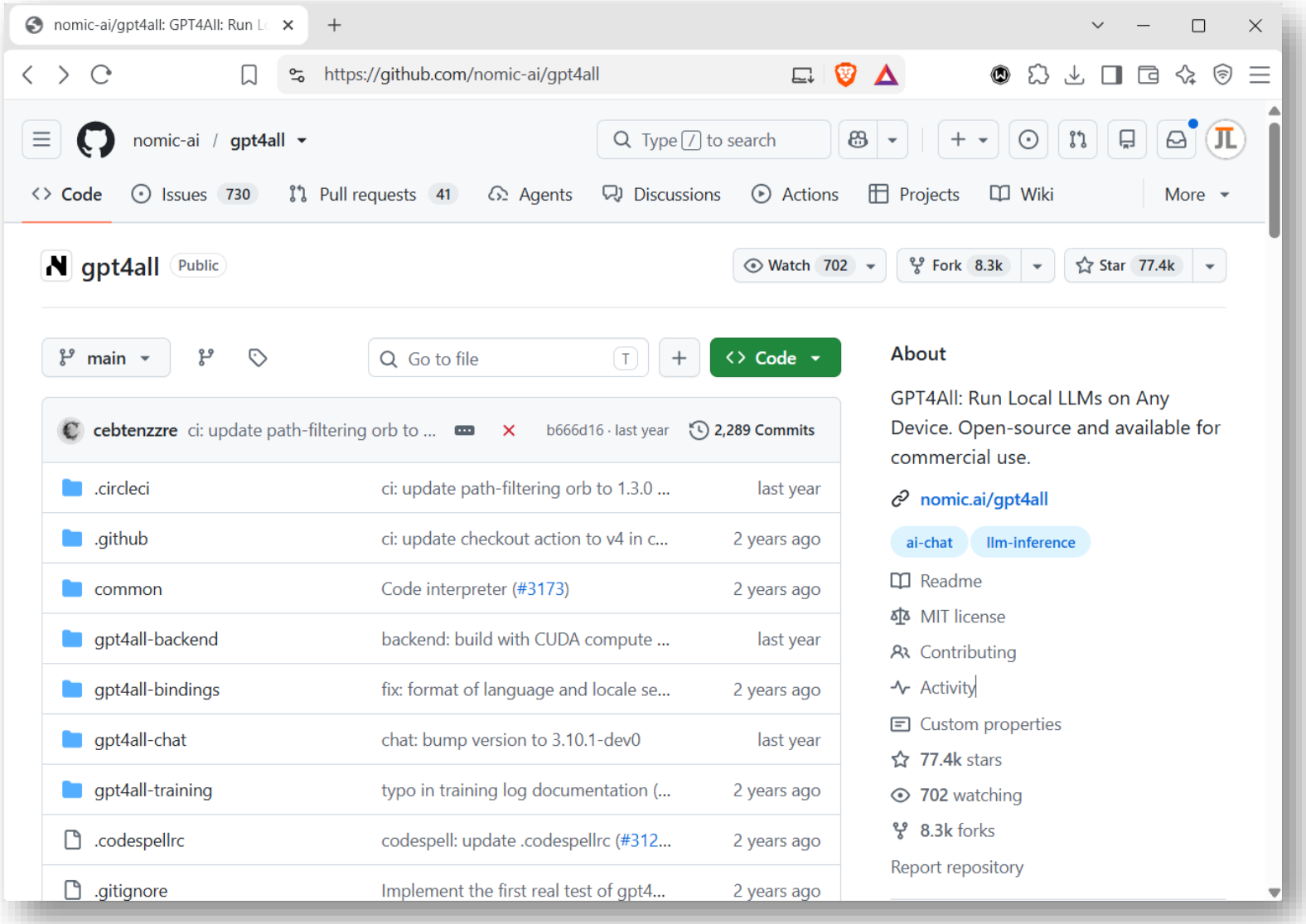
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Source: https://www.local-llm.net/tools/?category=desktop-ui

5. LOCAL AI CHATBOTS

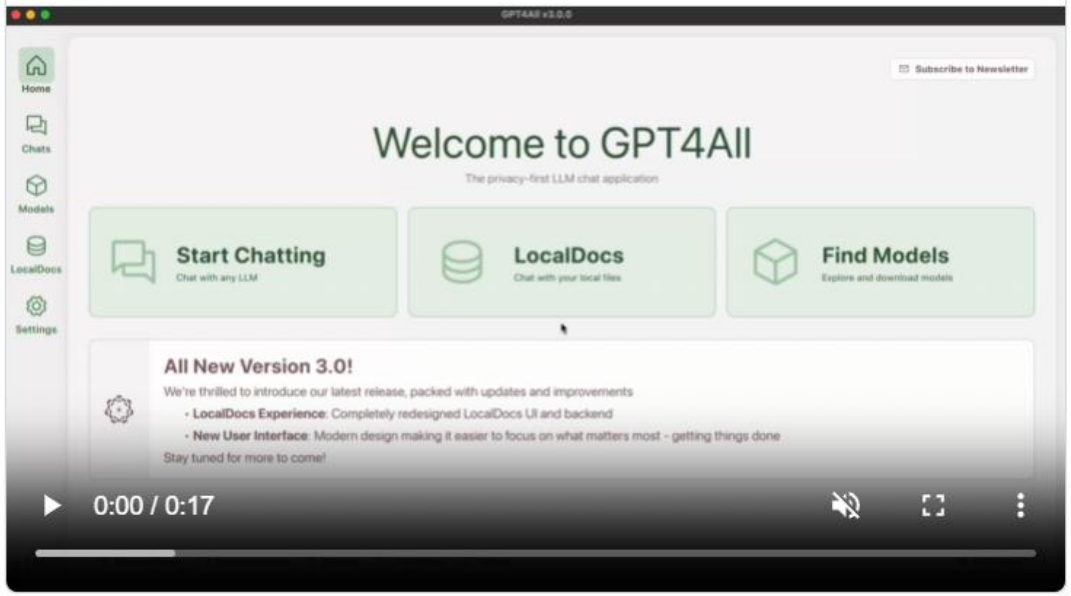




5. LOCAL AI CHATBOTS

gpt4all-installer-win64.exe

README Contributing MIT license



GPT4All is made possible by our compute partner Paperspace.

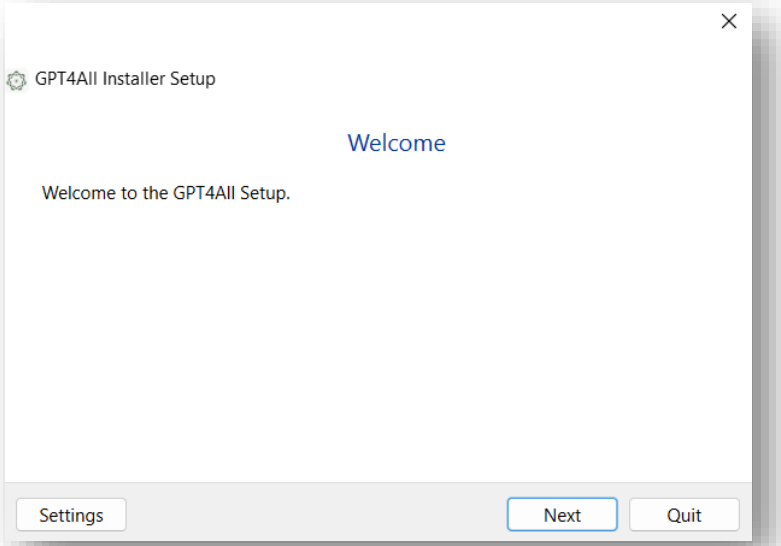
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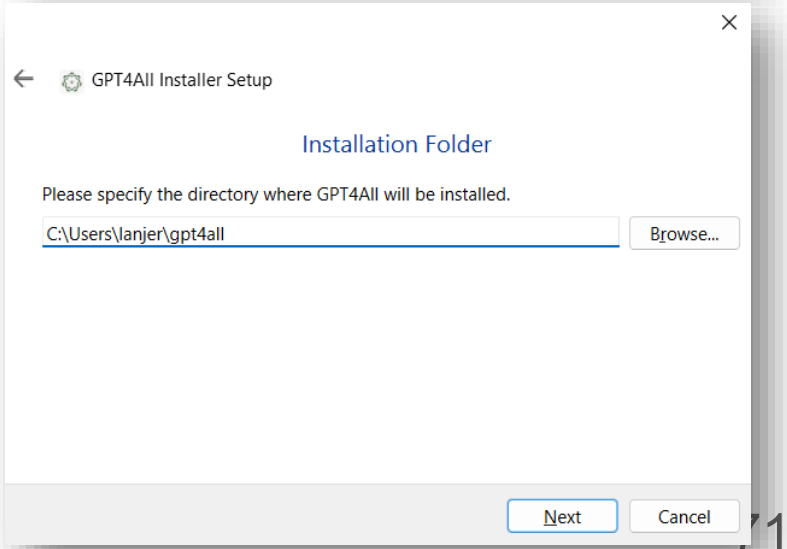
— Windows Installer —

— Windows ARM Installer —

— macOS Installer —

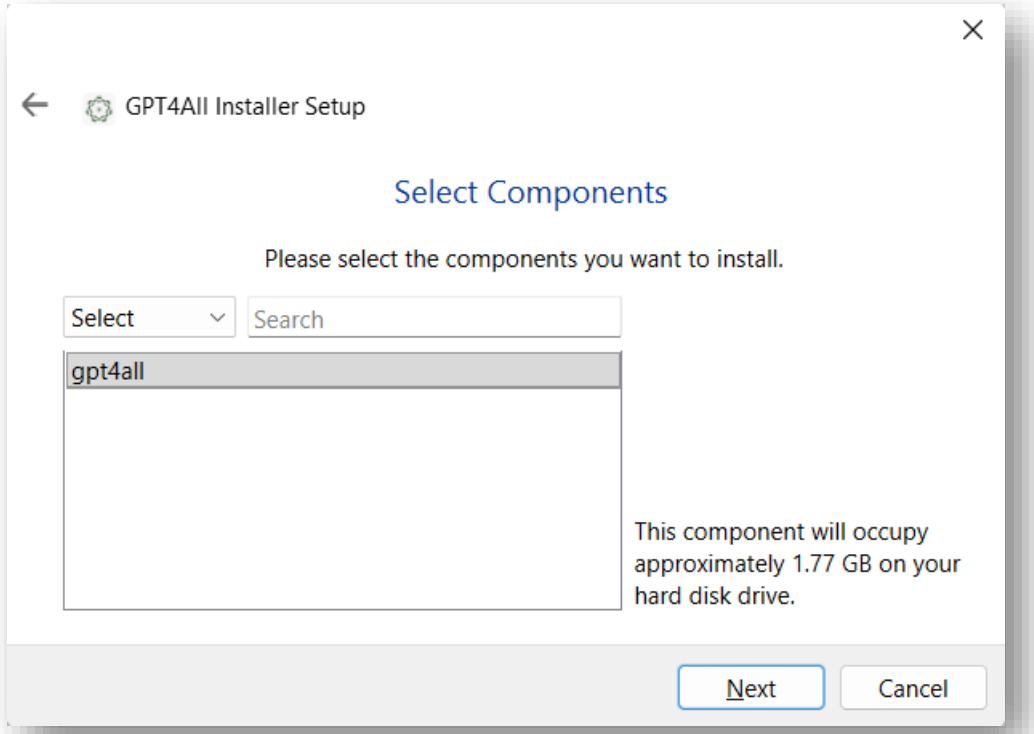
— Ubuntu Installer —

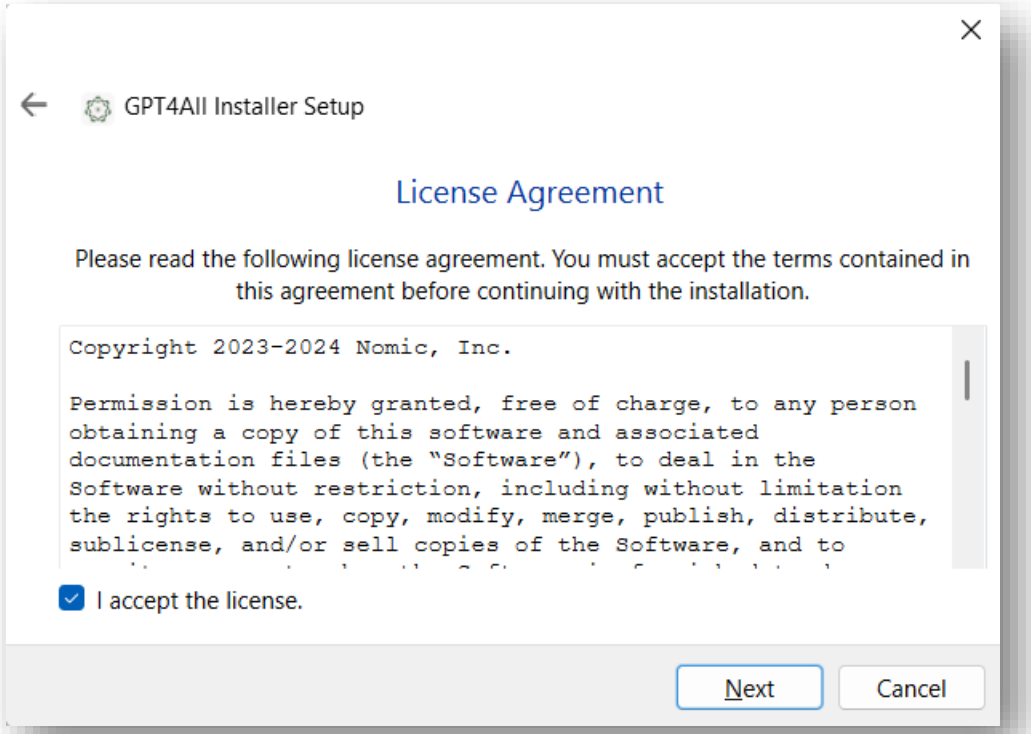




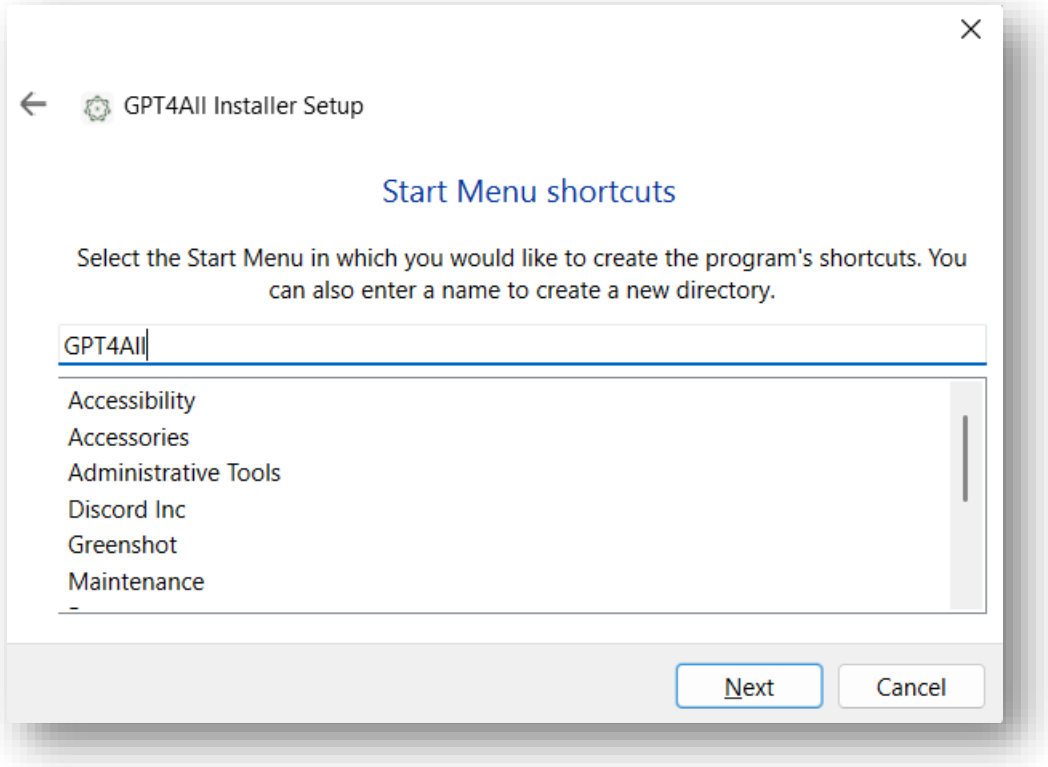
5. LOCAL AI CHATBOTS

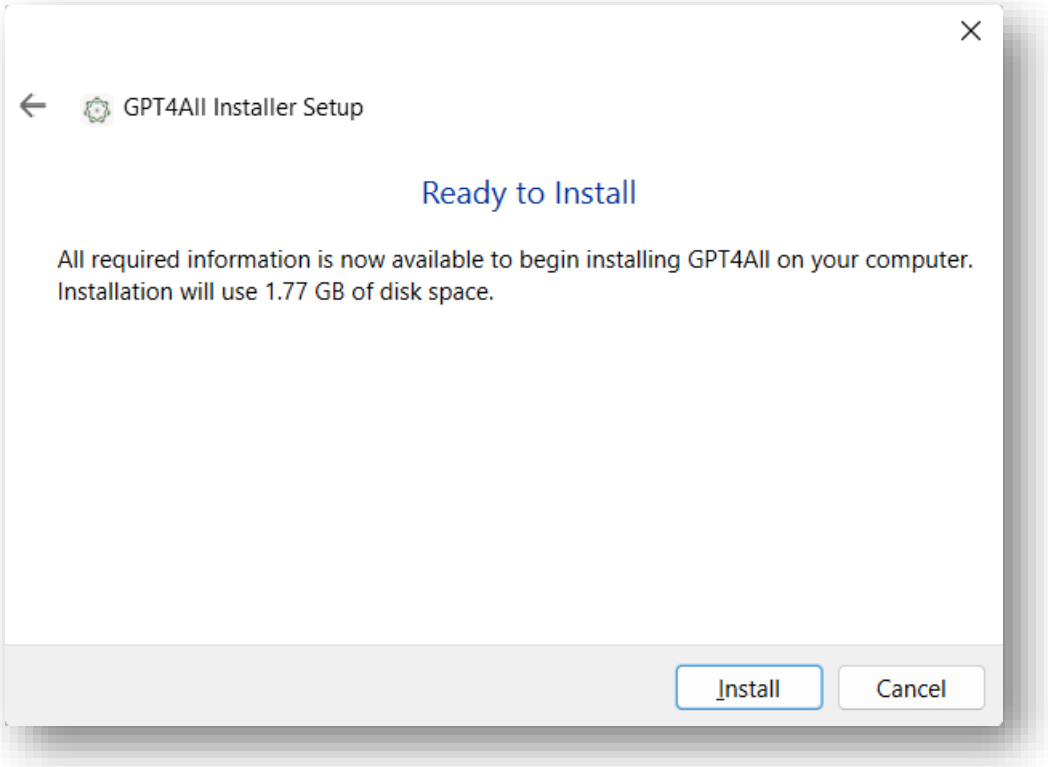
JÖNKÖPING UNIVERSITY
School of Engineering



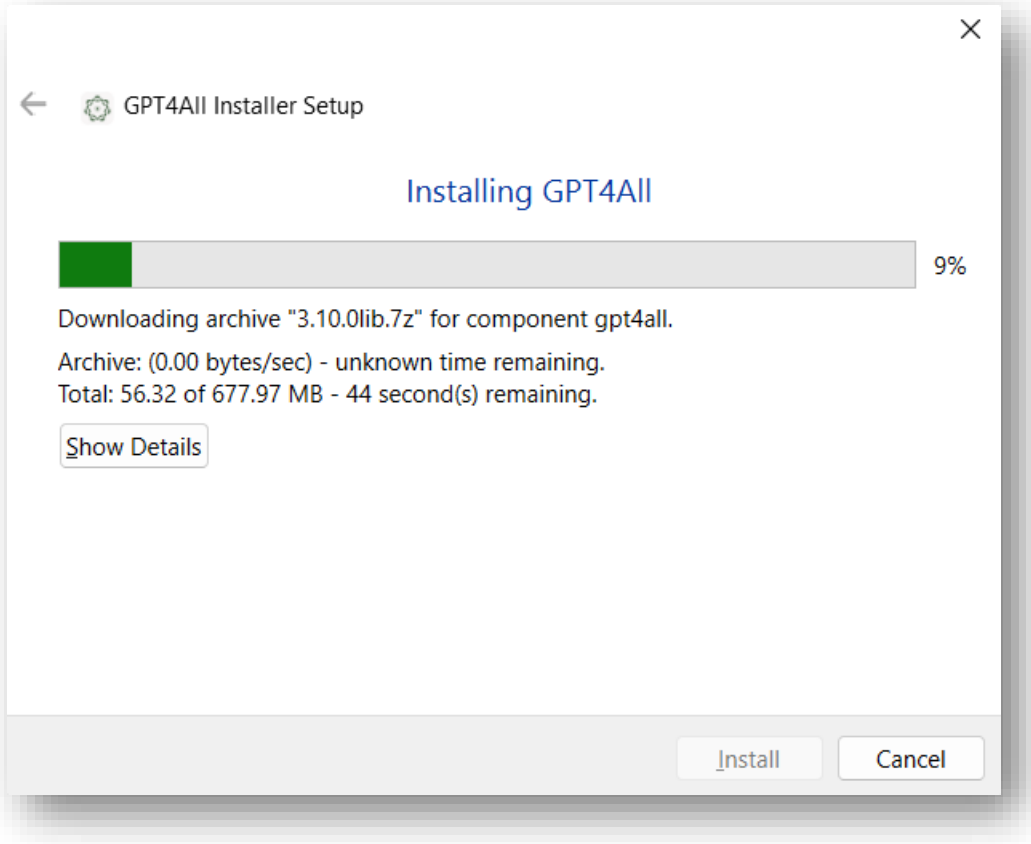


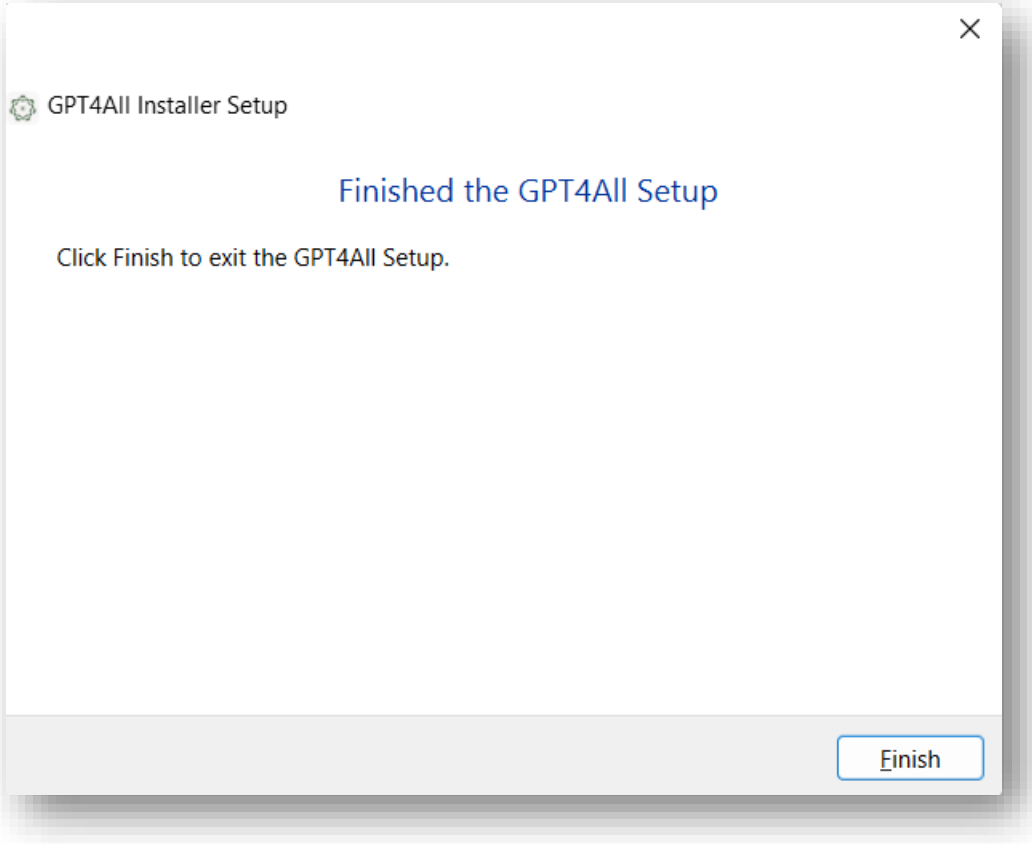
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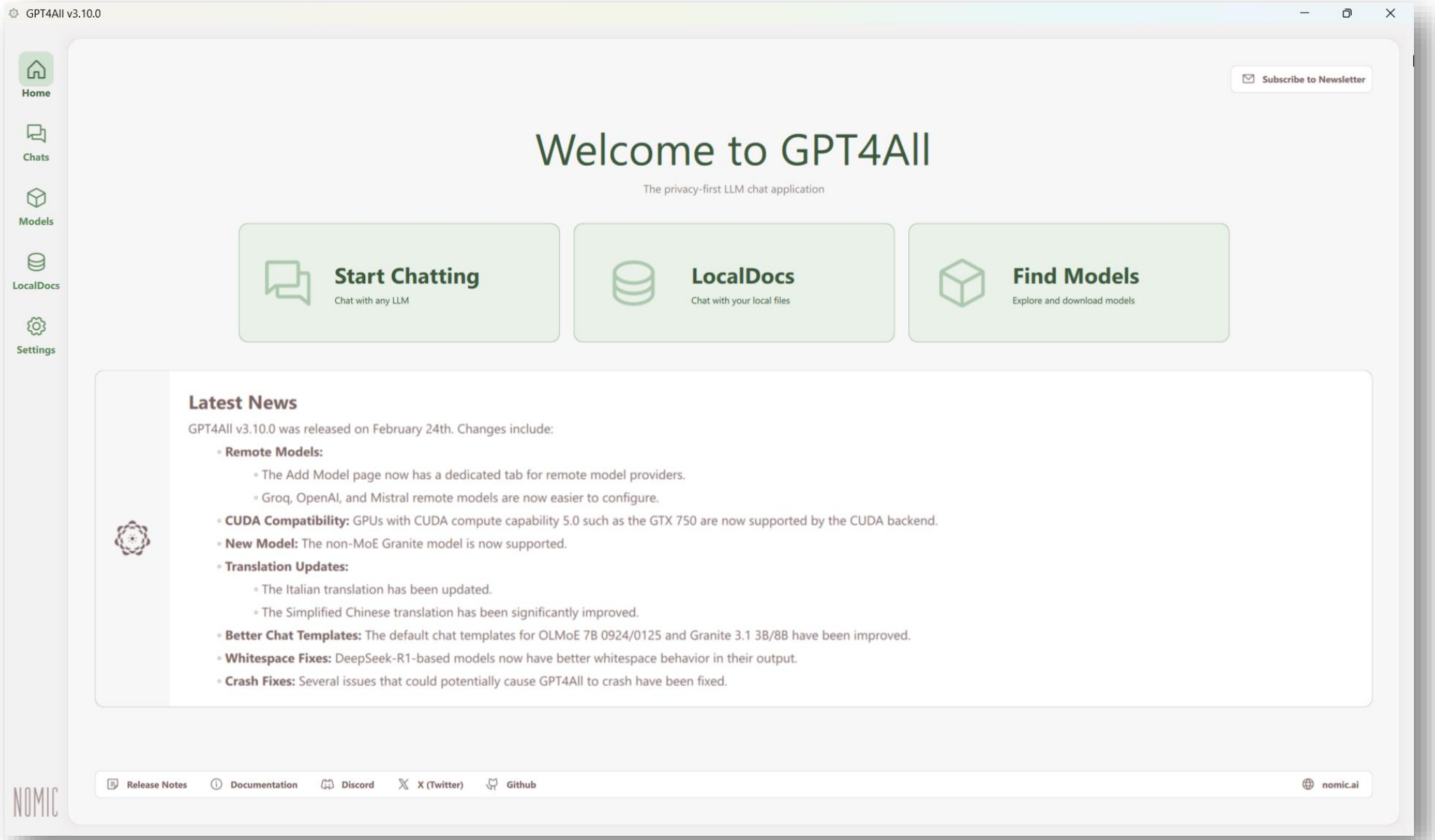


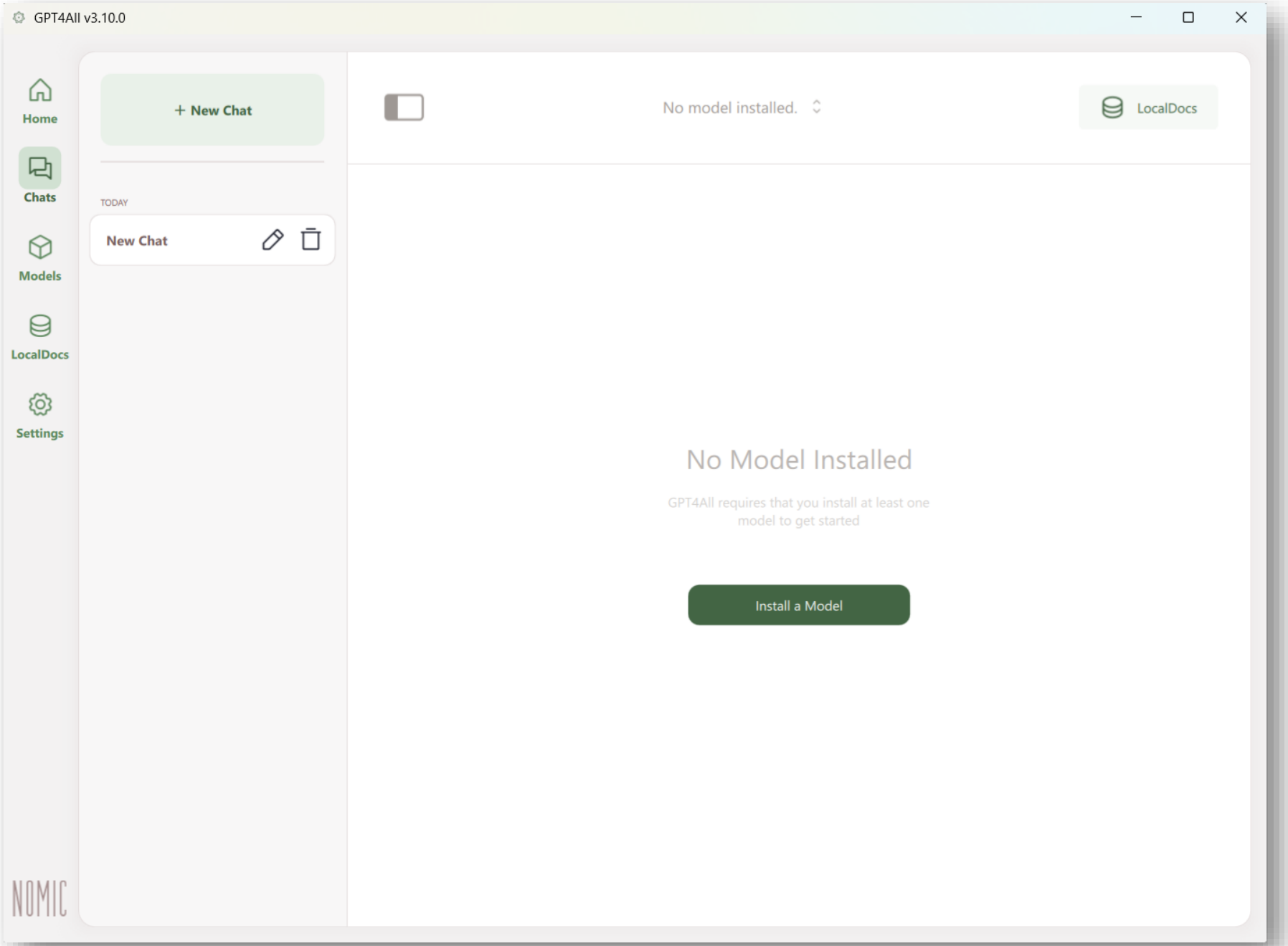


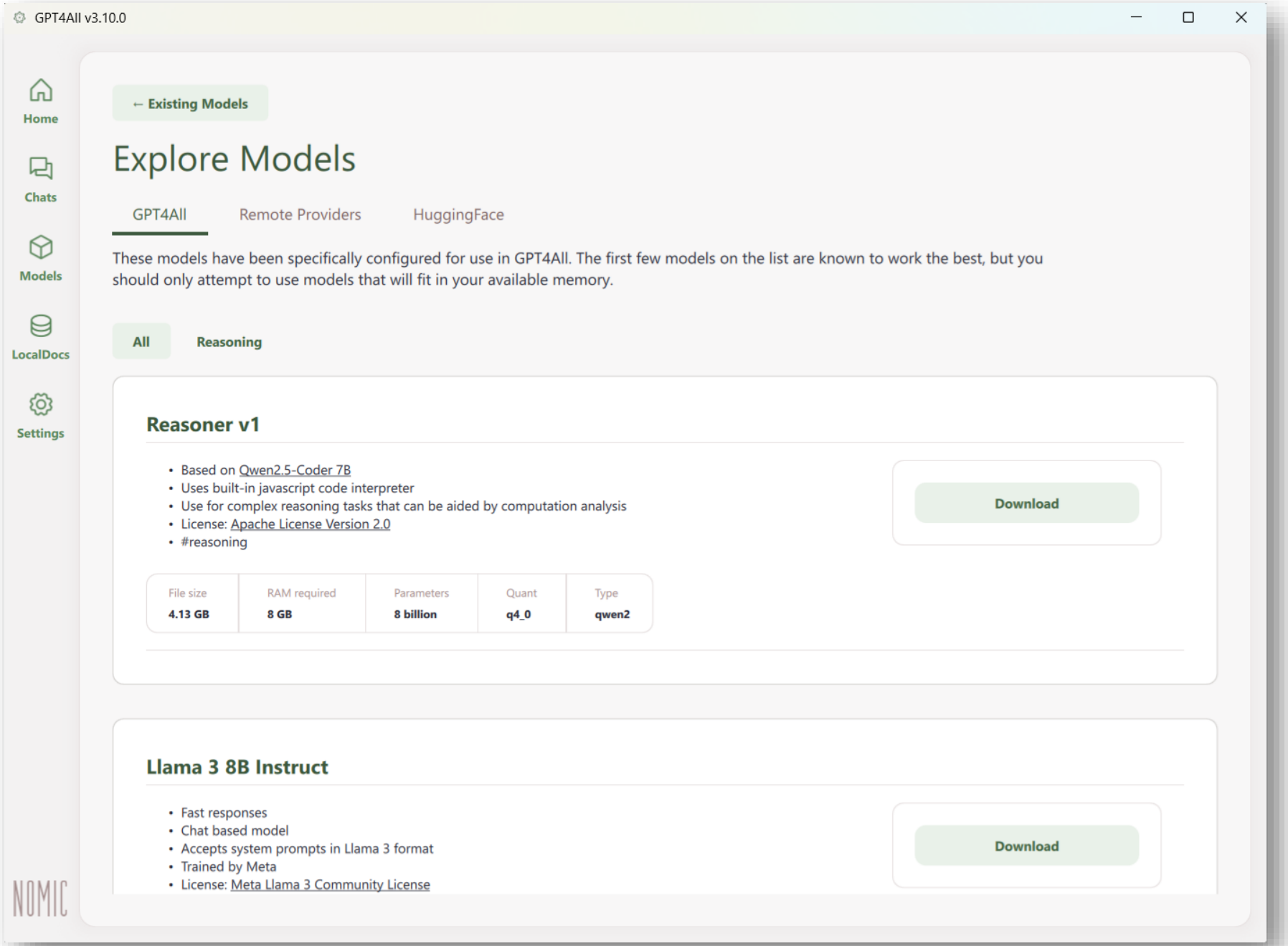
5. LOCAL AI CHATBOTS

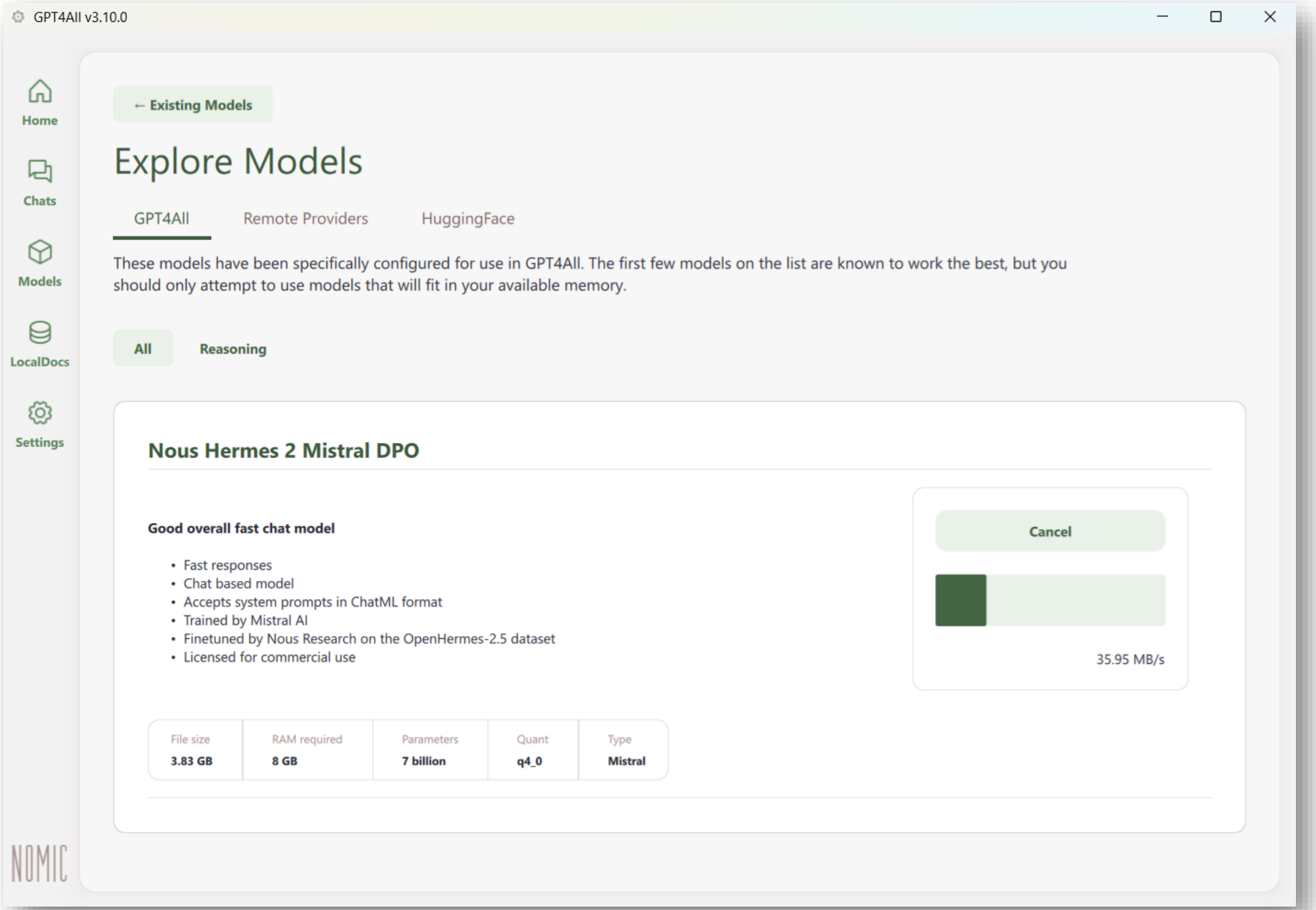




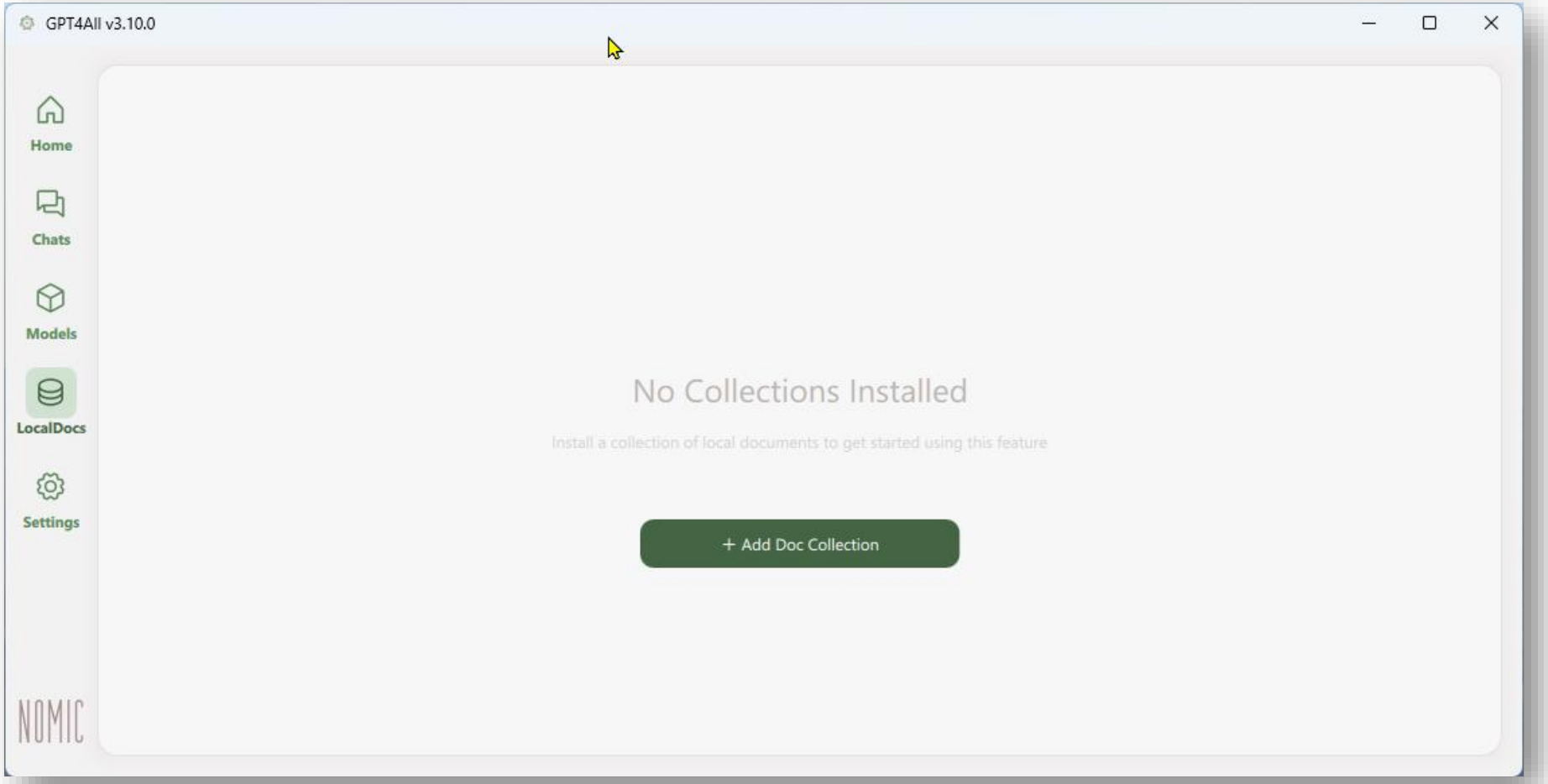

GPT4All v3.10.0
Home
Chats
Models
LocalDocs
Settings
Subscribe to Newsletter
Welcome to GPT4All
The privacy-first LLM chat application
Start Chatting
Chat with any LLM
LocalDocs
Chat with your local files
Find Models
Explore and download models
Latest News
GPT4All v3.10.0 was released on February 24th. Changes include:
Remote Models:
The Add Model page now has a dedicated tab for remote model providers.
Groq, OpenAI, and Mistral remote models are now easier to configure.
CUDA Compatibility: GPUs with CUDA compute capability 5.0 such as the GTX 750 are now supported by the CUDA backend.
New Model: The non-MoE Granite model is now supported.
Translation Updates:
The Italian translation has been updated.
The Simplified Chinese translation has been significantly improved.
Better Chat Templates: The default chat templates for OLMoE 7B 0924/0125 and Granite 3.1 3B/8B have been improved.
Whitespace Fixes: DeepSeek-R1-based models now have better whitespace behavior in their output.
Crash Fixes: Several issues that could potentially cause GPT4All to crash have been fixed.
Release Notes
Documentation
Discord
X (Twitter)
Github
nomic.ai
NOMIC


GPT4All v3.10.0
Home
Chats
Models
LocalDocs
Settings
+ New Chat
TODAY
New Chat
No model installed.
LocalDocs
No Model Installed
GPT4All requires that you install at least one model to get started
Install a Model
NOMIC

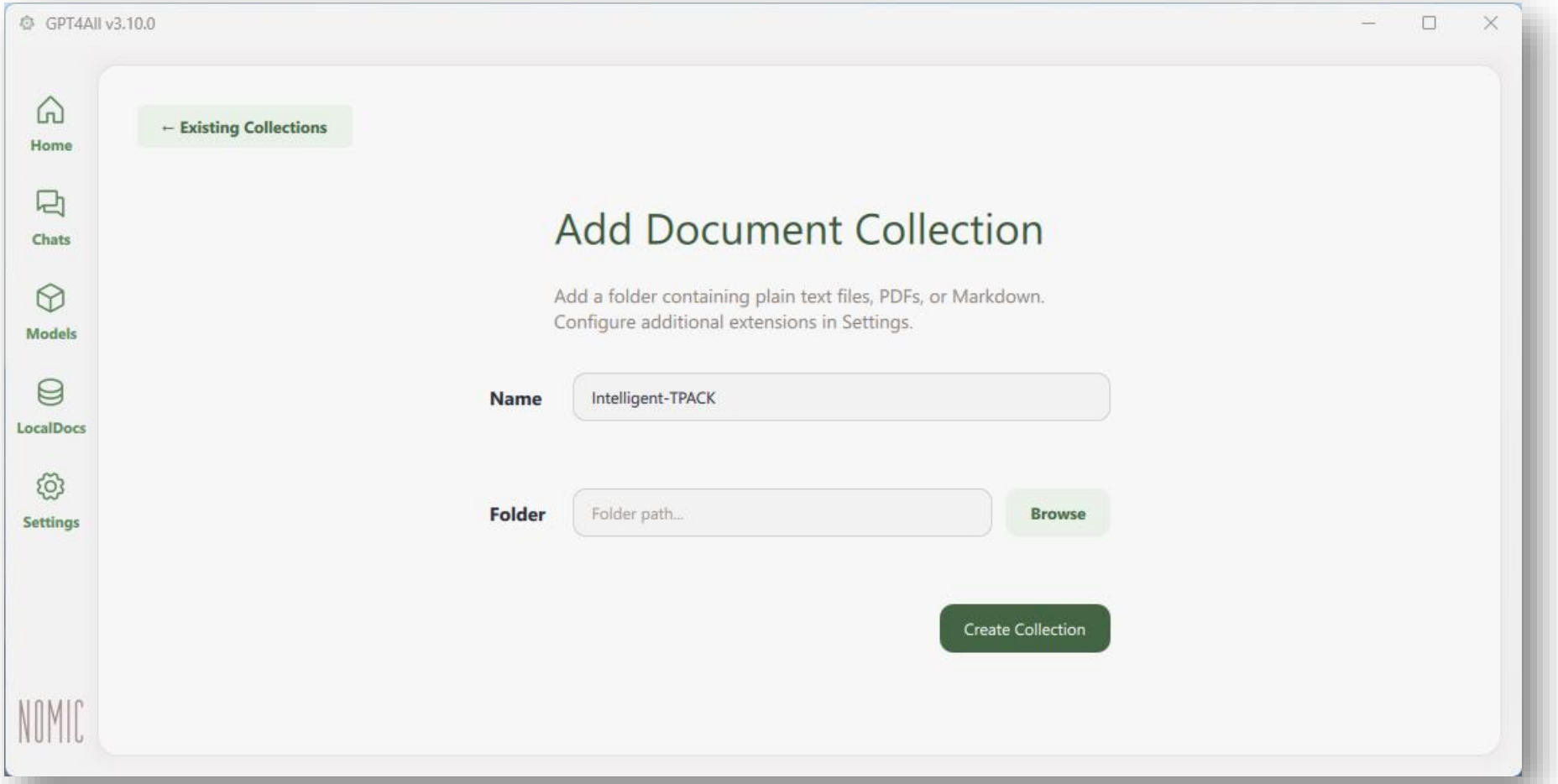

GPT4All v3.10.0
Home
Chats
Models
LocalDocs
Settings
← Existing Models
Explore Models
GPT4All
Remote Providers
HuggingFace
These models have been specifically configured for use in GPT4All. The first few models on the list are known to work the best, but you should only attempt to use models that will fit in your available memory.
All
Reasoning
Reasoner v1
• Based on Qwen2.5-Coder 7B
• Uses built-in javascript code interpreter
• Use for complex reasoning tasks that can be aided by computation analysis
• License: Apache License Version 2.0
• #reasoning
Download
File size
4.13 GB
RAM required
8 GB
Parameters
8 billion
Quant
q4_0
Type
qwen2
Llama 3 8B Instruct
• Fast responses
• Chat based model
• Accepts system prompts in Llama 3 format
• Trained by Meta
• License: Meta Llama 3 Community License
Download
NOMIC

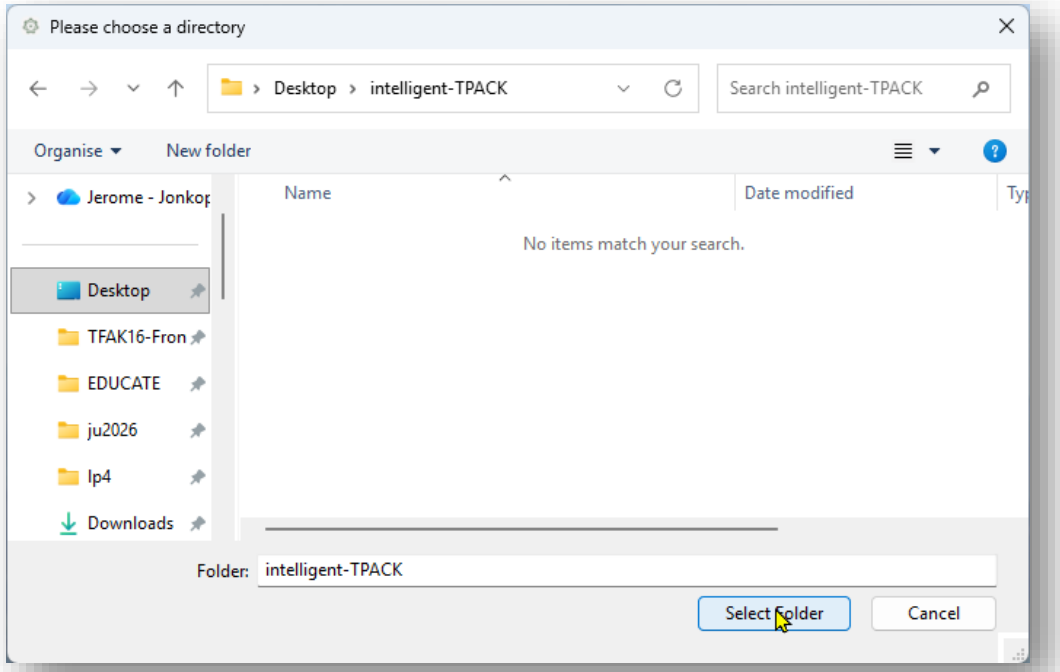

GPT4All v3.10.0
Home
Chats
Models
LocalDocs
Settings
← Existing Models
Explore Models
GPT4All
Remote Providers
HuggingFace
These models have been specifically configured for use in GPT4All. The first few models on the list are known to work the best, but you should only attempt to use models that will fit in your available memory.
All
Reasoning
Nous Hermes 2 Mistral DPO
Good overall fast chat model
• Fast responses
• Chat based model
• Accepts system prompts in ChatML format
• Trained by Mistral AI
• Finetuned by Nous Research on the OpenHermes-2.5 dataset
• Licensed for commercial use
Cancel
35.95 MB/s
File size
3.83 GB
RAM required
8 GB
Parameters
7 billion
Quant
q4_0
Type
Mistral
NOMIC

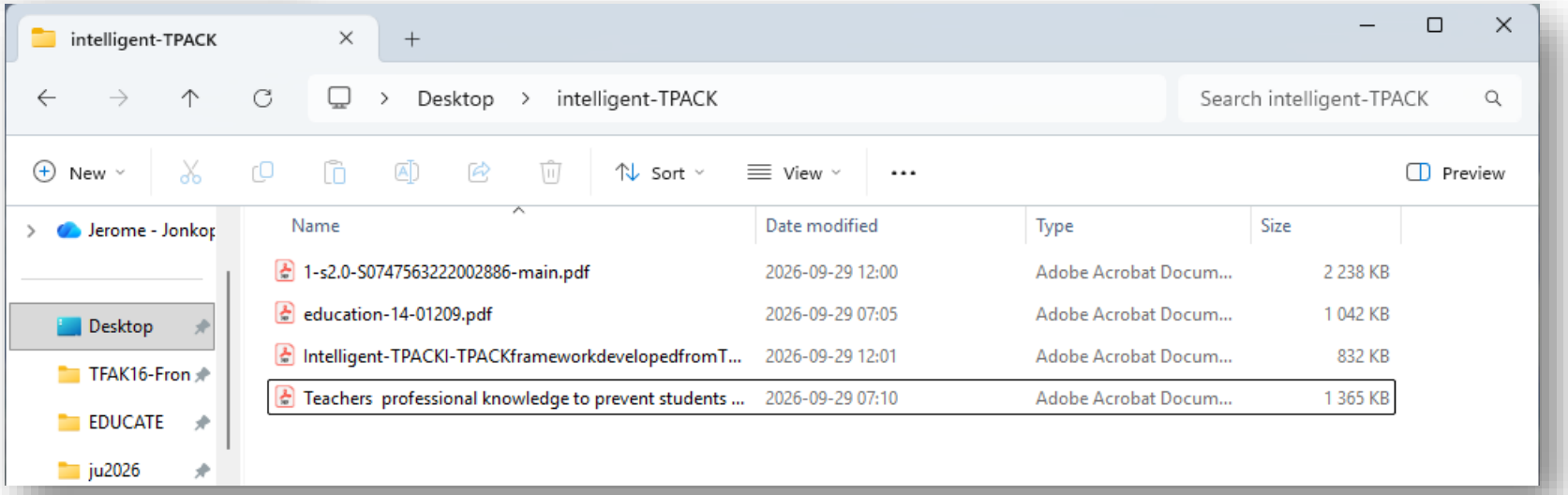
5. LOCAL AI CHATBOTS

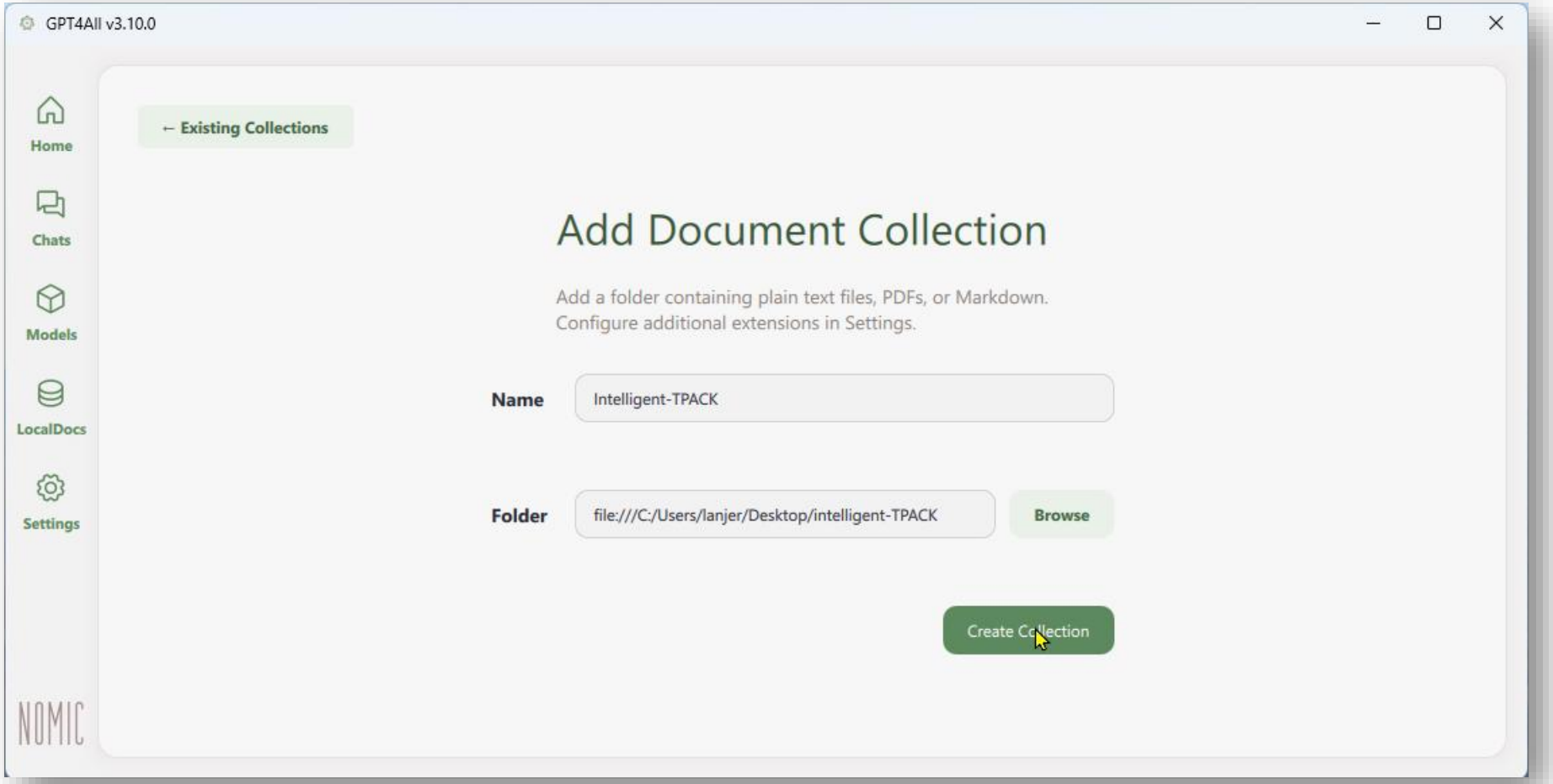


5. LOCAL AI CHATBOTS

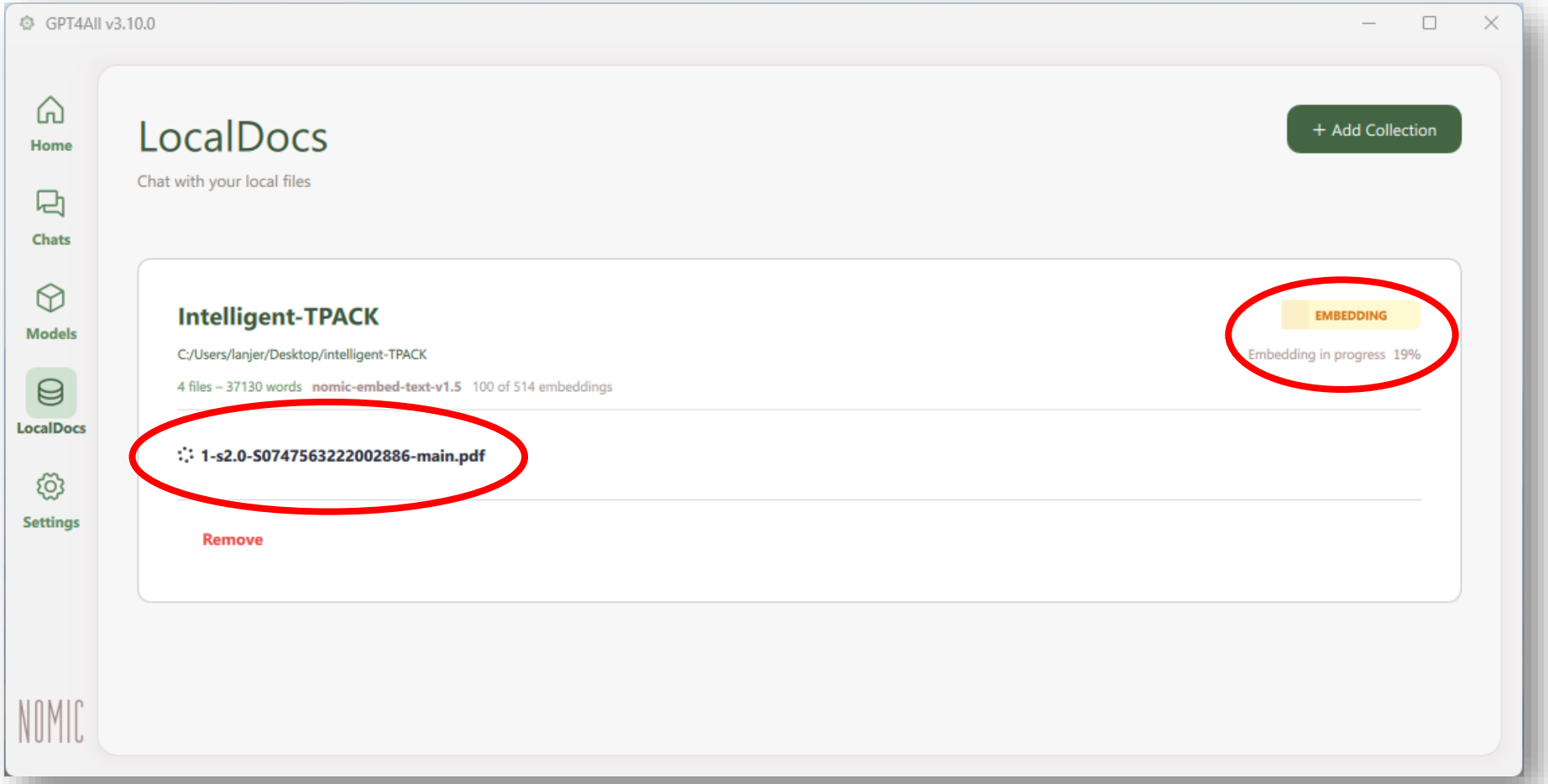



Please choose a directory
Desktop › intelligent-TPACK
Search intelligent-TPACK
Organise
New folder
Jerome - Jonkop
Name
Date modified
No items match your search.
Desktop
TFAK16-Fron
EDUCATE
ju2026
lp4
Downloads
Folder: intelligent-TPACK
Select Folder
Cancel

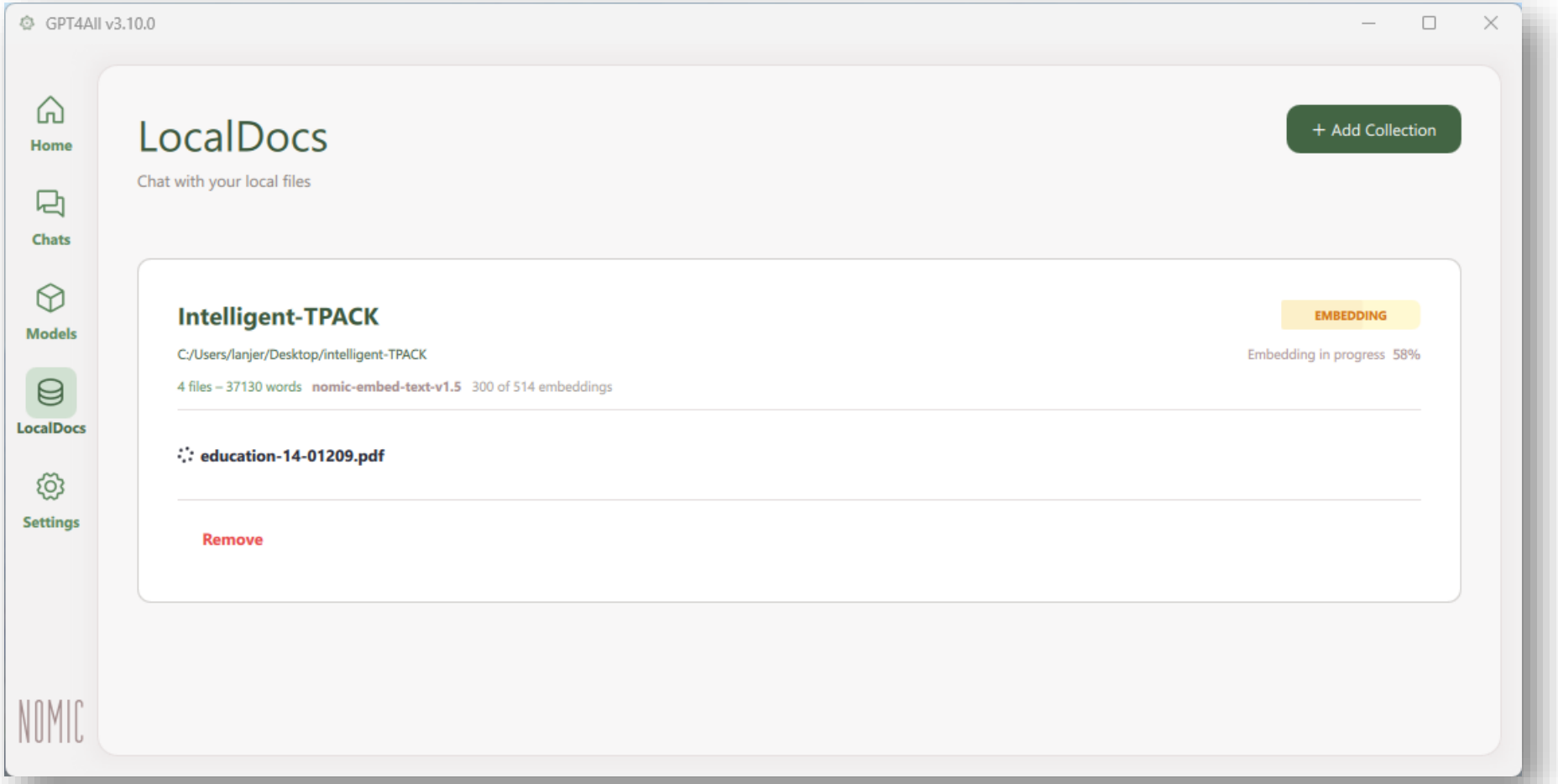

intelligent-TPACK
Desktop › intelligent-TPACK
Search intelligent-TPACK
New
Sort
View
Preview
Jerome - Jonkop
Name
Date modified
Type
Size
1-s2.0-S0747563222002886-main.pdf 2026-09-29 12:00 Adobe Acrobat Docum... 2 238 KB
education-14-01209.pdf 2026-09-29 07:05 Adobe Acrobat Docum... 1 042 KB
Intelligent-TPACKI-TPACKframeworkdevelopedfromT... 2026-09-29 12:01 Adobe Acrobat Docum... 832 KB
Teachers professional knowledge to prevent students ... 2026-09-29 07:10 Adobe Acrobat Docum... 1 365 KB
Desktop
TFAK16-Fron
EDUCATE
ju2026


GPT4All v3.10.0
Home
Chats
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LocalDocs
Settings
NOMIC
← Existing Collections
Add Document Collection
Add a folder containing plain text files, PDFs, or Markdown.
Configure additional extensions in Settings.
Name
Intelligent-TPACK
Folder
file:///C:/Users/lanjer/Desktop/intelligent-TPACK
Browse
Create Collection

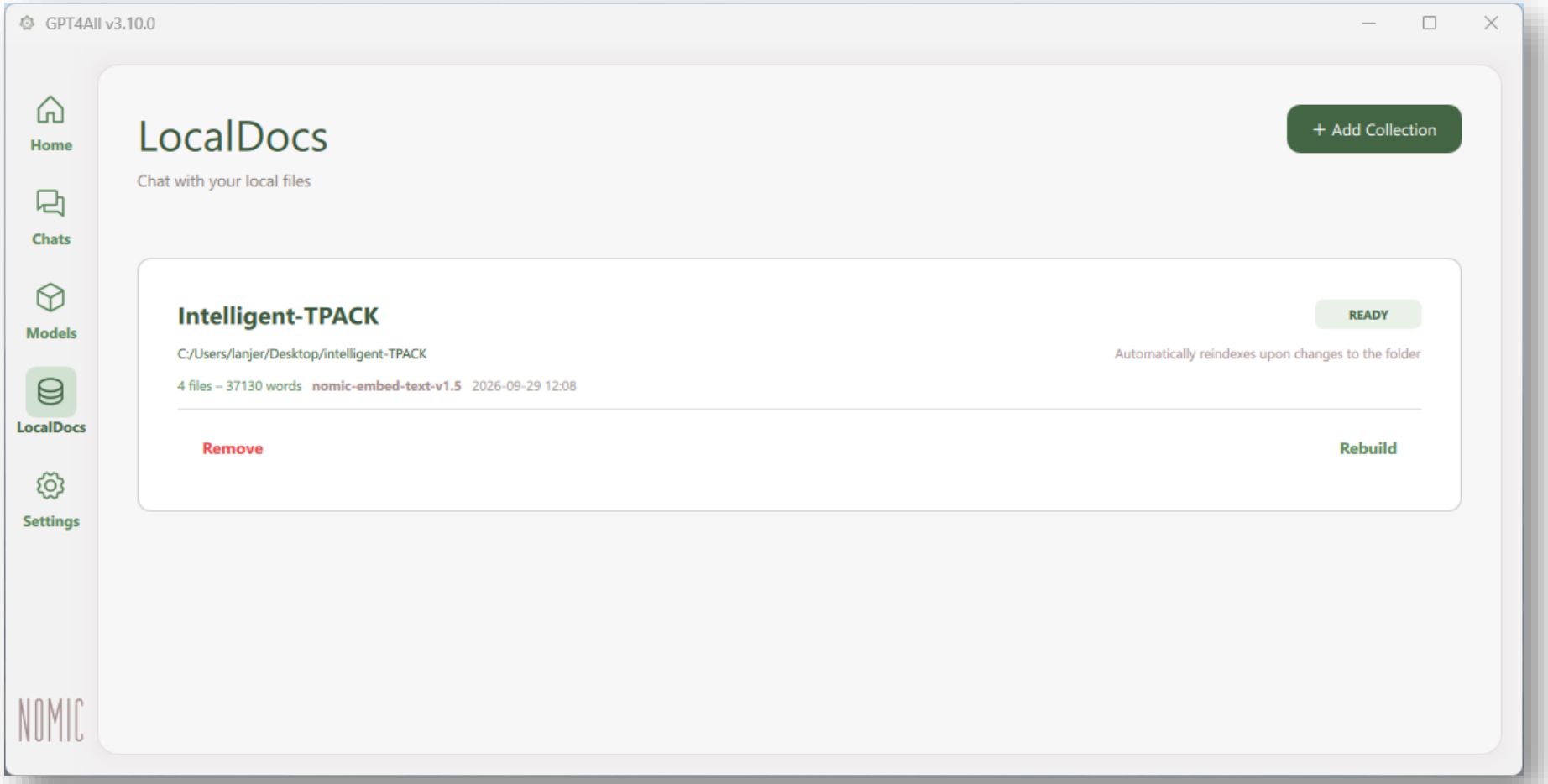
5. LOCAL AI CHATBOTS



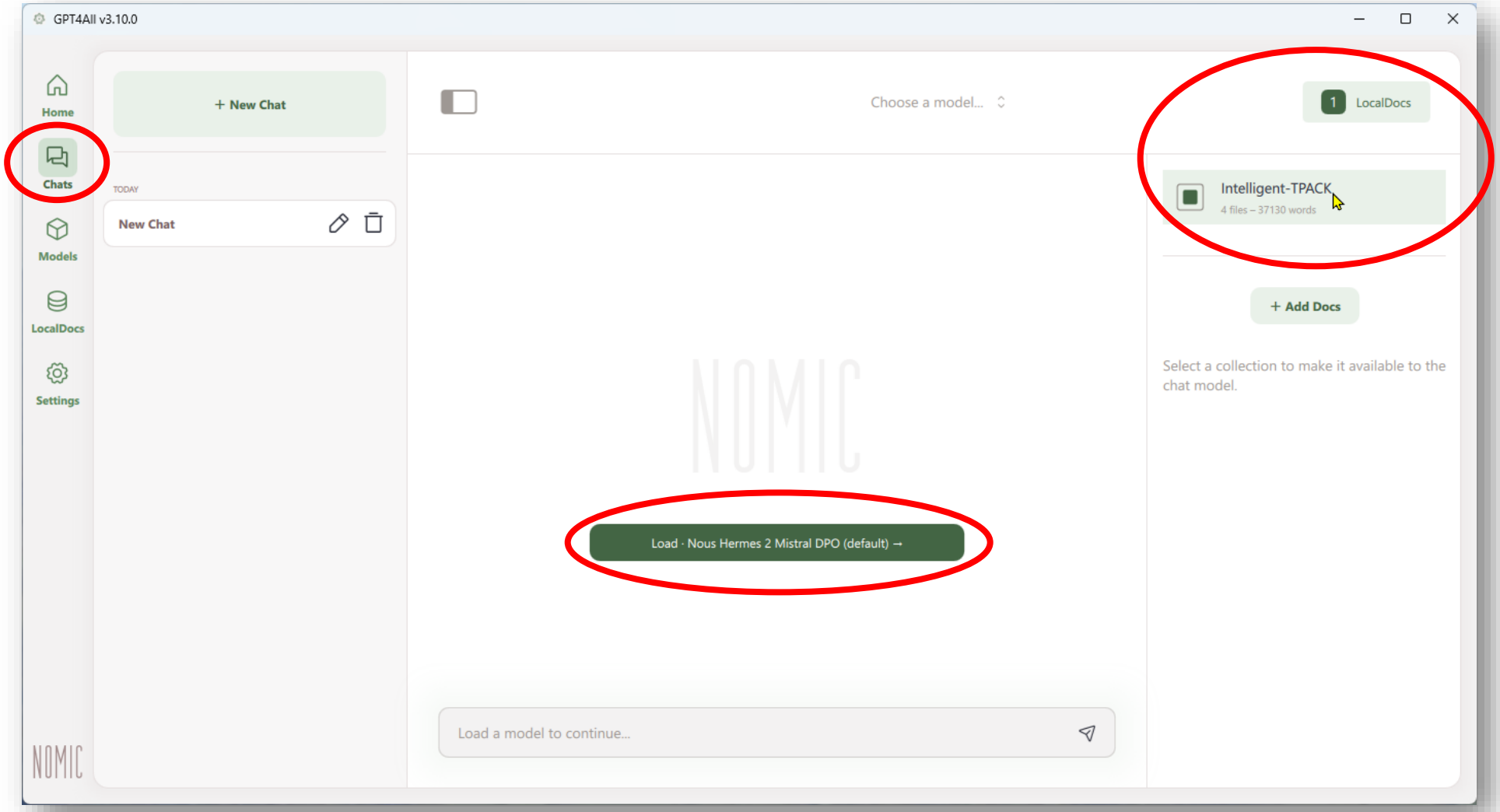
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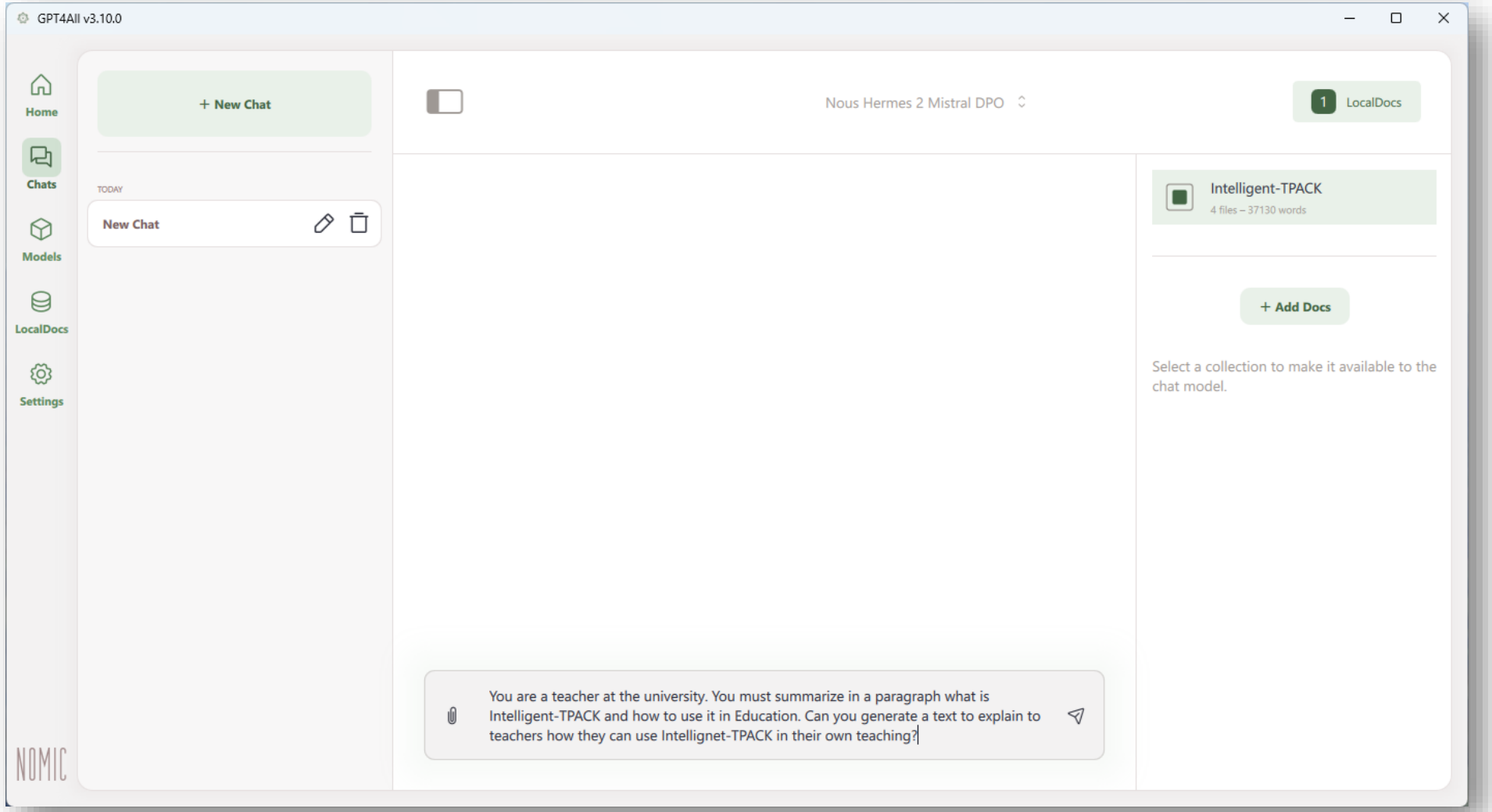
5. LOCAL AI CHATBOTS



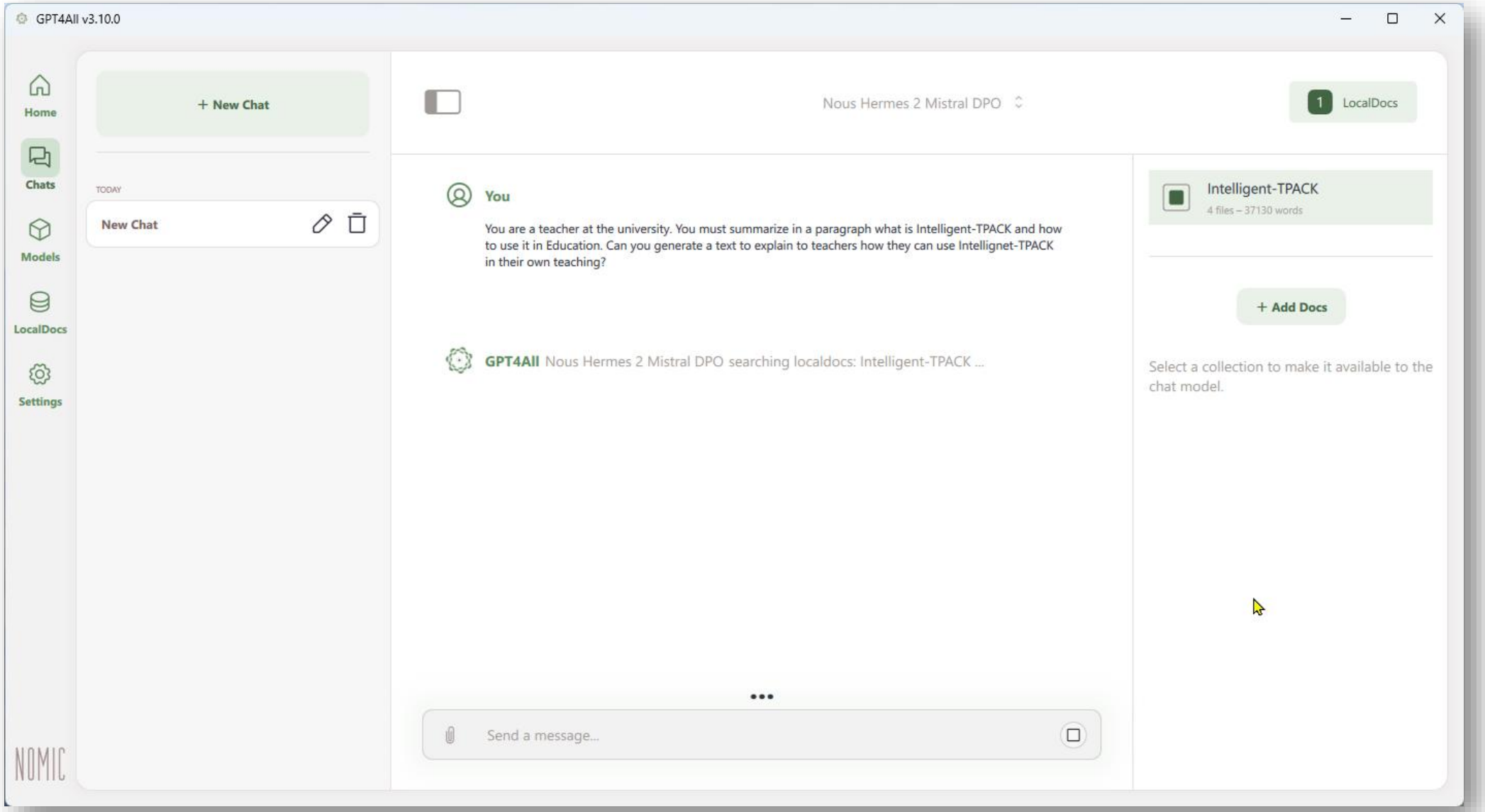
5. LOCAL AI CHATBOTS



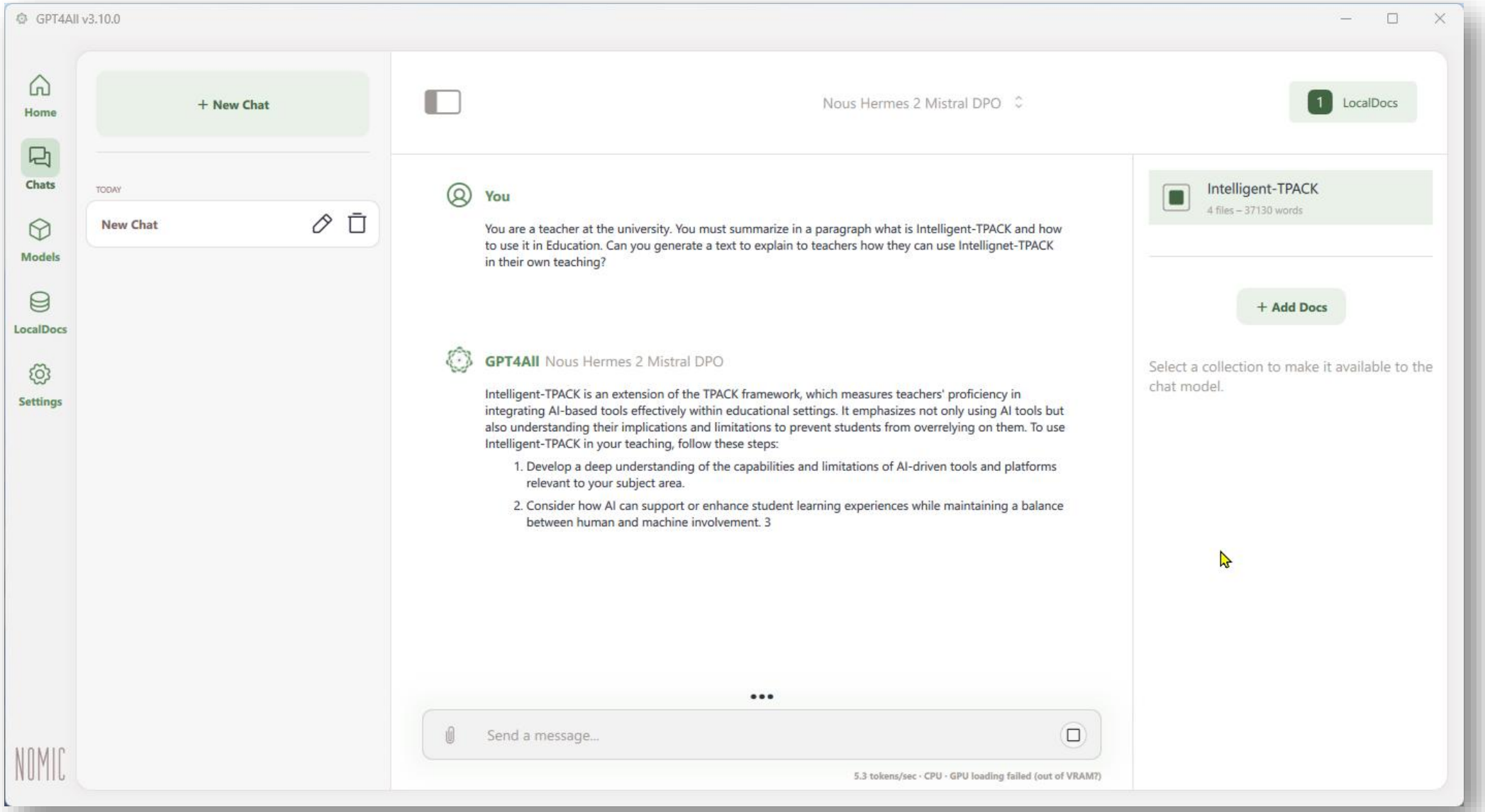
5. LOCAL AI CHATBOTS



5. LOCAL AI CHATBOTS



5. LOCAL AI CHATBOTS



You

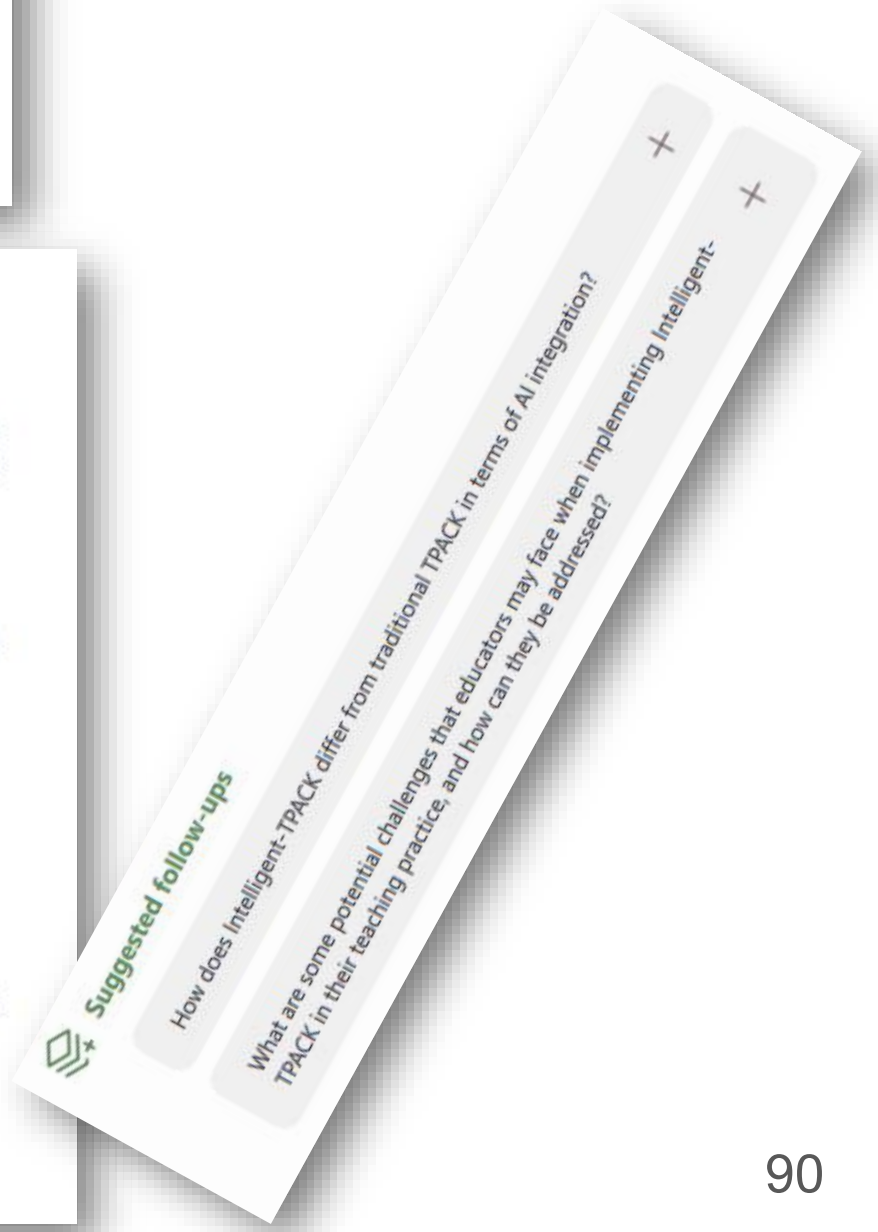
You are a teacher at the university. You must summarize in a paragraph what is Intelligent-TPACK and how to use it in Education. Can you generate a text to explain to teachers how they can use Intellignet-TPACK in their own teaching?

GPT4All Nous Hermes 2 Mistral DPO

Intelligent-TPACK is an extension of the TPACK framework, which measures teachers' proficiency in integrating AI-based tools effectively within educational settings. It emphasizes not only using AI tools but also understanding their implications and limitations to prevent students from overrelying on them. To use Intelligent-TPACK in your teaching, follow these steps:

1. Develop a deep understanding of the capabilities and limitations of AI-driven tools and platforms relevant to your subject area.
2. Consider how AI can support or enhance student learning experiences while maintaining a balance between human and machine involvement.
3. Design lessons that coherently integrate technology with pedagogy, considering both content knowledge (CK) and technological knowledge (TK).
4. Continuously evaluate the effectiveness of your use of AI-based tools in teaching and adjust accordingly to ensure responsible integration.
5. Encourage critical thinking and problem-solving skills among students by guiding them on how to interpret, analyze, and apply information from AI sources.
6. Foster a learning environment that promotes responsible AI usage and emphasizes the importance of human judgment and creativity in decision-making processes.

3 Sources >



6. CONCLUSION

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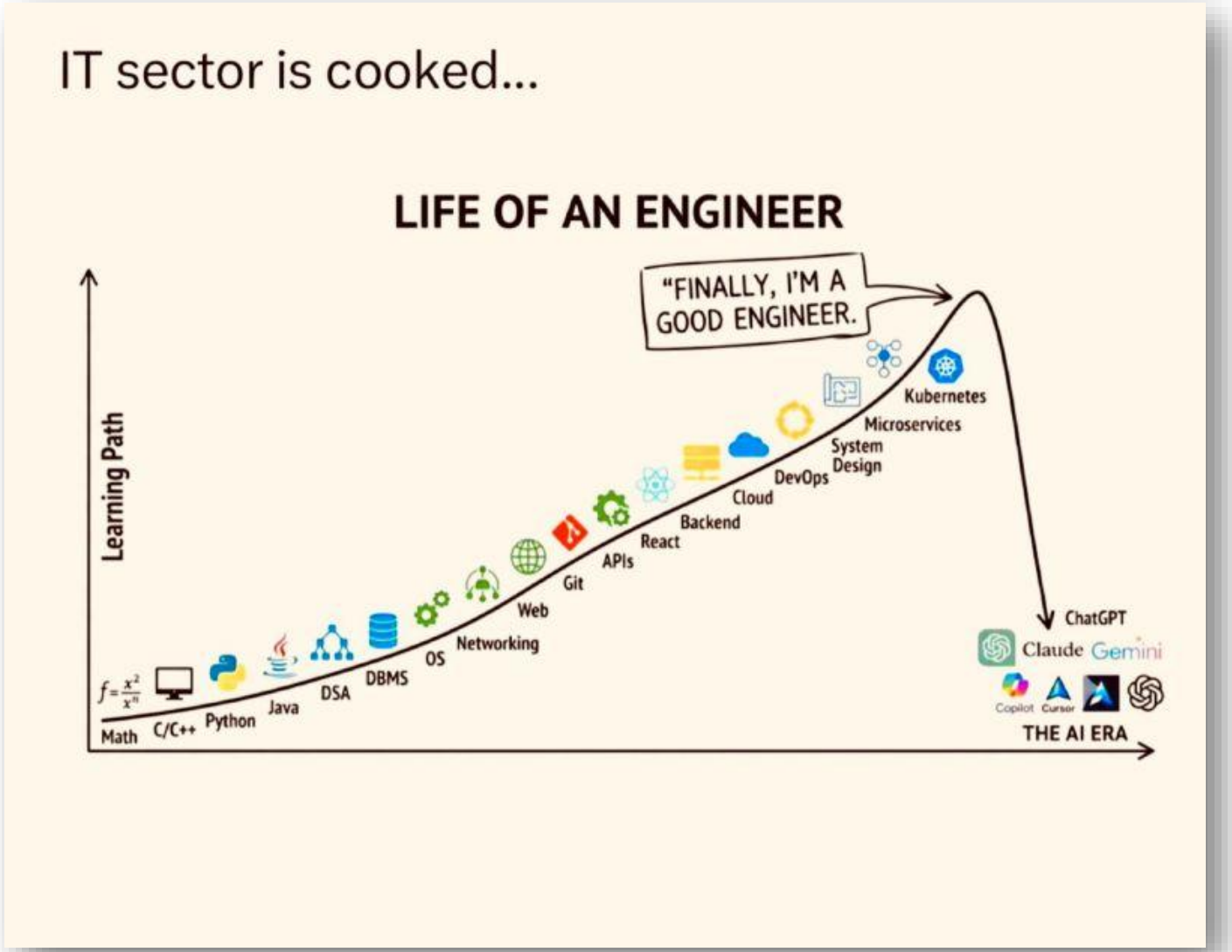
- AI is a game changer in Education/Society
- **Be positive!**
- Keep control over AI: ” I – AI – I ” (https://ju.se/portal/educate/en/guides/artificial-intelligence.html)
- And more, much more than this, 

your 



Thank you, Frank! Source: https://www.youtube.com/watch?v=w019MzRosmk

From LinkedIn:



Source: https://www.linkedin.com/feed/update/urn:li:activity:7506903844375904256?rcm=ACoAADjvy1EBNtqpefnbijtRW5UJVYvxXXvzbpo

Youcef Slimani • 3e et +
Daily Smile and Learn As A Service | Senior Software Engin...
1 j •

+ Suivre ···

🧠 WAKE UP, DEVELOPERS

👉 Big Tech needs adoption → New tools create revenue, users and lock-in.

👉 Every year needs a "revolution" → Cloud, microservices, Kubernetes, AI, agents... the cycle continues.

👉 Developers are the customers too → Your attention, subscriptions, certifications and tool usage have economic value.

👉 Convenience becomes dependency → Why understand something when a tool can generate it instantly?

👉 Generation replaces learning → "It works" slowly replaces "I understand why it works."

👉 Abstraction keeps rising → Developers can ship more while understanding less of the underlying system.

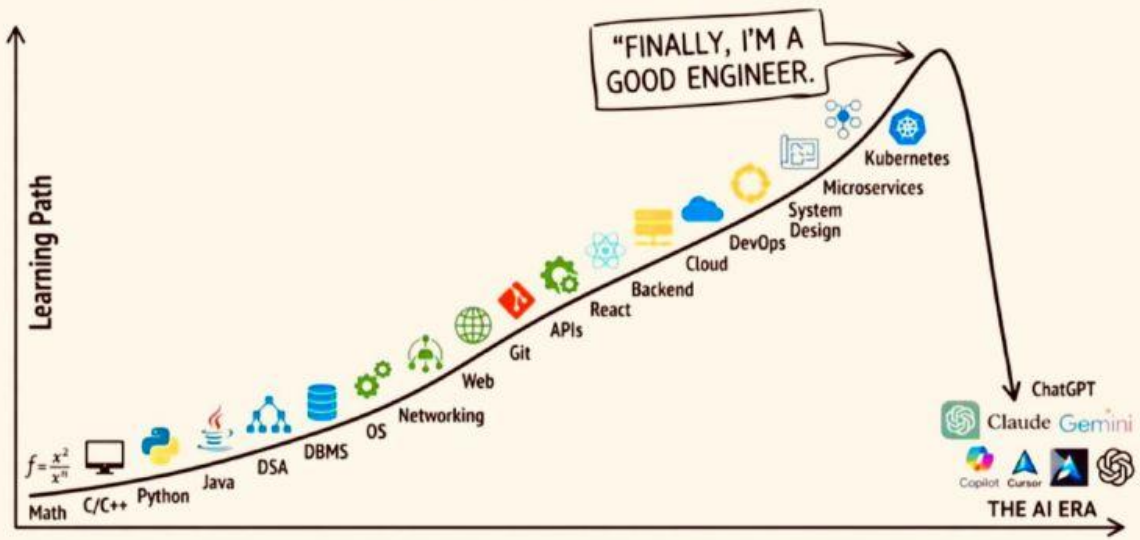
👉 Skills can atrophy → Debugging, algorithms, architecture, SQL, networking and fundamentals require deliberate practice.

👉 The dangerous outcome → Not an unemployed developer, but a developer who can produce code without being able to reason about it.

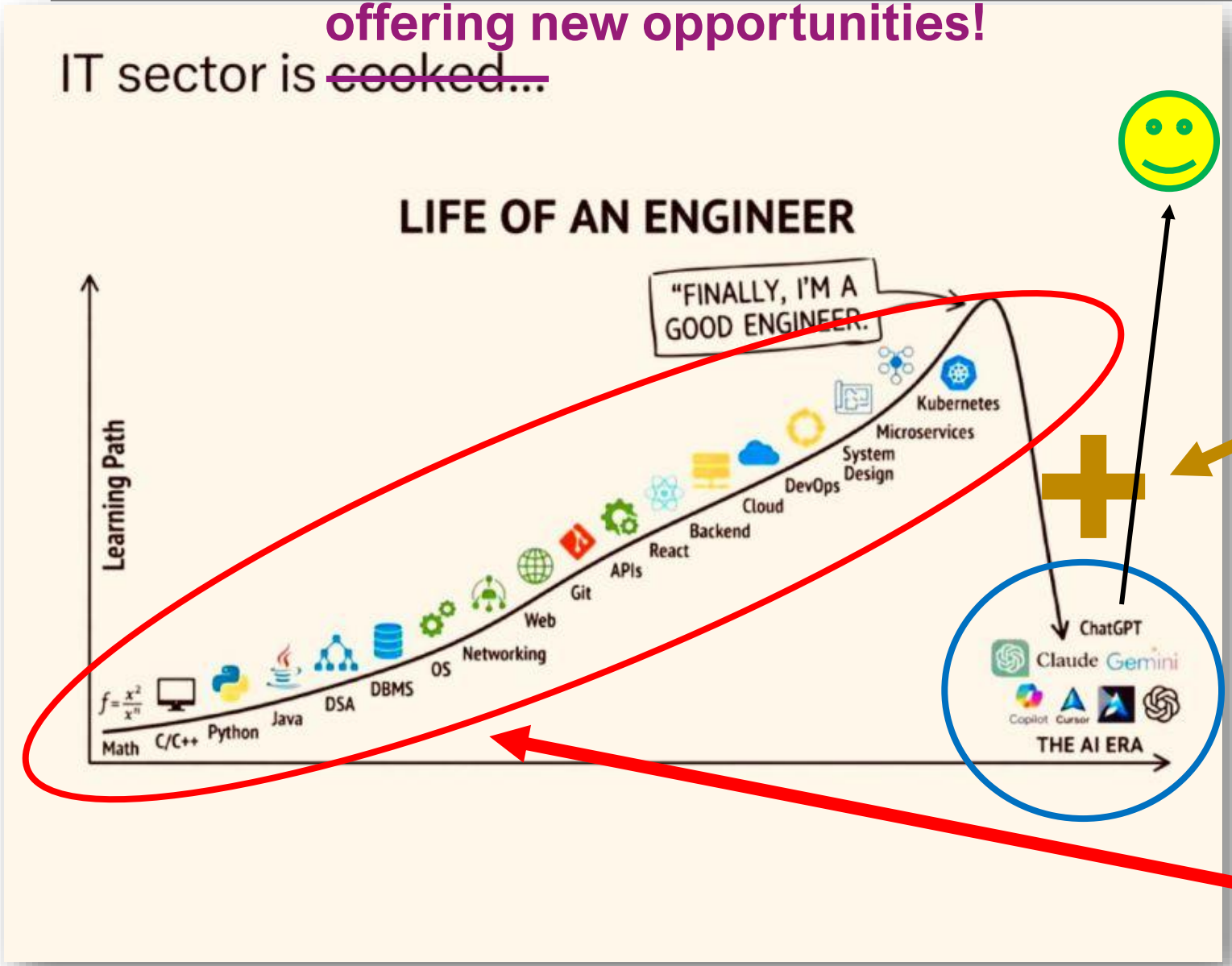
👉 Wake-up call → Use AI and Big Tech tools. Never outsource your brain to them.

The biggest danger isn't that AI will make developers obsolete.
It's that developers will voluntarily make themselves unnecessary by forgetting how to think.

IT sector is cooked...



offering new opportunities!



4. YOU WILL END UP THERE!

3. COMBINE THEM

2. LEARN AI TOOLS/TECHNOLOGIES

1. LEARN THE DEV TOOLS/TECHNOLOGIES

8. REFERENCES

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School of Engineering

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ANY QUESTIONS?


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